

Orbitrap Astral Software Manual

User Guide

Table of Contents

1. Orbitrap Astral

2. About this manual

- 2.1. Special notices and safety symbols
- 2.2. Typographical conventions
- 2.3. Access documentation
- 2.4. Reference documentation
- 2.5. Training

3. Configure the instrument in Thermo Foundation

4. Tune application

- 4.1. Overview of the Tune application
 - 4.1.1. Buttons and icons
- 4.2. Specify options for the Tune application
 - 4.2.1. Set preferences for the Tune application
 - 4.2.2. About dialog box
- 4.3. Display real-time chromatograms and spectra
 - 4.3.1. Control the chromatogram view
 - 4.3.2. Monitor the mass accuracy and signal stability
- 4.4. Zoom in on a spectrum
 - 4.4.1. Header information in the spectrum view
- 4.5. Control the ion source
 - 4.5.1. Parameters on the Ion Source page
 - 4.5.1.1. Emitter Optimization tab
 - 4.5.2. Optimize the ion source parameters
- 4.6. Set up the mass spectrometer for data acquisition
 - 4.6.1. Specify the scan type and the precursor settings
 - 4.6.2. Specify the HCD activation type settings
 - 4.6.3. Acquire a data file with the Tune application

4.7. Tune, calibrate, or check the mass spectrometer calibration

4.7.1. Calibration procedures

4.8. Generate reports

4.9. Use the panes at the right side

4.9.1. Favorites pane

4.9.2. Status pane and Control pane

4.9.3. History pane

4.9.4. Camera pane

4.10. Use the alerts console

4.11. Run diagnostics

5. Method Editor application

5.1. Menus

5.2. Global system parameters

5.2.1. Global MS settings

5.2.2. Ion Source Properties pane and timeline

5.2.3. Syringe Properties pane and timeline

5.2.4. Divert Valve Properties pane and timeline

5.2.5. Output Contact Closure Properties pane and timeline

5.3. Method Editor overview

5.3.1. Method Editor elements

5.3.2. Method timeline

5.3.3. Experiment workflow area

5.3.3.1. Experiment controls

5.3.3.2. Standard View workflow objects

5.3.4. Set the Method Editor preferences

5.3.5. Experiments and instrument methods

5.3.5.1. Create instrument methods

5.3.5.2. Add and delete experiments in a method

5.3.5.3. Save instrument methods

5.3.5.4. Add custom method templates

5.4. Standard view

5.4.1. Master scans

5.4.2. Full Scan Properties pane

5.4.3. Targeted Selected Ion Monitoring (tSIM) Scan Properties pane

5.4.4. Targeted MS2 (tMS2) Scan Properties pane

5.4.5. Data-Independent Analysis (DIA) Properties pane

5.4.6. All Ion Fragmentation (AIF) Scan Properties pane

5.4.7. Data-dependent scans

5.4.7.1. Data-Dependent Properties pane

5.4.7.2. Data-Dependent SIM Scan (ddSIM) Properties pane

5.4.7.3. Data-Dependent MS2 Properties pane

5.4.8. Filters

5.4.8.1. Precursor Selection Range filter

5.4.8.2. Monoisotopic Precursor Selection (MIPS) filter

5.4.8.3. Intensity filter

5.4.8.4. Precursor Fit filter

5.4.8.5. Charge State filter

5.4.8.6. Dynamic Exclusion filter

5.4.8.7. Targeted Inclusion - Targeted Mass filter

5.4.8.8. Targeted Inclusion - Targeted Mass Difference filter

5.4.8.9. Targeted Inclusion - Targeted Isotopic Ratio filter

5.4.8.10. Targeted Exclusion - Targeted Mass Exclusion filter

5.4.8.11. Apex Detection filter

5.4.8.12. Isotope Exclusion filter

5.4.9. Targeted Mass Trigger

5.4.10. Alternate precursor sorts

5.5. Use the experiment templates

5.5.1. Save custom templates

5.5.2. System templates

5.6. Summary page

- 6. Customer feedback**
- 7. Copyright and trademarks**
- 8. Glossary**
- 9. Index**

Orbitrap Astral

Mass Spectrometer

Software Manual

BRE0061125

Revision C

May 2025

About this manual

IMPORTANT

Export classification: not listed / Exportklassifizierung: nicht ausfuhrgenehmigungspflichtig

The Orbitrap Astral™ system is a member of the Thermo Scientific™ family of mass spectrometers that are powered by Orbitrap™ technology.

Contents

- Special notices and safety symbols
- Typographical conventions
- Access documentation
- Reference documentation
- Training

The Orbitrap Astral Software Manual describes software-related operating procedures of the Orbitrap Astral mass spectrometer.

To obtain a good understanding of the complete system, we recommend that you study the Orbitrap Astral Operating Manual. This manual also contains extensive information that concerns the safety of the personnel that operate the instrument. A basic knowledge of handling computers and of the Xcalibur™ software is assumed for the correct operation of the Orbitrap Astral system.

Designed, manufactured and tested in an ISO9001 registered facility, this instrument has been shipped to you from our manufacturing facility in a safe condition. This instrument must be used as described in this manual. Any use of this instrument in a manner other than described here may result in instrument damage and/or operator injury.

2.1 Special notices and safety symbols

Make sure you understand the special notices, symbols, and caution labels in this guide. Most of the special notices and cautions appear in boxes; those pertaining to safety also have corresponding symbols.

NOTICE

Highlights information necessary to prevent damage to software, loss of data, or invalid test results; or might contain information that is critical for optimal performance of the product.

NOTE

Highlights information of general interest.

TIP

Highlights helpful information that can make a task easier.

2.2 Typographical conventions

Different typographical conventions have been established for Thermo Fisher Scientific manuals.

Viewpoint orientation

The expressions *left* and *right* used in this manual always refer to the viewpoint of a person that is facing the front side of the instrument.

Data input

Throughout this manual, the following conventions indicate data input and output with the computer:

- Messages displayed on the screen are represented by capitalizing the initial letter of each word and by italicizing each word.
- Input that you enter by keyboard is identified by quotation marks: single quotes for single characters, double quotes for strings.
- For brevity, expressions such as “choose **File > Directories**” are used rather than “pull down the **File** menu and choose **Directories**.”
- Any command enclosed in angle brackets < > represents a single keystroke. For example, “press <F1>” means press the key labeled *F1*.
- Any command that requires pressing two or more keys simultaneously is shown with a plus sign connecting the keys. For example, “press SHIFT + F1” means press and hold the <Shift> key and then press the <F1> key.
- Any button that you click on the screen is represented in bold face letters. For example, “click **Close**.”

2.3 Access documentation

You can access product documentation using one of the following ways:

Visit the Help Portal

Full documentation (including release notes, installation instructions, operation manuals, and so on) is maintained on the Thermo Fisher Scientific Chromatography and Mass Spectrometry Help Portal. This documentation portal is regularly updated with the latest content, including video tutorials. Go to docs.thermofisher.com.

With a Thermo Fisher Scientific account, you can use features on the documentation portal, such as bookmarking documents, saving searches, and compiling documents into collections for easy reference.

Procedure

1. Go to docs.thermofisher.com/p/TechDocAccess to display a registration form.
2. Fill out the form to request access as a customer. Among other data, you have to provide the serial number (S/N) of your instrument. You can find the serial number of your instrument on the name plate. Refer to the Operating Manual of your instrument for information about its location.
3. Click **Submit Form**. The form contents will be sent using your default email client. You will receive an email with additional login information.

Result

Upon submitting the form, you will be directed to home page of the Help Portal.

View the product documentation on the data system computer

To view the product manuals, do one of the following:

- From the Windows™ taskbar, select **Start > Thermo Instruments**, and then open the applicable PDF file.
- From the Windows explorer, navigate to the following path:
C:\Program Files\Thermo Scientific\Instruments\TNG*instrument mode\version\system\Manuals*

2.4 Reference documentation

Reference documentation for the Orbitrap Astral MS includes the following:

- *Orbitrap Astral Operating Manual*

This manual contains precautionary statements that can prevent personal injury and instrument damage. It also describes the modes of operation and principle hardware components of the instrument. In addition, this manual provides instructions for cleaning and maintaining the instrument.

- *Orbitrap Astral Preinstallation Requirements Guide*

This manual contains information on the necessary environmental conditions in the intended location for the instrument.

- *Orbitrap Astral Performance Maintenance Manual*

This manual describes the user maintenance for the quadrupole and the bent flatapole.

You can access PDF files of the documents listed above and of this manual from the data system computer. Refer also to the user documentation that is provided by the manufacturers of third-party components:

- Forepump
- Turbomolecular pumps
- Syringe pump
- Data system computer and monitor
- Safety data sheets

2.5 Training

Thermo Fisher Scientific offers worldwide training on instruments and software. Experience has shown that maximum results can be obtained from a scientific instrument if the instrument operators receive an adequate training.

We recommend that the key users participate in a basic operator training. For information on training courses and enrollment, contact your local Thermo Fisher Scientific office.

Configure the instrument in Thermo Foundation

To establish control of the mass spectrometer from the data system computer, use the Thermo Foundation™ Instrument Configuration window to configure the instrument.

TIP

The Instrument Configuration window displays all changed set values of given parameters in bold type until you select **OK** or **Apply**.

To enable the new settings, restart the data system computer.

Procedure

1. From the *Windows* taskbar, select Start > All Apps > Thermo Foundation x.x > Instrument Configuration, where x.x. is the installed version.
2. Add the mass spectrometer's icon to the list of Configured Devices, and then configure the mass spectrometer parameters.
3. Select Done.

Table Parameters in the Instrument Configuration window

Parameter	Description
System Information	
Model	Displays the mass spectrometer model name.
Serial Number	Display the serial number of the instrument.
Instrument Name	(Optional) The name for the instrument.
Inlet	
Divert Valve	• None (for no divert valve) (Default)

Parameter	Description
	• A (for a system with one divert valve) • A and B (for a system with two divert valves)
Syringe Pump	Configured: Indicates that the system includes a data-system controlled syringe pump. Default: Not Configured
Contact Closure	Select the event type that triggers the contact closure: • Transition close-to-open (Default) • Transition open-to-close
Wait for Gas Flows to Stabilize	When the Wait for Gas Flows to Stabilize checkbox is selected, the instrument remains in Preparing for Run state until the following ion source gas pressures have stabilized depending on the used ion source: • Sheath Gas (Arb) • Aux Gas (Arb) • Sweep Gas (Arb) When the ion source gases have reached the values defined in the instrument method, data acquisition starts.
Wait for Temperatures to Stabilize	When the Wait for Temperature to Stabilize checkbox is selected, the instrument remains in Preparing for Run state until the following ion source temperatures have stabilized depending on the used ion source: • Ions Transfer Tube Temp (°C) • Vaporizer Temp (°C) When the ion source temperatures have reached the values defined in the instrument method, the data acquisition starts.
Ion Source	
Default Source	• NSI (optional) • Heated ESI (H-ESI) • APCI (atmospheric pressure chemical ionization) (optional) • ESI

Parameter	Description
	NOTE: Although atmospheric pressure photoionization (APPI) is not a configuration option, the mass spectrometer automatically detects if the ion source contains the APPI vacuum ultraviolet (VUV) lamp through the lamp's USB connection.
Default Source Type	Default Source Type: Heated ESI or APCI • Dedicated Heated ESI or Dedicated APCI • Combination Source
Enable Sweep Gas for NSI Source	Provides sweep gas flow to the NSI source. IMPORTANT: Sweep gas can help in the desolvation of the nanospray, but excessive sweep gas can deflect the nanospray from the orifice of the ion transfer tube, which decreases sensitivity. You must have the ion sweep cone installed for the sweep gas to work.
Enable Minimum Gas Flow Requirement for Sealed Sources (H-ESI, ESI, APCI, and APPI)	Provides minimum background flow of ion source gases to keep the source housing pressurized. IMPORTANT: When the ion source is in off mode or standby mode, it is pressurized with background nitrogen gas to avoid back-streaming from the drain and introducing contaminants.
Optional Hardware	
Internal Calibration (EASY-IC) Source	Activates or deactivates the EASY-IC™ ion source.
Analog Inputs	
Channel 1 in Use	Reads the 0–10 Vdc analog input signal from an external device and writes the data into the raw data file. For additional information, refer to the instrument manual.
Channel 1 Label	The name you provide for channel 1.
Channel 2 in Use	Reads the 0–10 Vdc analog input signal from an external device and writes the data into the raw data file. For additional information, refer to the instrument manual.
Channel 2 Label	The name you provide for channel 2.
Historical Data Retention	

Parameter	Description
	Once a week on the specified day and time, the data purging process will initiate. This process only occurs if the instrument is in an idle state. In addition to these settings, automatic purging managed by the TNG database service can occur nightly at 11pm if the instrument is in an idle state.
Day	Use the list box to select the weekday for the data retention. Default: Sunday
Time	Use the list box to select the hour for the data retention. Default: 11 PM
Buttons	
OK	Saves the instrument configuration settings and closes the window.
Cancel	Closes the Instrument Configuration window without saving your changes.
Apply	Applies the changes you made in the Instrument Configuration window.

Tune application

Use the Tune application to control, monitor, tune, and calibrate the Thermo Scientific™ Orbitrap Astral mass spectrometer (MS). You can also optimize the instrument for your specific compounds, perform method development, and run real-time mass spectrometry experiments one analysis at a time.

Contents

- Overview of the Tune application
- Display real-time chromatograms and spectra
- Zoom in on a spectrum
- Control the ion source
- Set up the mass spectrometer for data acquisition
- Tune, calibrate, or check the mass spectrometer calibration
- Use the panes at the right side
- Generate reports
- Specify options for the Tune application
- Run diagnostics

4.1 Overview of the Tune application

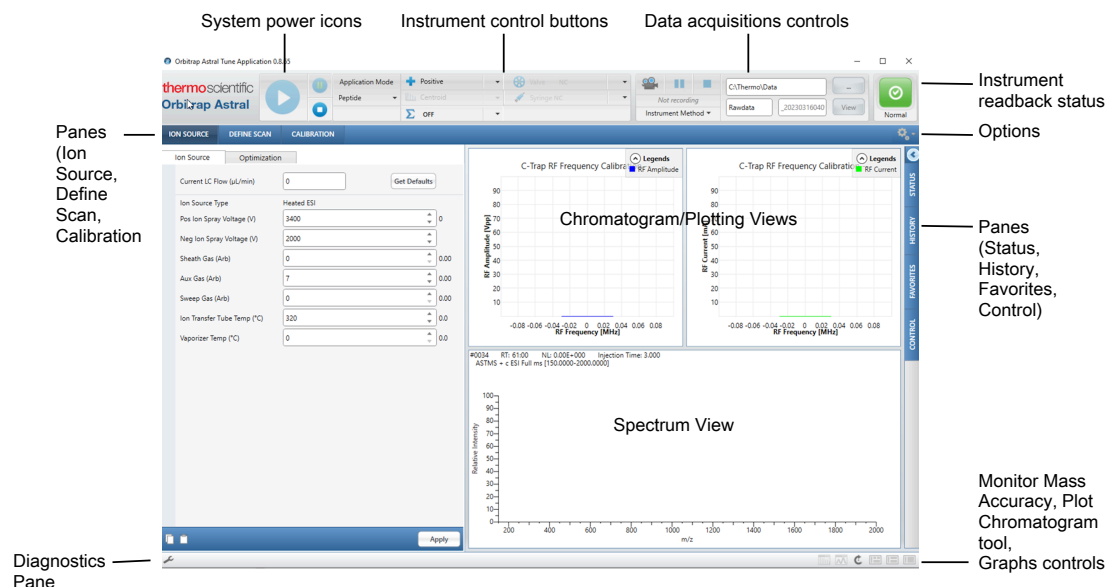
Use the Tune application to operate the Orbitrap Astral MS with the data system computer. The Tune application organizes its functions in panes and on pages. To display the panes and pages, select the associated buttons and tabs. If information about setting the parameters for a specific view, page, or dialog box is available, select **Options > Help** or press the F1 key.

The Tune application has the following main components:

- Three system power icons to set the MS's power mode (on, standby, and off).
- The system readback icon to indicate the various readback states.
- Three panes on the left to define parameters for scanning, ion source, and calibration.

- Four panes on the right to display information about instrument status, history and favorite settings.
- Up to three graphs to show real-time information about the current measurement.

Figure Tune application



Display the Tune application

Select **Start > All Programs > Thermo Instruments > Orbitrap Astral Tune**.

TIP

You can set a few preferences for how the Tune application works. Select the Options icon, , and then select **Preferences**.

4.1.1 Buttons and icons

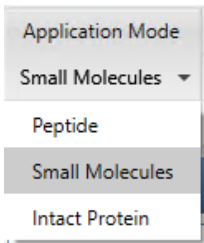
For complete descriptions of features, see the applicable table of the Tune icons and buttons.

System power icons

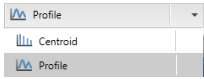
Button	Description
A green button indicates the active instrument status.	
On mode	In the On state, all voltages, the sheath gas, the auxiliary gas, and the sweep gas are on.

Button	Description
	
Standby mode	In the Standby state, the high voltages and the vaporizer heater are off. The RF voltages and the DC offset voltages remain on. The sheath gas and the auxiliary gas remain on at a low flow rate (6 arbitrary units).
	
Off mode	In the Off state, the RF and high voltages, ion transfer tube heater, and vaporizer heater are off. In addition, the offset voltages are off. The sheath gas, auxiliary gas, and ion sweep gas remain on at a low flow rate to flush vapors from the ion source.

Application mode

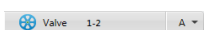
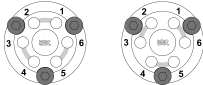
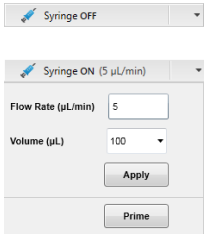
Button	Description
	See the menu item descriptions below.
Peptide	This application mode provides default instrument settings for the analysis of complex peptide mixtures.
Small Molecule	This application mode optimizes instrument settings for the analysis of small molecules with focus on quantification.
Intact Protein	This application mode optimizes instrument settings for the analysis of large molecules such as intact proteins, monoclonal antibodies, and other macromolecules. You can use the pressure mode settings for optimized analysis of intact proteins under denatured or native conditions. NOTE: The Application Mode Intact Protein is available only when the BioPharma license is installed. NOTE: The Application Mode Intact Protein does not support the Detector Type Astral.

Instrument and spectrum buttons

Button	Description
	Indicates the ion polarity mode (positive or negative).
	Indicates the data type mode (centroid or profile).
	Indicates whether the spectrum scan averaging is off or on and has the number of scans to average. Range: 2–1000 NOTE: The Tune application displays the averaged spectrum in a brown color. The spectrum header displays the number of averaged scans in the scan filter (for example, AV:2).
	Indicates the mass spectrometer pressure mode. Available options are: • Standard Pressure: Optimal detection for intact protein mass and top-down detection of large fragment ions. • Low Pressure: Optimal detection for small molecules, bottom-up and top-down protein experiments. • High Pressure: Optimal detection for native protein analysis experiments. NOTE: This feature is available only when the BioPharma license is installed. NOTE: This feature is not available for Astral detector.

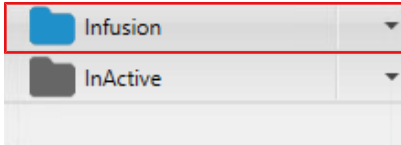
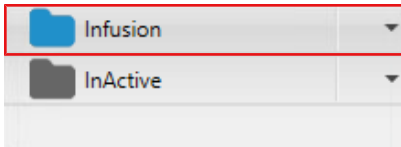
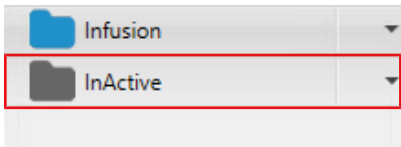
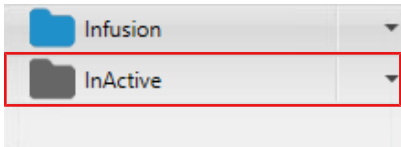
Modular valve and syringe pump buttons

The Tune application displays these buttons for H-ESI, APCI, APPI, and NSI modes.

Button	Description
	Indicates the position (1-2 or 1-6) for two divert/inject valves (A and B) on the button left side. The down arrow opens two choices: valve A and optional valve B. 
	Indicates whether the syringe pump is pumping. The syringe pump settings box defines the pump flow rate and the syringe volume. You can also prime the syringe by pressing and holding Prime. Syringe pump flow rate (µL/min): 0.05–500 Syringe volumes: 50, 100, 250, 500, 1000, 5000, and 10 000 Syringe priming (µL/min): 100

OptiSpray cartridge controls

(Available when the OptiSpray ion source is connected to the MS.)

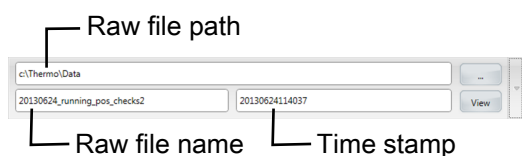
Button	Description
	Cartridge type Select this button to display the OptiSpray cartridge details.
	
	Cartridge status Indicates the OptiSpray cartridge's current status. Select this button to switch the OptiSpray cartridge between the Inactive and Active positions. This button is unavailable during emitter optimization. Default active: Emitter is positioned in front of the ion transfer tube (default: 44.50 mm x, 5.66 mm y, 4.80 mm z) NOTE: After an emitter optimization (quick, full,
	

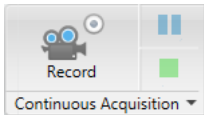
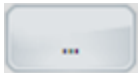
Button	Description
	or manual) has been performed, the optimized position is saved as the new active position. Inactive: Emitter is positioned away from the ion transfer tube (30.00 mm x, 5.66 mm y, 4.80 mm z) NOTE: The inactive emitter position cannot be changed.

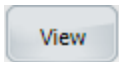
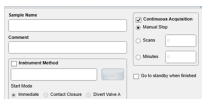
Data acquisition buttons

The Tune application saves the acquired data to a raw data file (.raw extension).

Figure Raw file data



Button	Description
Select the label below the Record button to expand the acquisition settings box. 	Select the Record toggle button to acquire data to a new raw data file. The name of the file—if not changed—is a combination of the base file name and a time stamp, which consists of the year (YYYY), month (MM), day (DD), and time (HHMMSS). To stop the data acquisition, select Stop (blue square). The color of the square then changes to green. To pause the data acquisition, select Pause (two blue bars). The color of the bars then changes to orange. To resume the data acquisition, select Paused (shown as an orange circle). NOTE: If the Instrument Method checkbox in the data acquisition settings box is selected, the Pause feature is not available.
	(Optional) To change the destination folder for the raw data, select the Browse button. The default folder location is in drive:\Thermo\Data.

Button	Description
	You must manually set FreeStyle as the default application to view the RAW files. (To specify the default application, right-click the RAW file, then navigate to ThermoFisher.FreeStyle.App.exe (C:\Program Files (x86)\Thermo\FreeStyle).
Sample information and data acquisition method	
	Select the button to the left of the View button to expand the acquisition settings box. As shown in this figure, you can select only one checkbox. 
Sample Name	The name of the sample.
Comment	A comment that describes the acquisition.
Instrument Method	Indicates to load an instrument method and start the data acquisition with one of these options: • Immediate: Starts the acquisition immediately when you select Record. • Contact Closure: Starts the acquisition when the mass spectrometer receives a contact closure signal from the connected analog autosampler. • Divert Valve: Starts the acquisition when the position of the divert/inject valve changes from its current position.
Continuous Acquisition	Indicates how long the mass spectrometer acquires data as follows: • Manual Stop: The mass spectrometer acquires data until you select Stop. • Scans: The number of scans acquired by the mass spectrometer before the data acquisition stops. • Minutes: The time (in minutes) that the mass spectrometer acquires data.
Go to Standby When Finished	The mass spectrometer goes in standby mode when the current acquisition is completed.

Button	Description
Enable FAIMS CV Scan	This parameter is available only when the system has detected an FAIMS source. Indicates the use and configuration of the FAIMS compensation voltage (CV) scan tool. The Start and Stop values defining the FAIMS CV range. The Step value defines the step size in volts within the range. The Estimated Duration read-only value counts down the estimated completion time (in seconds) for the CV scan. Range (start/stop): -300 to 300; default: -100 to 100 Range (step): 0.1–2; default: 1

Instrument Status Icons

Icon	Description
NOTE: When you start data acquisition, the instrument status icon does not change.	
Normal 	The system parameters are within tolerance.
Initializing 	The system is initializing.
Changing Settings 	One or more settings are changing.
Source Off 	The ion source is off.
Disconnected 	There is no communication between the mass spectrometer and the data system.
Error	An error has occurred.

Icon	Description
	
Miscellaneous Icons	
Icon	Description
This icon appears at the far right side of the Tune window, below the Instrument Status icon.	
	Displays the Options menu.
The Copy and Paste icons appear at the bottom of the Ion Source page.	
Copy	(Ion Source page)
	Copies the ion source parameters to the clipboard.
Paste	(Ion Source page)
	Pastes the ion source parameters from the clipboard.
TIP: You can use the clipboard to transfer the ion source parameters between the Tune application and the Method Editor application.	
Diagnostics	Displays the Diagnostics pane.
	
Monitor Mass Accuracy	Displays the Monitor Mass Accuracy dialog box. Plots the difference between the measured peak position and the theoretical peak positions in the mass list. To start or stop monitoring the mass accuracy, select the icon.
	
Plot Chromatogram	Displays the Plot Chromatogram dialog box. Plots a chromatogram for the TIC/Base Peak or User Defined <i>m/z</i> . Additionally, the Spray Stability as %RSD of the ion current can be plotted. The %RSD must be less than 15% for the spray stability to pass the evaluation in the beginning of a calibration procedure. To start or stop the ion current display, select the icon.
	
Rotate Plots	Rotates the displayed graphics.
	

Icon	Description
Graph mode 	Divides the screen into up to three areas.

4.2 Specify options for the Tune application

The Options menu of the Tune application has the following commands:

Command	Description
Preferences	Displays the Tune Preferences dialog box
Load standard diagnostics	Displays the Diagnostics pane (left pane).
View calibration reports	Opens the folder that contains the calibration reports.
View other reports	Opens the folder that contains the other reports.
View Instrument	Displays a window that shows a schematic drawing of the instrument ion optics and analyzer.
Tune Help	Displays the Tune Help.
Instrument Manuals	Opens the folder that contains the instrument manuals that are installed with the Tune application.
Instrument Web page	Displays the website of the Orbitrap Astral mass spectrometer.
About Tune	Displays the About dialog box with information about the instrument, the current Tune version, and the active licenses.

Display the Options menu

Select the Options icon, .

4.2.1 Set preferences for the Tune application

Use the Tune Preferences dialog box to specify the behavior of the Tune application.

Figure Tune Preferences dialog box



Tune Preferences



General

- ☒ Clear Calibration check boxes when complete
- ☒ Clear Diagnostics check boxes when complete
- ☐ Enable Hotlink for Define Scan and Ion Source Panel

Report Options

Calibration Reports

- ☒ Automatically generate reports

Path:

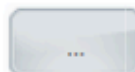


- ☐ Show Report Generation Options dialog box
- ☐ Do not generate reports

Other Reports

- ☐ Automatically generate reports

Path:



- ☒ Show Report Generation Options dialog box
- ☐ Do not generate reports

Report Content Options

- ☒ Show Console
- ☒ Show graph
- ☒ Show spectrum
- ☒ Show system configuration
- ☒ Show embedded system configuration

Alerts Console Options

- ☒ Show warnings
- ☐ Show recovered errors and warnings
- ☐ Show information
- Minutes included in the log file before error detected
- Minutes included in the log file after error detected

OK

Cancel

The Tune Preferences dialog box has the following parameters:

Parameter	Description
General	• Clear Calibration checkboxes when complete • Clear Diagnostics checkboxes when complete • Enable Hotlink for Define Scan and Ion Source Panel Select the Hotlink checkbox to allow the Tune application to send any changes immediately to the instrument.
Report Options	You can select different options for calibration reports and other reports: • Automatically generate reports To change the destination folder for the reports, select the Browse button. The default folder location is in C:\ProgramData\Thermo\Astral\Instrument\Reports\. • Show Report Generation Options dialog box • Do not generate reports
Report Content Options	• Show Console • Show graph • Show spectrum • Show system configuration • Show embedded system configuration
Alerts Console Options	Use these parameters to specify what is displayed in the alerts console. The log files are stored in C:\ProgramData\Thermo\Astral\Log. • Show warnings • Show recovered errors and warnings • Show information • Minutes included in the log file before error detected • Minutes included in the log file after error detected

Display the Tune Preferences dialog box

Procedure

1. Select the Options icon, .
2. Select Preferences.

4.2.2 About dialog box

Use the About dialog box to display information about the instrument, the current Tune version, and the active licenses. To copy the instrument identification to the clipboard, select the  button.

The About dialog box has the following parameters:

Parameter	Description
End User License Agreement	Display various dialog boxes with legal information and information about third party licenses used by the Tune application.
Add license	Displays the License dialog box.
Close	Saves your changes and closes the dialog box.

Display the About dialog box

Procedure

1. Select the Options icon, .
2. Select About Tune.

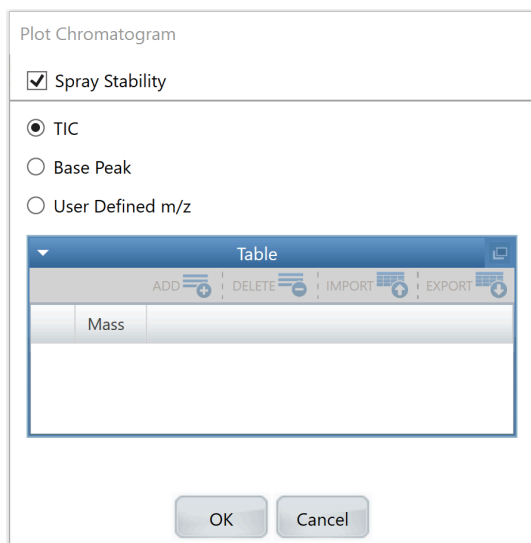
4.3 Display real-time chromatograms and spectra

See these topics and the button descriptions in Buttons and icons.

4.3.1 Control the chromatogram view

Use the Plot Chromatogram dialog box to plot the chromatogram from the TIC, the base peak, or one or more user-defined m/z values. For the user-defined mass table, you can import the settings from (or export them to) a CSV, a TXT, or an XML file.

Figure Plot Chromatogram dialog box



NOTE

The Tune application always displays the Chromatogram view. Observe the chromatogram trace to see how system parameter changes affect the ion signal intensity so that you can optimize the parameter settings as needed.

Display the TIC or the ion current for a select mass as a function of time

Procedure

1. Select the Plot Chromatogram icon, , at the bottom of the Tune window.
2. Select the chromatogram type, and then select **OK**.
Chromatogram data and spray stability data (if selected) are shown in separate windows during the monitoring.
3. To stop the plot, select the Plot Chromatogram icon again.

Parameters in the Plot Chromatogram dialog box

Parameter	Description
Spray Stability	Plots the spray stability. Plots a chromatogram for the TIC/Base Peak or User Defined m/z . Additionally, the Spray Stability as %RSD of the ion

Parameter	Description
	current can be plotted. The %RSD must be less than 15% for the spray stability to pass the evaluation in the beginning of a calibration procedure.
TIC	Plots the TIC chromatogram.
Base Peak	Plots the base peak chromatogram.
User Defined m/z	Plots the signal intensities of the ions you enter in the mass table as a function of time.
Mass table	
Mass	(User Defined m/z) Indicates up to 10 m/z values for the ion signal intensities that you want to plot. Range: 40–6000, default: 195.0877

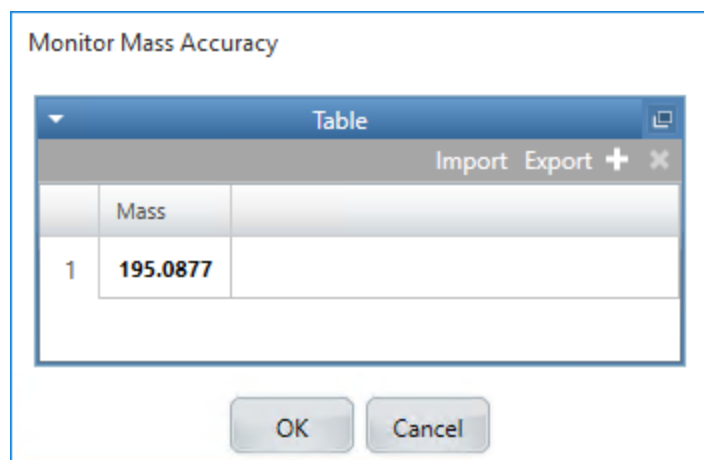
Shortcut menu commands in the chromatogram view

Command	Description
Copy	Copies the chromatogram to the Clipboard.
Print	Sends a copy of the chromatogram to the printer.

4.3.2 Monitor the mass accuracy and signal stability

Use the Monitor Mass Accuracy dialog box to plot the difference between the measured and theoretical peak positions in the mass list. For the mass table, you can import the settings for the mass table from (or export them to) a CSV, a TXT, or an XML file.

Figure Monitor Mass Accuracy dialog box



Procedure

1. Select the Monitor Mass Accuracy icon, , at the bottom of the Tune window.
2. Enter one or more m/z values with up to four decimal places accuracy within the allowed scan range. Then select **OK**.

NOTE

When you enter masses in the table, include at least two decimal places.

A plot “Monitoring Mass Accuracy” starts in the upper view panel: mass list entries are identified within a mass tolerance of ± 25 ppm. Mass accuracy (ppm) is plotted against the scan number. The monitored mass traces are shown in the legend of the plot.

3. To stop the plot, select the Monitor Mass Accuracy icon again.

Parameters in the Monitor Mass Accuracy dialog box

Parameter	Description
Mass	Indicates up to 10 m/z values for the peaks that you want to monitor in the mass list. Range: 40–6000; default: 195.0877
Buttons	
OK	Displays the mass accuracy plot. To stop monitoring the mass accuracy, select the  icon.
Cancel	Closes the dialog box without saving the entered data.

4.4 Zoom in on a spectrum

In the Spectrum view, you can zoom in on one or both axes as needed.

Procedure

1. To zoom in on both axes, drag the cursor diagonally across the spectrum to make a rectangle where the corners define the displayed axes ranges.
2. To zoom in on one axis, drag the cursor horizontally or vertically along the axis or the spectrum to define the displayed range.
3. To go back to the default view, select Reset Scaling in the shortcut menu.

Shortcut menu commands in the Spectrum View

Command	Description
Reset Scaling	Resets the axes to the default intensity and mass ranges.
Copy to Clipboard	Copies a screen capture of the mass spectrum to the Clipboard.
Pan	To use this tool, you must first zoom in on the spectrum. Then you can select this tool to slide the spectrum back and forth along the x axis by holding down the left mouse button and moving the mouse cursor.
Print	Sends a copy of the mass spectrum to the printer.
Show Frequency	Displays the frequency spectrum (in kHz) in blue. Clear this option to display the m/z spectrum.
Show Absolutes	Changes the y-axis scale from percentages to ion flux (ions/s).
Display	• Normalized: The y-axis scale of the mass spectrum normalizes so that the most intense peak equals 100. • Fixed: The y-axis scale of the mass spectrum does not change. • Creep: The y-axis scale of the mass spectrum automatically increases if the peak intensity increases, but does not decrease if the peak intensity decreases.

Command	Description
Profile Label	Adds the resolution (R) and charge state (z) values next to the peak labels.
Profile Label Setting	- Resolution: Calculates the resolution (R) and labels the peaks with it. - Charge: Labels the peaks with the charge state (z).

4.4.1 Header information in the spectrum view

This table defines the information at the top of the real-time spectrum.

Header information	Description
Scan number (#)	The scan count, which only resets to zero when you start data acquisition.
RT	The retention time (in minutes), which also resets to zero when you start a data acquisition.
NL	The normalization level, which is a measure of the ion signal intensity.
Injection time	The injection time (in milliseconds) per scan.
Scan settings (filter)	The detector type, polarity mode (+ or –), data type (profile or centroid), source type, scan type, and other settings.

4.5 Control the ion source

The Ion Source *pane* uses the following real-time indicators to identify the readback status for each parameter:

- Green square ()—Indicates that the parameter has reached the specified value.
- Orange square ()—Indicates that the parameter is trending toward the specified value.
- Red square ()—Indicates that the parameter could not reach the specified value.

NOTE

A red border around a parameter box indicates that the value is out of range.

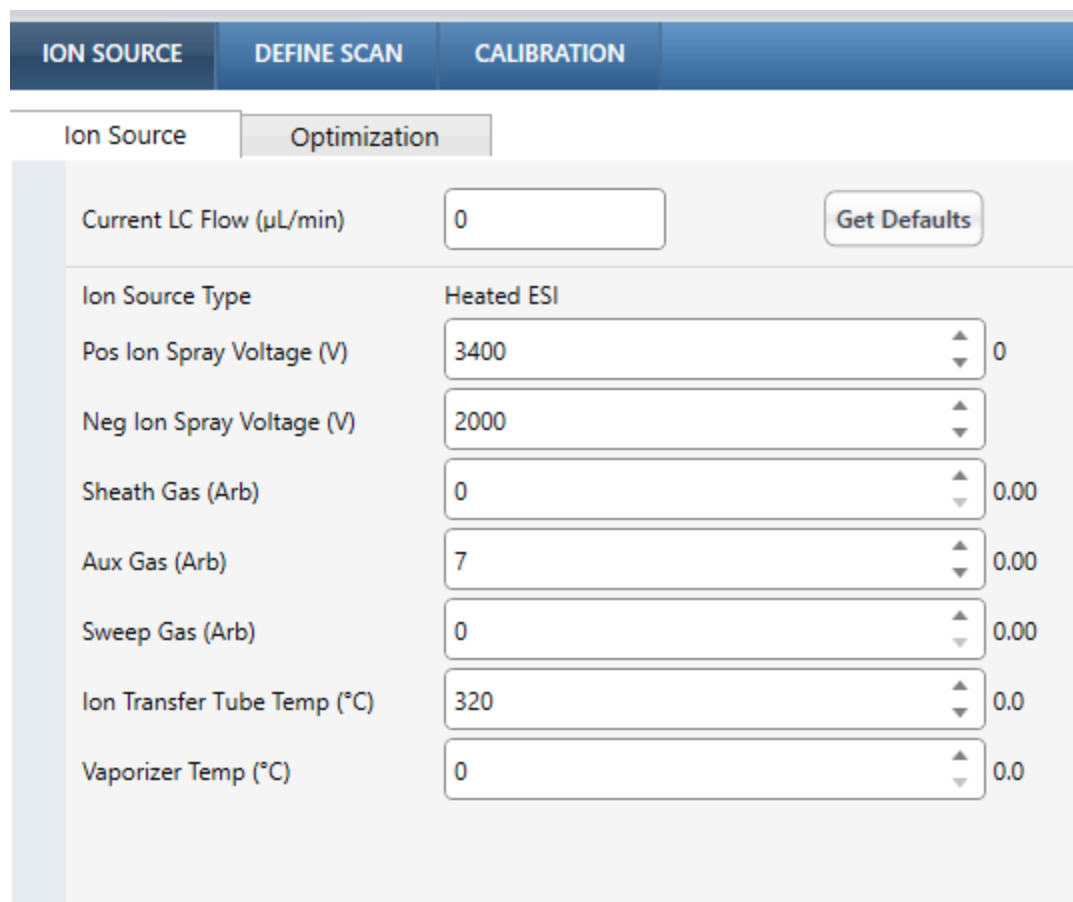
TIP

After setting the parameters on the Ion Source pane, you can use the  button to copy the parameters settings for pasting into an instrument method.

4.5.1 Parameters on the Ion Source page

Use the Ion Source page to set the ion source parameters.

Figure Tune application—Ion Source page



ION SOURCE		
DEFINE SCAN		
CALIBRATION		
ION SOURCE		
Optimization		
Current LC Flow (µL/min)	0	Get Defaults
Ion Source Type		
Pos Ion Spray Voltage (V)	3400	0
Neg Ion Spray Voltage (V)	2000	
Sheath Gas (Arb)	0	0.00
Aux Gas (Arb)	7	0.00
Sweep Gas (Arb)	0	0.00
Ion Transfer Tube Temp (°C)	320	0.0
Vaporizer Temp (°C)	0	0.0

Parameters for the H-ESI, APCI, and NSI ion source types

Parameter	Description
Current LC Flow (µL/min)	(Ion Source Type: H-ESI or APCI) The LC flow rate (µL/min).

Parameter	Description
	Select Get Defaults, and then select Apply to update the ion source parameters. The Tune application sets the applicable ion source parameters to the appropriate default values for the specified LC flow rate. Range: 0–3000; default: 0 NOTE: • Selecting Apply does not start the LC. • When the ion source is in off mode or standby mode, it is pressurized with background nitrogen gas to avoid back-streaming from the drain and introducing contaminants.
Ion Source Type	The configuration for the ion source housing. The instrument automatically recognizes the source type and displays the corresponding parameters for the detected source type. In case of a OptaMaxNG ion source (Heated ESI and APCI), you can select the source type that you want to use. Although atmospheric pressure photoionization (APPI) is not a configuration option, the mass spectrometer automatically detects if the ion source contains the APPI vacuum ultraviolet (VUV) lamp through the lamp's USB connection. NOTE: For some ion source types (MALDI, for example), you can specify additional settings in the Define Scan subtree of the Diagnostics pane.
Pos Ion Spray Voltage (V) Neg Ion Spray Voltage (V)	(Ion Source Type: H-ESI or NSI) The source spray voltage (absolute value, in volts). Positive polarity range: 0–6000 (H-ESI) or 0–3000 (NSI); default: 3400 (H-ESI) or 1800 (NSI) Negative polarity range: 0–5500 (H-ESI) or 0–2300 (NSI); default: 2300 (H-ESI) or 1500 (NSI)
Sheath Gas (Arb)	(Ion Source Type: H-ESI or APCI) The sheath gas flow rate (arbitrary units). Range: 0–80; default: 0 (or determined by the Tune application based on the LC flow rate)

Parameter	Description
Aux Gas (Arb)	(Ion Source Type: H-ESI or APCI) The auxiliary gas flow rate (arbitrary units). Range: 0–25; default: 2 (or determined by the Tune application based on the LC flow rate) NOTE: If the vaporizer temperature is set to values above 100 °C, the auxiliary gas is automatically set to 5 unless it is not set higher manually.
Sweep Gas (Arb)	The sweep gas flow rate. Range: 0–20; default: 0 (or determined by the Tune application based on the LC flow rate) In the Instrument Configuration, you can switch on the sweep gas for the NSI source.
Ion Transfer Tube Temp (°C)	The temperature of the ion transfer tube (capillary) in °C. Range: 0–400; default: 320 (or determined by the Tune application based on the LC flow rate)
Cartridge Temp (°C)	(Ion Source Type: OptiSpray) Specify the temperature of the OptiSpray cartridge (in degrees Celsius). Range: 35–60; default: 45 NOTE: The cartridge temperature appears but is not applied when a cartridge is inserted. To apply the temperature, you must select Apply.
Vaporizer Temp (°C)	(Ion Source Type: H-ESI or APCI) The temperature (in °C) of the APCI vaporizer, which vaporizes sample and solvent, or of the H-ESI source heater, which heats the auxiliary gas. Range: 0–550; default: 0 (off) (or determined by the Tune application based on the LC flow rate) NOTE: You must set the auxiliary gas to 5 or greater for vaporizer temperatures above 100 °C.
Pos Ion Discharge Current (µA)	(Ion Source Type: APCI) The APCI source discharge current (microamperes). Range: 0–100; default: 4 (positive polarity mode) and 10 (negative polarity mode) IMPORTANT: The actual possible maximum value of the discharge current is limited by the maximum value of the
Neg Ion Discharge Current (µA)	

Parameter	Description
	spray voltage of 5000 V.
APPI Lamp	(Ion Source Type: H-ESI or APCI) Switches on or off the optional APPI lamp. Tune displays the readback value for the current lamp state next to the list box, where 0 is off and 1 is on. Default: Off NOTE: The Tune application displays this parameter only when the ion source contains the optional APPI lamp. The lamp state shown in the application might not match the lamp state shown in Method Editor if an instrument method uses APPI mode.

Parameters for the FAIMS Pro system

For additional information, refer to the *FAIMS Pro System User Guide*.

Parameter	Description
Total Carrier Gas Flow (L/min)	The flow rate for the FAIMS and carrier nitrogen gas. Range: 3.5–7.7; default: 4.6
FAIMS Mode	The temperatures for the FAIMS electrodes are as follows: - Standard Resolution: Provides the best transmission mode by setting the inner and outer electrodes to 100 °C. - High Resolution: Provides twice the resolution of the standard resolution mode by setting the inner electrode to 80 °C and the outer electrode to 100 °C. - User Defined: Displays the fields to enter the electrode temperatures. NOTE: The electrode temperatures affect ion separation. Although ion transmission might be lower in FAIMS high resolution mode, the peaks are narrower due to the electrodes' temperature differential.

Parameter	Description
FAIMS Inner Electrode Temp (°C)	(FAIMS Mode: User defined) The maximum temperature for the inner electrode in the FAIMS Pro system (in °C). Range: 70–100; default: 100
FAIMS Outer Electrode Temp (°C)	(FAIMS Mode: User defined) The maximum temperature for the outer electrode in the FAIMS Pro system (in °C). Range: 70–100; default: 100

4.5.1.1 Emitter Optimization tab

When the OptiSpray ion source is connected to the MS, the Emitter Optimization tab becomes available in the Ion Source pane.

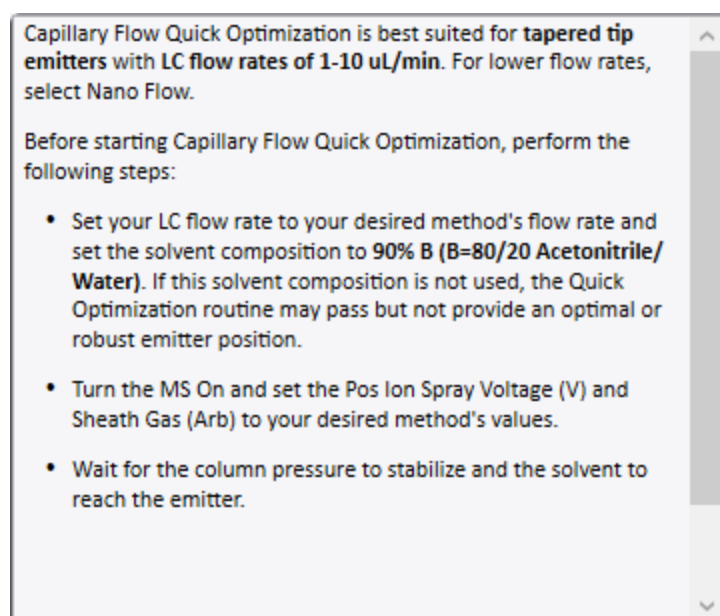
The following table describes the different optimization methods that you can use to set to emitter's active position. For additional information, refer to the *OptiSpray Ion Source User Guide*.

Parameter	Description
Optimization Mode	Select the mode for optimizing the emitter position: • Quick: Applies sheath gas and scans a m/z range (positive mode) containing acetonitrile solvent solution and cluster ions to set the emitter position. If FAIMS is installed, the FAIMS voltages are set to Off for this procedure. • Full: Averages the sum of up to 10 entered ions in SIM mode with an isolation of 1 m/z (range: 50–2,000 m/z) and uses the result as the emitter position. Can be performed in positive or negative mode. If FAIMS is installed, the FAIMS voltage status that was set prior to full optimization is maintained throughout the procedure. • Manual: Uses manual movement commands to move the emitter in increments of 0.1 mm to 0.5 mm on the x (left/right) axis and along the user's choice of the y (up/down) axis, z (toward, away) axis, or combined yz (diagonal) axis. This mode is independent of ion signal.

Parameter	Description
	NOTE: For manual optimization, the cartridge must be in the Active position.
Optimization Flow	(Quick optimization mode) Select the flow mode to use for the quick emitter optimization routine: • Nano Flow: Flow rate less than 1 $\mu\text{L}/\text{min}$. Requires 20% B (80/20 ACN/Water) solvent. • Capillary Flow: Flow rate between 1-10 $\mu\text{L}/\text{min}$. Requires 90% B (80/20 ACN/Water) solvent.

The Emitter Optimization tab also contains an Instructions area that provides a quick set of instructions for the capillary flow and nano flow optimization routines.

Figure Instructions area in the Emitter Optimization tab



4.5.2 Optimize the ion source parameters

Follow this procedure for the H-ESI and APCI source types to maximize the ion signal for your application.

NOTE

A red border around a parameter box indicates that the value is out of range.

Figure Tune application—Optimization page

The screenshot shows the 'Optimization' tab of the 'Ion Source' application. It features a 'Select source condition to optimize' section with four radio buttons and their corresponding value ranges. Below this is an 'OPTIONS' section with four input fields: 'Start Value' (0), 'Stop Value' (15), 'Step Size' (3), and 'Signal Type' (TIC). At the bottom, a status box displays 'Select Accept or Reject to continue.', 'User clicked Accept.', and 'Optimization In Progress...'. An 'Abort' button is located at the bottom right.

During the optimization, the Tune application displays the progress. While the optimization procedure is running, the upper right graph window plots Intensity RSD values per optimizing condition (10 scans).

After the optimization procedure is finished, the upper left graph window displays a smoothed optimization curve, the optimized parameter value in the legend box, and highlights this value with a vertical line in the graph.

You can abort the optimization. When the optimization completes, you can select to generate a report. Then, an Ion Source Optimization Report is generated and presented to you.

Procedure

1. On the Ion Source - Optimization page, select the parameter that you want to optimize (for example, start value, stop value, step size, and signal type options).
2. To enter the value of a parameter, select the arrow in the spin box to increase [up arrow] or to decrease [down arrow] the value. Or, enter the value in the spin box text field.
3. If applicable, select the required option in the list box.
4. Select **Optimize**.

Parameters for the ion source - Optimization page

Parameter	Description
Spray Voltage (V)	(H-ESI mode) Optimizes the source electrospray voltage (absolute value, in volts).
Sheath Gas (Arbitrary)	(H-ESI and APCI modes) Optimizes the sheath gas flow rate (in arbitrary units).
Aux Gas (Arbitrary)	(H-ESI and APCI modes) Optimizes the auxiliary gas flow rate (in arbitrary units).
Sweep Gas (Arbitrary)	Optimizes the sweep gas flow rate (in arbitrary units).
Discharge Current (μ A)	(APCI mode) Optimizes the APCI source discharge current (in microamperes).
Options	
Start Value	The initial value of the parameter being optimized.
Stop Value	The final value of the parameter being optimized.
Step Size	The step size of the parameter being optimized.
Signal Type	- TIC (default): Maximizes the total ion current (TIC) signal. - Base Peak: Maximizes the base peak signal. m/z: Maximizes the signal of the ion with the m/z value that you specify in the m/z box.
m/z	(Signal Type: m/z) The m/z of the ion to optimize on.
Buttons	
Optimize	Starts the optimization based on the ion source parameters.
Abort	Stops the optimization based on the ion source parameters.
Accept	Accepts the optimized ion source parameter.
Reject	Rejects the optimized ion source parameter.

4.6 Set up the mass spectrometer for data acquisition

Set up the mass spectrometer for data acquisition by using the Define Scan pane. For the mass table, you can import the settings from (or export them to) a CSV, TXT, or XML file.

Figure Tune application—Define Scan pane (Full Scan)

4.6.1 Specify the scan type and the precursor settings

To specify a scan type and the precursor settings, select from the following parameters.

Parameter	Description
Scan Type	(Detector Type: Orbitrap) - Full Scan: One stage of mass analysis. - MS² Scan: MS/MS mass analysis. - SIM Scan: Selected ion monitoring (SIM) scan type. - EASY-IC™ Calibrant: Displays the reagent ion spectrum for <i>m/z</i> 202 for the internal EASY-IC™ ion source. IMPORTANT: When you select EASY-IC™ Calibrant, the isolation width and all other <i>m/z</i> scan parameters are automatically set to allow for the detection of the fluoranthene reference mass.
Precursor (<i>m/z</i>)	(Scan Type: MS ² Scan or SIM Scan) The precursor mass is the center mass of the isolation range. The isolation range is one half of the isolation width applied to either side of the precursor mass. Allowable range of values for the first mass of the isolation range: 40–2500; default: 524.2649

Parameter	Description
Precursor Charge State	(Scan Type: MS² Scan) The charge state of the precursor ion. The mass spectrometer uses the charge state of the ion to adjust its acceleration into the IRM/Ion Processor region. Higher charge states require less acceleration to achieve the same level of fragmentation. Range: 1–100
Isolation Width (<i>m/z</i>)	(Scan Type: MS² Scan or SIM Scan) The isolation width (as an <i>m/z</i> value) for the precursor ions. The mass range for the peak is centered at the precursor mass and ranges from one-half of the isolation window to either side of the precursor mass. Range: MS²: 0.4–2000, default: 2 (Orbitrap and Astral) SIM: 0.4–2000, default: 2 (Orbitrap); 0.4–10, default: 2 (Astral)
Collision Energy Type	(Scan Type: MS² Scan) • Absolute: Collision energy is set independent of the mass and charge of the precursor ion. • Normalized: Collision energy is set in percentage. This value is a dimensionless number that is approximately equivalent to the absolute collision energy for a reference ion of a mass 500 and a charge of 1. The actual HCD energy is calculated on basis of mass and charge of either the selected precursor ion (Data Dependent MS² scans) or the center mass of an isolation window (targeted MS² or AIF scans).
HCD Collision Energy (% or V)	(Scan Type: MS² Scan) The MS injects isolated precursor ions into the IRM and the ion processor (for Astral when using single Collision energy value) with an HCD collision energy (as a percentage or independent of mass and charge).
HCD Collision Energies (% or V)	The MS injects isolated precursor ions into the IRM in one to five steps, each with a different HCD collision energy (as a percentage or independent of mass and charge). The MS then mass analyzes all groups of ions in a single scan. A high HCD collision energy results in more ion activation, which generally leads to more fragmentation. A low HCD collision energy results in less ion activation, which generally leads to less fragmentation. With a single value, only one collision energy is applied. If more than one value is defined, the instrument will perform Stepped Collision Energy. Use commas to separate several energy values. The ion processor and the IRM are calibrated to generate the same MS² spectra. Range: 1–200; default: 30

Parameter	Description
Detector Type	(Scan Type: MS² Scan or SIM Scan) The available options are: • Astral: Select this detector type for highest sensitivity and speed. • Orbitrap: Select this detector type for highest resolution, mass accuracy and dynamic range.
Orbitrap Resolution	(Detector Type: Orbitrap) The mass resolution of the Orbitrap analyzer, which is proportional to 1/sqrt of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200. The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000 (default), 180 000, 240 000, and 480 000.
Scan Range Mode	(Detector Type: Orbitrap) Select the scan range mode to determine the scan range. The available options are: • Auto: The last mass (l_m) is automatically set to $(\text{Precursor} + \text{IsoWidth} / 2) \times \text{Charge} \times 1.02 + 20.0$. The first mass ($f_m$) is set such as to satisfy the Factor 5–10–15 rule for the $l_m:f_m$ ratio to ensure the best retainment of all masses in the IRM. It is subject to the corresponding m/z lower limit:

Last mass m/z [Th]	Factor of 5–10–15 Rule	First mass m/z [Th]
≤ 200	5	40
200–250	5	$l_m / 5$
250–500	10	50
500–1500	10	$l_m / 10$
1500–2250	15	150
≥ 2250	15	$l_m / 15$

- Define First Mass: Explicitly define the first mass and the system automatically sets the last mass.
- Define m/z Range: Explicitly define the first and last mass.

Because the system optimizes the transfer of ions for the defined scan range, it is recommended to restrict the scan range to the region of interest.

Parameter	Description
	(Application Mode: Small Molecule) Auto-Extended: The instrument calculates the scan range starting from the precursor m/z. This scan range mode complies with the Factor 15 rule with the first mass m/z range 40–2500.
First Mass (m/z)	(Scan Range Mode: Define First Mass) The scan range's First Mass m/z value. Range: 40–2500; default: 120
Scan Range (m/z)	(Scan Range Mode: Define m/z Range) The values for the Scan Range (m/z). Range: 40–6000 (Astral); 40–6000 (8000 with BioPharma option) (Orbitrap); default: 100–1000
Mass Range	The available options are: - Normal: Last mass $\leq m/z$ 3000 - High: Last mass $\leq m/z$ 6000 NOTE: The application of high mass range mode will eventually cause scan speed limitation.
RF Lens (%)	Specify the value to control the RF amplitude applied to the ion funnel. Range: 0-200; default: 50 This value is a scaling factor applied to the nominal mass-to-voltage relationship. Decreasing the RF level decreases the transmission of high m/z ions through the ion funnel, increases the transmission of the low m/z ions, and might decrease the amount of fragmentation of fragile ions in the ion funnel. Increasing the RF level has the opposite effect. The applied amplitude resulting from the entered level is first mass dependent. TIP: For optimal transmission, the default value provides a good starting point. A fine tuning of the MS is possible.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. - Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. - Custom: You can set the AGC Target in percent. TIP: Please refer to the system templates for application-specific suggestions.
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (MS2, DIA: 3000); default: 100

Parameter	Description
	This is the normalized AGC Target value, it is represented as a percentage to help in calculating the desired AGC Target. The base normalized value is different for scan types. The values are as follows: Orbitrap analyzer: Full Scan, DIA: Normalized Base (100%) = 1e6 SIM and tMS2: Normalized Base (100%) = 1e5 Astral analyzer: MS2, DIA: Normalized Base (100%) = 1e4 SIM: Normalized Base (100%) = 1e3
Time (ms)	Maximum time that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 50 (Orbitrap), 10 (Astral)
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. NOTE: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral acquisition.
Source Fragmentation	Select the checkbox to switch on ion source CID. An offset voltage in the ion source accelerates the ions into background gas. Collisions with the background gas might aid in the desolvation of the ions and might increase sensitivity.
Energy (V)	(Source Fragmentation: On) Specifies the collision energy voltage for ion source fragmentation. Range: 1–135; default: 35 If you set the source fragmentation energy too high, fragmentation of ions may occur.

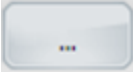
4.6.2 Specify the HCD activation type settings

Procedure

1. Select between the following settings for the Collision Energy Type:
 - Normalized: Enter 1–5 values separated by commas; each value as a range 1–200%)
 - Absolute: Enter 1–5 values separated by commas; each value as a range 1–200 V).

4.6.3 Acquire a data file with the Tune application

Procedure

1. Open the Data Acquisition pane, and then do the following:
2. (Optional) To change the destination folder for the raw data, select the  button. The default folder location is in C:\Thermo\Data.
3. In the File Name box, type the name of the analyte (for example, reserpine).
If the base file name already exists in the save location, the Tune application adds a time-stamp suffix that consists of the year (YYYY), month (MM), day (DD), and time (HHMMSS).
4. In the Sample Name box, type the name of the analyte (or another suitable label).
5. In the Comment box, type a comment about the experiment.
For example, describe the ionization mode, scan type, scan rate, sample amount, or method of sample introduction. The data system includes the comment in the header information for the raw data file.
6. Under Continuous Acquisition, select the **Manual Stop** checkbox (acquires data until you stop the acquisition).
7. Select **Record** to start data acquisition.
After the Tune parameters reach their specified settings, the data acquisition process begins and the small circle on the Record button changes to red ().
8. When you are finished, select **Record** again to stop the acquisition.
The small circle on the Record button changes to gray (not recording).

4.7 Tune, calibrate, or check the mass spectrometer calibration

You can tune, calibrate, or check the mass spectrometer calibration with the special calibrant probe (Ion Source Type: ESI).

Use the FlexMix™ calibration solution (Pierce™ P/N A39239) to calibrate the instrument:

- It is designed for both positive and negative mass calibration in the range from m/z 40–3000.

- It is stable at room temperature. Do not expose it to light or heat.

IMPORTANT

After a bakeout, the system must be re-calibrated. The instrument recognizes that it has been vented and adds certain calibrations steps to the system calibration that only need to be run after a bakeout.

NOTE

You can set a few preferences for how the Tune application works. Select the Options icon, , and then select Preferences.

If a System Calibration fails, restarting the calibration applies the values from the previously successful sequence and continues from that previous point.

4.7.1 Calibration procedures

The Calibration pane shows the last time you calibrated the mass spectrometer. Generally, you should calibrate the mass spectrometer every month of operation for optimum performance over the entire mass range.

Figure Tune application—Calibration pane

ION SOURCE

DEFINE SCAN

CALIBRATION

Status

Polarity (+)

Polarity (-)

Mass

▶ Recommended Calibration: 6/9/2021

System

▶ Recommended Calibration: 7/8/2021

Calibration

Mode

Calibrate

Polarity

Negative

Type

Mass

Mass in progress...

Check spray stability.

Check calibration solution quality.

Consider running FlexMix Solution Evaluation procedure in Diagnostics pane.

Check syringe and syringe pump.

Click "Start" button again to resume from the latest checkpoint.

If this does not help, check manual or start subprocedures from Diagnostics pane.

⋮

Instructions

Spray is unstable. Please perform following tasks and retry:

• Adjust HESI probe position (typically near position 1 with syringe flow rate at 3-5 $\mu\text{l}/\text{min}$)

• Adjust the HESI needle height

• Optimize spray voltage (starting with 3.5 kV; increasing or decreasing it while **keeping spray current $\leq 0.5 \mu\text{A}$**)

• Change sheath and aux gas settings (typically 3 or lower, no sweep gas, with syringe flow rate at 3-5 $\mu\text{l}/\text{min}$)

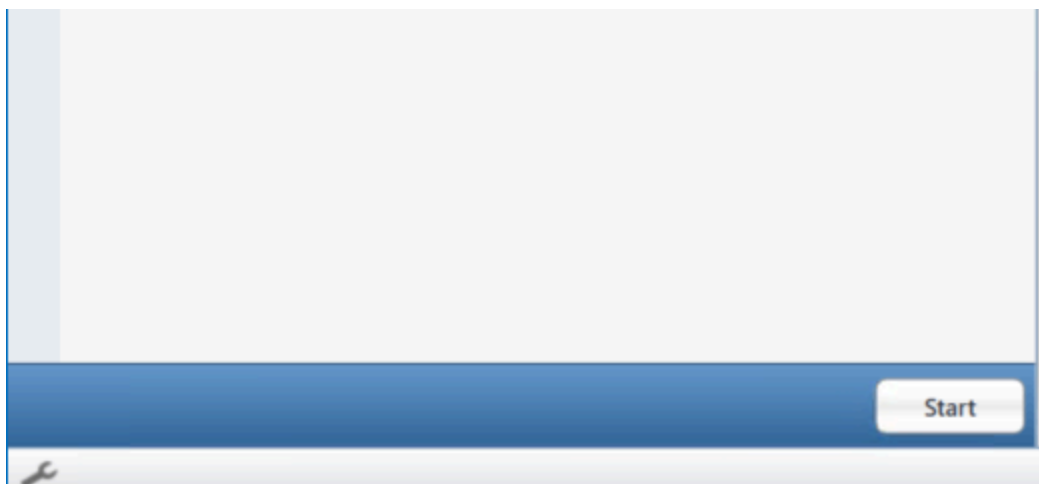
• If there are repeated spikes in the spray stability graph

• Check if it is hard to push the syringe, if so, back flush the tubing and HESI needle insert with methanol

• Snip off a small piece at both ends of the PEEK tubing

• Inspect the tip of the needle insert (preferably under a microscope) for dirt or damage. Replace the HESI needle insert if necessary

⋮



NOTE

We recommend to use the calibrant probe (Ion Source Type: ESI) for all calibrations. In the course of the ion source optimization, monitor the Spray Current in the Status pane (Ion Source > Spray Current). Make sure that the Spray Current is not higher than 0.5 μA in positive mode and not higher than 0.3 μA in negative mode.

IMPORTANT

While running the calibration procedures, you cannot run methods in the Xcalibur data system.

Calibration parameters are instrument parameters that affect the efficiency, sensitivity, mass accuracy, and resolution of the ion accumulation and detection process.

Tune parameters are instrument parameters that affect the magnitude of the ion signal. There are two types of tune parameters:

- Mass-dependent—Affects the RF voltage of the ion funnel and the DC offset voltages of the lenses and multipoles. Use the Calibration Options pane to optimize the mass-dependent tune parameters.
- Compound-dependent—Affects the ion source parameters, such as the spray voltage or spray current, and the sheath, auxiliary, and sweep gas flow rates. See Optimize the ion source parameters to optimize the compound-dependent tune parameters.

The Calibration pane has the following parameters:

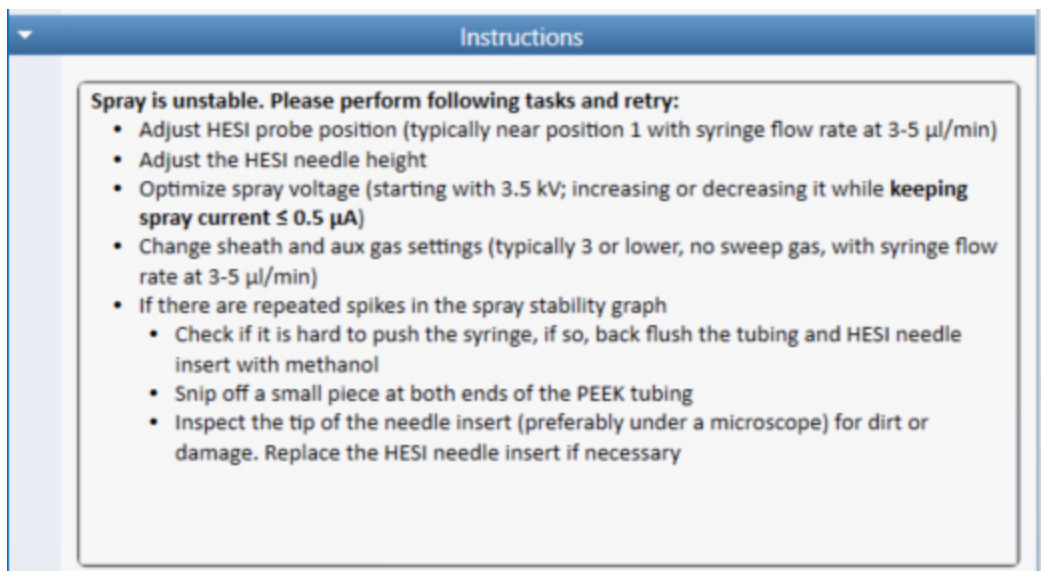
Parameter	Description
Status	
Polarity	Select the polarity for which you want to see the calibration data.

Parameter	Description
Recommended Calibrations	This is the recommended date for recalibration. The recommendations are based on the dates of the last successful calibrations and predetermined time intervals. Dates displayed in bold and italic font indicate that the recommended date has been exceeded. If the calibration failed, the recommended date is the current date.
Last Successful Calibration	This is the date of the last successful calibration.
Calibration	During the calibration, the Tune application displays event messages in the gray window: • Started/finished subprocedures with results • Reached/resumed checkpoints • Calibration result The system reports all results in the gray window. The result is either “pass” or “failed.” If the result is “failed,” the system displays a recommended action based on the failure type.
Mode	• Calibrate The system runs all selected mass calibration or system calibration procedures with selected polarity in the correct order. • Check The system runs all selected mass check or system check procedures with the selected polarity. IMPORTANT: The check procedures do not update the recommended date for next full system calibration or update the calibration parameters.
Polarity	Select the polarity for which you want to perform the calibration or check.
Type	• Mass: Uses the masses of a stable FlexMix spray to calibrate or check the masses. • Mass & System: Performs a whole system calibration or check. This calibration/check requires FlexMix solution.

Parameter	Description
	- Customized Mass: Uses external masses from a list for the calibration. Select Load user-defined mass lists to get access to a directory where you can - recall/load available mass list files, - save new mass list files, - save updated mass list files, - delete user-defined mass list files. To do so, delete the mass list file manually in the Windows Explorer. You can select provided mass lists and user-defined mass lists. A mass list can contain both positive and negative m/z values. For each polarity, you can specify up to ten theoretical masses. The Tune application shows only the reference masses whose polarity matches the selected polarity. If you want to use a mass in a list, select the corresponding checkbox.
Instructions	This field displays appropriate notifications about various routine states, for example spray stability, calibration mixture evaluation, contamination, or degradation. It gives recommendations about next steps to solve issues.

Parameter

Description



IMPORTANT

To minimize the possibility of cross-contamination, do not use the same syringe and PEEK tubing for the calibration and the sample measurement. Before you start using your analyte, flush the inlet components again.

After the calibration, you see either a green check (✓) adjacent to the calibration name to indicate a successful calibration or a red X mark (✗) to indicate a failed calibration. A date appears in the Last Calibrated column for each successful calibration test. A date does not appear for failed calibrations.

Calibration checkpoints

When a calibration fails, a regular report is generated. When you start the calibration again, all previous sub-procedures up to the checkpoint are replayed at a higher speed—including the graphs displayed.

A checkpoint becomes invalid:

- After 24 hours (after the calibration was initially started).
- On instrument reboot.

After four unsuccessful restarts of the calibration, the calibration starts from the beginning.

After each calibration, you can choose to generate a report. A report generated after the finished calibration contains all successful parts of the entire calibration sequence, even those that came before the checkpoint.

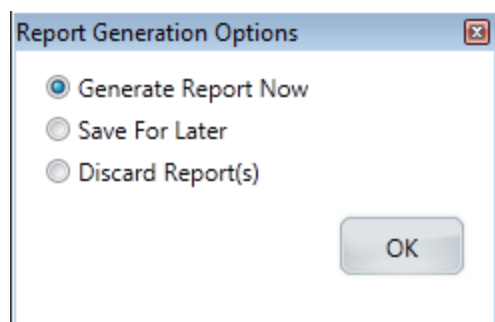
IMPORTANT

An unsuccessful or partially successful sequence does not change the instrument calibration. The calibration results are stored only when the whole system calibration has passed.

4.8 Generate reports

You can generate a report after running the procedures on the Calibration and Diagnostics panes. Depending on the settings in the Tune Preferences dialog box, no report is created, the report is created automatically, or the Report Generation Options dialog box is displayed.

Figure Report Generation Options dialog box



Select one of the following options:

- **Generate Report Now:** The PDF report is generated and saved in the following default folder: C:\Thermo\Instruments\Reports.
- **Save For Later:** You can specify a folder to save the PDF report.
- **Discard Report(s):** No PDF report is saved.

The PDF report has the following content:

- The title distinguishes between Calibration Reports and Calibration Check Reports.
- The header contains Date & Time, Instrument Model name, Instrument Serial, and Software Version.
- The PDF report comprises a section header for every (sub-)calibration run.
- The table summarizes all calibration results or calibration check results:

The column “Name” includes name of Calibration; sub-steps are indented. A unit is shown if required.

The column “Result” comprises test result of a calibration/check shown as “Passed” highlighted in green or “Failed” highlighted in red. Sub-step results are shown as “Passed” or “Failed,” too.

The columns “Value”, “Minimum” and “Maximum” are restricted to calibration/check values. If no relevant value is available, the field will be filled with “-”.

The column “Comment” comprises further important information about passed results or reasons for failure.

- The mid section summarizes calibration/check plots and additional information for each Calibration/Check procedure; each procedure is highlighted in bold letters and a larger font.
- The PDF report is finalized by a general “System Configuration” report containing graphs of some of the sub-procedures.

Procedure

1. In the Report Generation Options dialog box, select **Generate Report Now** and select **OK**.

NOTE

If you save the report for later, a Report button appears next to the Start button until you view the report. You cannot run methods in the Xcalibur data system until you view or cancel the saved report.

2. Save the report as a PDF file.

Result

After completing the procedures, the Tune application prompts you to generate a report with the results.

4.9 Use the panes at the right side

Four panes on the right of the Tune application window display information about instrument

status, history and favorite settings. To expand or collapse the panes, select the



buttons.

- Favorites pane
- Status pane and Control pane
- History pane
- Camera pane

4.9.1 Favorites pane

The parameter panel of the Favorites pane shows the active ion source and several scan parameters.

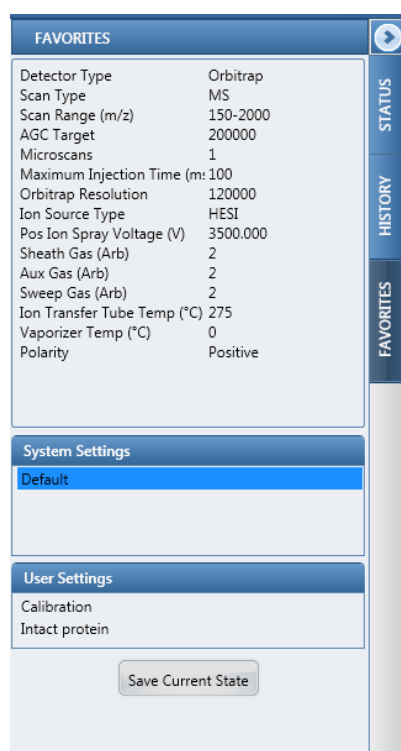
The System Settings pane provides default Ion Source and Define Scan settings for the ESI Source recommended for calibrating the Orbitrap Astral MS.

The User Settings pane allows you to save user-defined Ion Source and Define Scan settings.

NOTE

You cannot rename or delete the System Settings.

Figure Favorites pane



Save a favorite state

You can save the current key parameters of the mass spectrometer by using the Favorites pane. A validation procedure checks if the Ions Source settings and the Define Scan settings were applied before saving these parameters as Favorite settings. If this condition is not fulfilled, an error message appears that reminds you to apply the active settings.

Procedure

1. Modify parameters in one of the Ion Sources or Define Scan panes.
2. Select **Apply**.

3. Select the Favorites tab to display the Favorites pane.
4. In the Favorites pane, select **Save Current State**.
5. Type a unique name in the box that appears, and then select **Save Current State**.

Result

The newest favorite state appears first in the Favorites list. You may enter up to 100 states.

Load a favorite state

You can load the current or previously saved key mass spectrometer parameters by using the Favorites pane. When you *load* a favorite state, its key parameters appear in the Ion Source and Define Scan panes, but the Tune application does not automatically submit them to the mass spectrometer. Changed properties are shown in bold in the parameter panel.

Procedure

- Select the name in the Favorites list, or right-click it and select **Load**.

Result

The Tune application displays the key parameters in the Parameters box.

Apply a favorite state

You can apply previously saved key mass spectrometer parameters by using the Favorites pane. When you *apply* a favorite state, the Tune application submits the parameters to the mass spectrometer and also displays them in the parameters panel.

Procedure

- Double-click the name in the Favorites list, and then select **Apply** in the Ion Source or Define Scan pane. Or, right-click the name and select **Apply**.

Delete a favorite state

You can delete a saved favorites entry by using the Favorites pane.

Procedure

- Right-click the name in the Favorites list and select **Delete**.

Rename a favorite state

You can rename a saved favorites entry by using the Favorites pane.

Procedure

1. Right-click the name in the Favorites list and select **Rename**.
2. Type a different name in the box that appears.

4.9.2 Status pane and Control pane

Use the Status pane and the Control pane to monitor the working condition of your instrument.

View the Status pane

The Status pane displays real-time status information in a tree view for many mass spectrometer parameters and components. The status readback indicators update every few seconds and are as follows:

- Green square ()—Indicates that the parameter is within its expected range
- Orange square ()—Indicates that the parameter is trending toward the expected range
- Red square ()—Indicates that the parameter is outside of its expected range

Figure Status pane—By Function

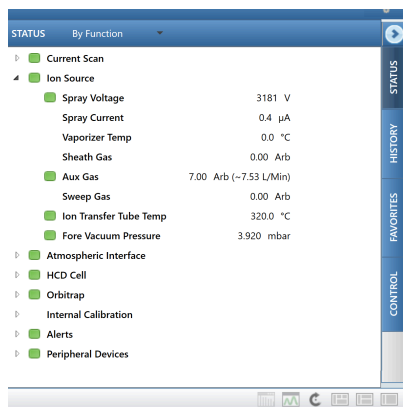
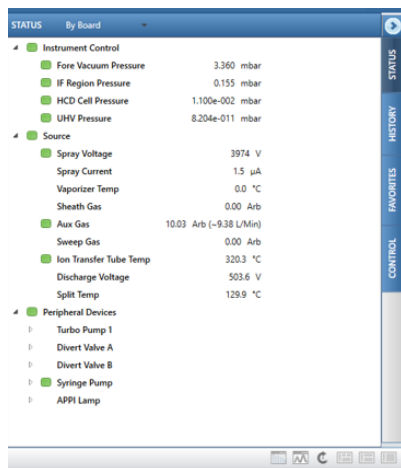


Figure Status pane—By Board



View the Status pane options

Procedure

1. Select the Status button at the right side of the Tune window.
2. In the header of the Status pane, select the downward arrow, and then select By Function (default) or By Board.
3. Select the arrow next to the category that you want to expand.

View the Control pane

The Control pane displays real-time status information in a tree view for many mass spectrometer parameters and components. The status readback indicators update every few seconds.

Procedure

1. Select the Control button at the right side of the Tune window.
2. Select the symbol next to the category that you want to expand.

4.9.3 History pane

The History pane records all parameter changes made in the Ion Source and Define Scan panes as “change records,” which appear as sub-items under the date they were created. A change record is inactive if the ion source type of the change record differs from the currently installed ion source type. The maximum number of change records is 100.

When you select a given time, the pane indicates what changes were made (shown in bold black) compared to current Ions Source settings and Define Scan Settings.

By right-clicking, you can either apply the settings directly or just load the settings into the parameter panel where you can then make further changes and apply them. By double-clicking on the change

record, settings are directly applied into the parameter panel for further changes.

Figure History pane

HISTORY	
Scan Type	MS Scan
Scan Range (m/z)	200-2000
Normalized AGC Target	100
Time (ms)	100
Microscans	1
Orbitrap Resolution	15000
Ion Source Type	Heated ESI
Pos Ion Spray Voltage (V)	4500
Sheath Gas (Arb)	5
Aux Gas (Arb)	0
Sweep Gas (Arb)	0
Ion Transfer Tube Temp (°C)	275
Vaporizer Temp (°C)	20
Polarity	Positive

History Logs	
▲ Today - Wednesday, October 4, 2021	
2:15:27 PM: FT MS	
2:11:50 PM: FT MS	
▲ Tuesday, October 3, 2021	
12:48:51 AM: FT MS	

View or modify a change record

Procedure

1. Select from the following:

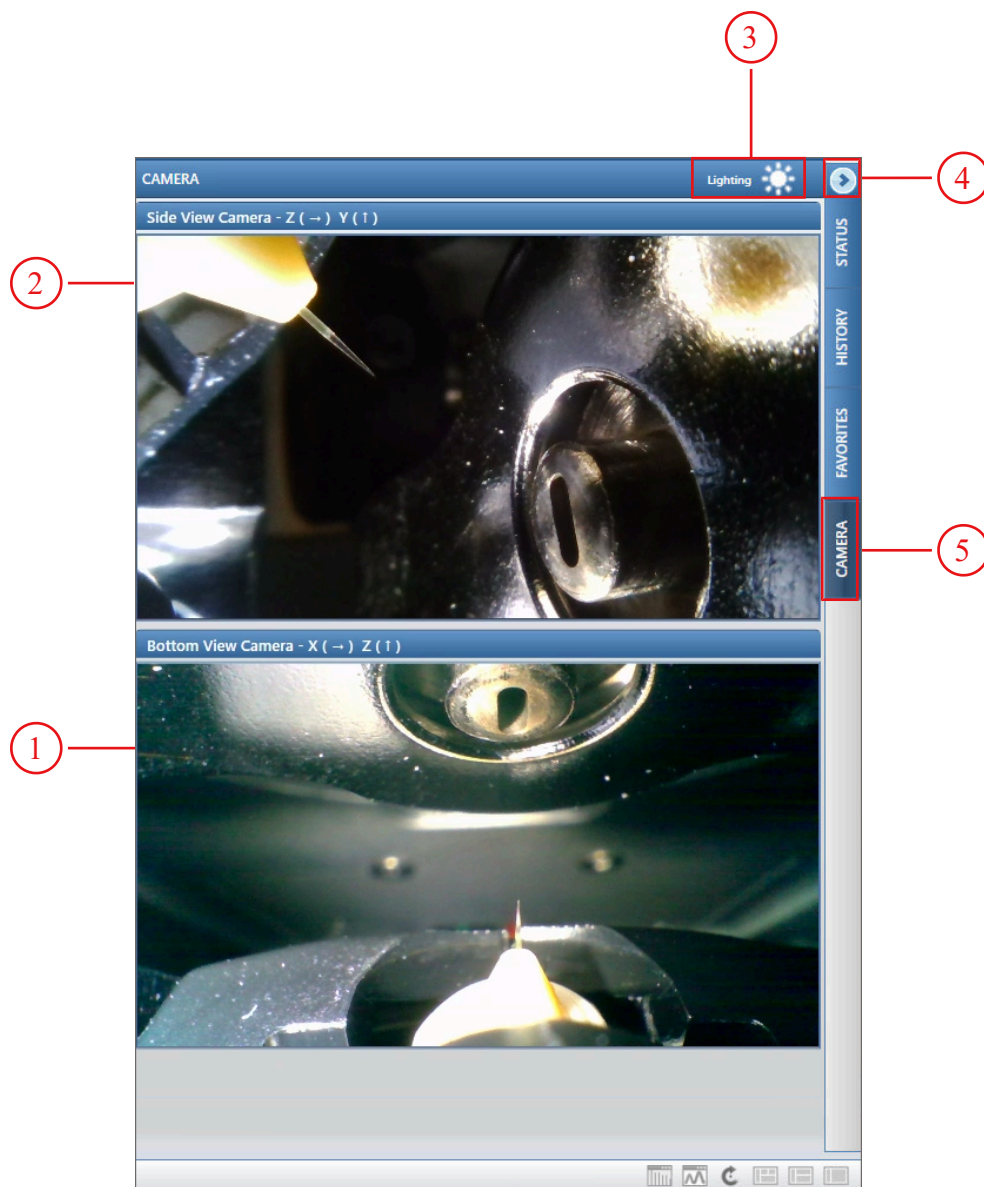
- Select a change record to display its parameters, or right-click it and select Load. Parameters shown in bold differ from their default values.
- Double-click a change record, and then select Apply in the Ion Source pane or Define Scan pane to submit its parameters to the mass spectrometer. You can also right-click the change record and select Apply in the shortcut menu.

4.9.4 Camera pane

The Camera pane is available in the Tune application when the OptiSpray ion source is installed and powered on. Two camera views allow you to view the emitter from the bottom and the side, as well as two lighting modes for visualizing the emitter and the spray. These lighting modes function when the mass spectrometer is in the On or Standby mode.

The following figure shows the camera controls.

Figure Camera pane



No.	Description	No.	Description
1	Bottom-view of the emitter	4	Expand/collapse the Camera pane
2	Side-view of the emitter	5	Access the camera views
3	Change between camera lighting modes		

Procedure

- Perform any of the procedures in the following table to control the OptiSpray camera.

Table Tasks for controlling the OptiSpray camera

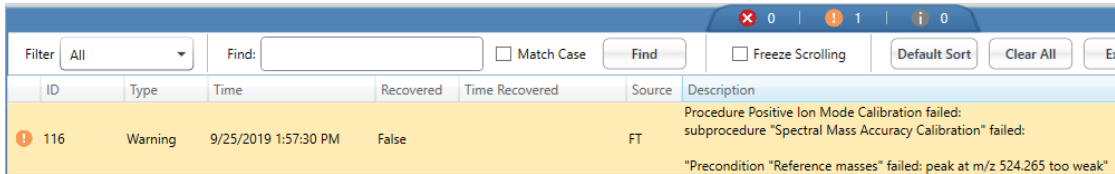
Task	Procedure
Open the camera	From the right side of the Tune window, select Camera .
Adjust the lighting mode	From the header bar of the Camera pane, select Lighting to change the lighting mode to one of the following:  : Emitter visualization lighting (white) is active  : Spray visualization lighting (green) is active
Expand the camera window	Select  .
Collapse the camera window	Select . The camera lighting turns off.
Resize the camera window	Select and hold the left-side corner or the middle bar of the camera window, then drag to the size you want.

4.10 Use the alerts console

The alerts tab appears at the bottom of the screen only when the Tune application generates

an alert message (for example, , which indicates one [1] error). The application shows one or more icons that indicate the types of messages. You can filter the messages by type and sort their order by selecting any of the column headings.

Figure Alerts console



ID	Type	Time	Recovered	Time Recovered	Source	Description
116	Warning	9/25/2019 1:57:30 PM	False		FT	Procedure Positive Ion Mode Calibration failed: subprocedure "Spectral Mass Accuracy Calibration" failed: "Precondition "Reference masses" failed: peak at m/z 524.265 too weak"

To open or close the alerts console

When the alerts tab appears at the bottom of the screen, select it.

Table Parameters in the alerts console

Parameter	Description
Alert type icons	 Errors

Parameter	Description
	•  Warnings •  Recovered errors and warnings •  Information The number after the icon indicates the total number of messages for this type.
Filter	Select the type of message to show.
Find	Searches the displayed messages by entering one or more keywords. To search all messages, set the Filter to All.
Match case	Makes the search case sensitive.
Freeze Scrolling	Prevents the list of messages from scrolling as new messages appear.
Buttons	
Default Sort	Resets the order of the alert types to show the errors first, then the warnings, and then the remaining messages sorted by time.
Clear All	Removes all of the messages, except for the active error and warning messages.
Export	Exports all alert messages to a CSV file.

4.11 Run diagnostics

Only Thermo Fisher Scientific field service engineers can run the complete diagnostics on the mass spectrometer subsystems. As a customer, your access to the diagnostics is limited.

TIP

You can set a few preferences for how the Tune application works. Select the Options icon, , and then select Preferences.

Procedure

1. Select the Diagnostics icon, , at the bottom of the Tune window.
2. Select the arrow next to the category that you want to expand.
3. Select the components that you want to test.
4. If necessary, enter new values for the selected parameters.
5. Select **Start**.

Category	Description
The descriptions for the individual parameters are built into the application. To see a tooltip description, point to the parameter name, and for additional information, select Learn More.	
System	A manual bake-out can be started, different diagnostic checks can be performed. Also, you can evaluate the purity of the FlexMix calibration solution and burn in the internal calibration source.
Mass Calibration	
Partial Calibration	When you know that certain subsystems might be out of specs, you can calibrate the healthy subsystems from here.
Optional Calibration	Procedures that provide additional calibrations to prepare the instrument for specific applications. For example, a dedicated low m/z mass calibration can be performed. It improves the mass accuracy for low m/z applications. However, the standard m/z mass accuracy (150–2000 m/z) may shift in a sub-ppm level after running this procedure.
FAIMS	Contains procedures for the FAIMS source performance. The checkboxes in this area are available only when the system has detected an FAIMS source.
OptiSpray	(Available when the OptiSpray ion source is installed.) Evaluates the hardware interlocks, internal electrical boards, and the motor function of the OptiSpray ion source. It also tests the heater, sheath gas, and spray current of the OptiSpray cartridges.
Tools	The Tools section contains tools for different functions. Not all are described here.

Category	Description
Calibration > Skip Spray Stability Evaluation for Mass Calibration	This function allows to skip the spray stability check, which is run before the mass calibration. This allows to run the mass calibration with a less stable spray, for example when a mass calibration needs to be performed with the HESI sprayer. However, we strongly recommend the low-flow setup with the calibrant sprayer. Generally, the spray stability evaluation generates real-time graphs of the TIC and of the relative standard deviation (RSD) of the TIC. The %RSD must be less than 15% for the spray stability evaluation to pass. For the system calibration, the %RSD should be less than 10%. TIP: If the spray stability test fails, see Optimize the ion source parameters. For instructions, refer to the instrument manual.
Calibration > Change Mass Calibration Due Time	By default, the recommended calibration time is 25 hours after the last successful mass calibration. Here, the due time setting ("Recommended Date") for the next mass calibration can be customized. You may consider a longer time period between two succeeding mass calibrations. Mass accuracy is ≤ 3 ppm within 25 hours (specification). Please evaluate any change in the due time setting. A change in the due time will adjust the recommended date and thus the time point when a warning message appears in the resulting raw data files. To change the calibration time

Procedure

1. In the Diagnostics pane, select **Change Mass Calibration Due Time**.
2. In the table below, the parameter value becomes visible and shows the current setting.
3. Enter a new validity period (for example, 72—for 3 days).
4. Select **Start** to store the new setting.

Category	Description
	Result A green check will appear next to the Parameter in the tree.
Define Scan > Set Multi RF Injection Threshold Ratio	This parameters allows to widen the mass range injected through the ion funnel (interface) by one single injection. The factor describes the mass range width (between first mass and last mass) that is transferred through the interface by a single injection. With a factor of 4 (default), this allows to transfer from m/z 100 to m/z 400 or from m/z 250 to m/z 1000, for example. Range: 1.2–1000 You can access a wider mass range with one single injection by entering a larger value. For example, type “6” and press <Enter>. You can now access from m/z 250 to m/z 1500 (factor of 6 from first mass setting) with one injection, for example. The settings are applied by selecting the Start button. TIP: The system is optimized for a factor of 4. Keep in mind that the change of the factor to larger values will (usually) sacrifice ion transfer through the interface, particularly for the higher m/z values in the interrogated mass ranges transferred by this factor.
Define Scan > External Handshake	Three parameters are available for the external handshake. You can set specific values in the Parameter Value column. • Scan Handshake mode 0: Off (default) 1: Handshake 2: Handshake (MS only) Select Handshake to insert a hardware handshake mechanism between scans. In this mode, trigger lines are used to establish a mechanism of scan synchronization with an external device. There is a Ready output signal from the MS and a Trigger input signal to the MS. The external device can start an MS scan (acquisition of a single spectrum)

Category	Description
	self-timed, but it needs to adhere to the Ready output signal coming from the MS, to not overrun the mass spectrometer timing. - Handshake N-th Counter Input range: 1–1000000, default 1 This parameter is available only when Scan Handshake mode is active. A set value of greater than 1 will insert a handshake only every N-th scan. A set value of, for example, 10 will enable for 10 scans to be subsequently acquired for a single trigger input. The trigger input needs to be set for the first scan of a series only. The other scans (in this example, 9 scans) will also run in absence of the trigger input. This is applied in a continuous MS application like DESI or MALDI in a line scan. - Handshake min. Trigger Delay (msec) Input range: 0–500, default 10 This parameter is available only when Scan Handshake mode is active. It provides an option to reduce the jitter between an external trigger input and the actual injection of ions for the MS scan, in case exact timing between the external trigger input and the actual ion injection event is needed. The settings are applied by selecting the Start button.
Define Scan > Diagnostic Trigger out	To synchronize external events (for example, laser pulsing) with scan events that are not individually triggered by a trigger input signal, various states of a scan event can be reported to the external device. 0: Off (default) 1: Inject Event 2: HCD Event; syncs with HCD trapping time, the contact is closed when ions are in the HCD cell 3: Acquisition

Category	Description
	• 4: PreTrigger; a contact closure is registered on the peripheral panel Digital Out 1B. The pretrigger signal can be measured with an oscilloscope using a circuit with a pull up resistor. The cycle duration of the measured signals fits to the cycle time of the chosen resolution. The settings are applied by selecting the Start button.

Method Editor application

Use the Method Editor application with the Thermo Scientific™ MS to create an MS instrument method (combined with the optional autosampler and liquid chromatograph instrument methods) by defining the experiment type and setting various parameters. These include settings for the MS, syringe pump, and divert valve; and the mass ranges and fragmentation transitions for the experiments.

NOTE

This topic describes the parameters of the Method Editor standalone application. Additional parameters are displayed when you start the Instrument Setup application from within the Xcalibur™ mass spectrometry data system.

TIP

The parameter descriptions for the scan types are built into the application. To see a tooltip description, point to the parameter name, and for additional information, select **Learn More**.

Contents

- Menus
- Global system parameters
- Method Editor overview
- Standard view
- Use the experiment templates
- Summary page

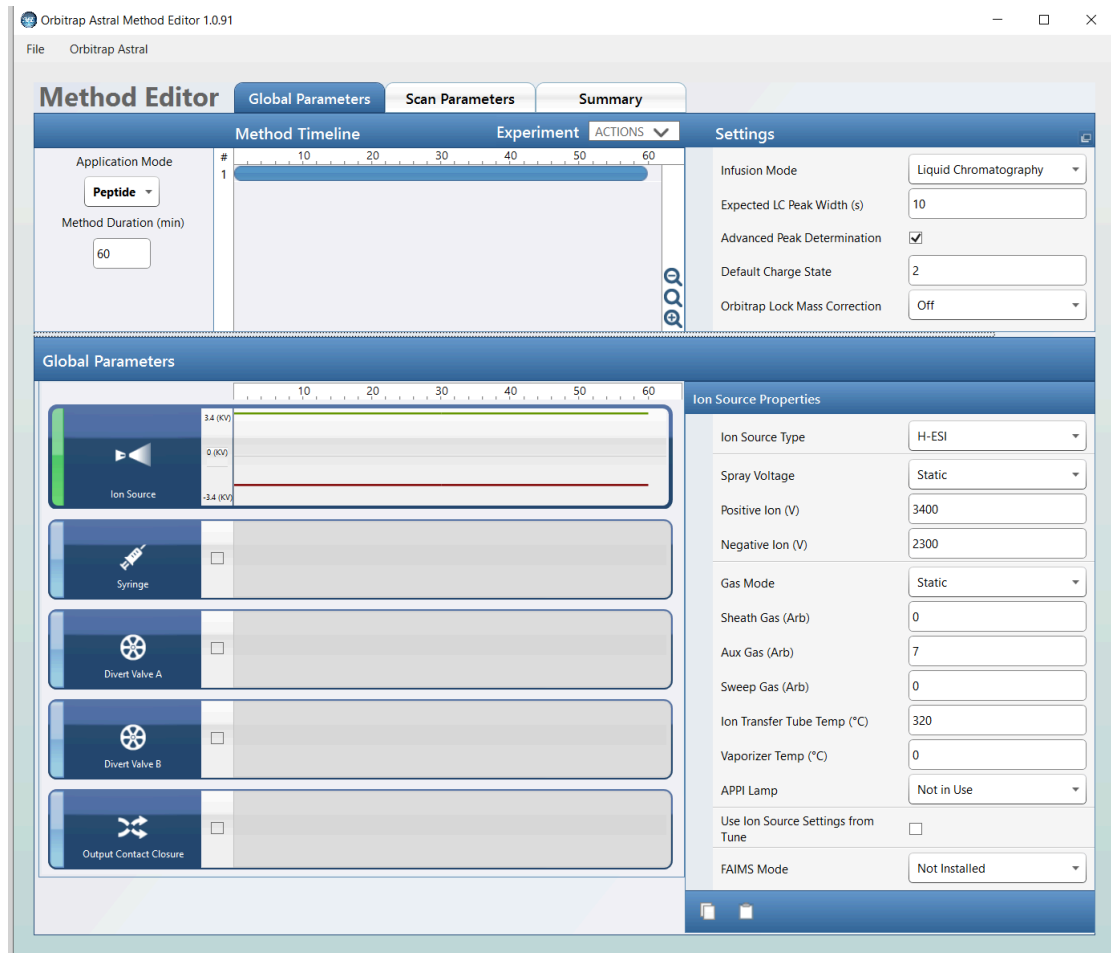
5.1 Menus

The Method Editor application window has the following menus:

- File menu

- menu

Figure Method Editor application window



File menu

The File menu has the following commands:

Command	Description
New	Creates a new method.
Open	Displays the Open dialog box. Use the Open dialog box to find and open an instrument method file (*.meth) that already exists.
Save	Saves the (changed) settings of the active method. If a new method is created, the Save As dialog box opens.
Save As	Opens the Save As dialog box. Use the Save As dialog box to save the settings of the active method into a new file and to select the location (disk and folder) where you want to save it.

Command	Description
Is Method Modified	Checks if the parameters of the active method are different from the saved file. The result is displayed in a message box.
Validate	Checks if the active method has all parameter settings that are required for it to be executed successfully.
Create Method Summary	Creates a text file that contains all parameter settings of the active method.

menu

The menu has the following commands:

Command	Description
Import Method from Raw Data File	Reads the data stored in an raw data file (of an Orbitrap Exploris Series MS or an Orbitrap Astral MS) and creates a method from the information therein. IMPORTANT: Importing the method from a raw file will replace the contents of the current method. If you do not want to replace the contents of the current method, open a new method before you import the raw file data.
Import Mass Lists from Q Exactive Method File	Imports inclusion or exclusion mass lists into a Method Editor mass list table from a method that was created on an Exactive Series instrument. If the method contains multiple mass lists, you can select the list that you want to import. NOTE: The Method Editor application does not support the list import for lock mass, neutral loss, and mass tag.

5.2 Global system parameters

On the Global Parameters page, you can configure the settings for the ion source, syringe pump, divert valves, and any connected devices controlled by the contact closure signal. The Method Editor application uses these settings for all scans.

NOTE

A red border around a parameter box indicates that the set value is out of range. A yellow border around a parameter box indicates that the set value is outside of the recommended range. Point to

the parameter box to display the valid range. The Method Editor displays all changed set values of given parameters in bold type until you save them.

TIP

Select the Summary tab to view the global parameter settings.

5.2.1 Global MS settings

On the top left side of the Method Editor, you specify the experiment application mode and the method duration. On the top right side, you select the infusion mode and how to perform the Orbitrap Lock Mass Correction.

NOTE

To use the standard internal calibration setting, you must activate the internal calibration in the Foundation Instrument Configuration window. See [Configure the instrument in Thermo Foundation](#).

Table Parameters for the global MS settings

Parameter	Description
NOTE: These parameters appear on both the Global Parameters and Scan Parameters pages.	
Application Mode	Displays a list of application modes where each mode has predefined experiment templates. The modes are as follows: • Peptide This application mode provides default instrument settings for the analysis of complex peptide mixtures. • Small Molecule This application mode optimizes instrument settings for the analysis of small molecules with focus on quantification. • Intact Protein This application mode optimizes instrument settings for the analysis of large molecules such as intact proteins, monoclonal antibodies, and other

Parameter	Description
	macromolecules. You can use the pressure mode settings for optimized analysis of intact proteins under denatured or native conditions. NOTE: The Application Mode Intact Protein does not support the Detector Type Astral. NOTE: The Application Mode Intact Protein is available only when the BioPharma license is installed.
Method Duration (min)	Defines the method duration in minutes. Range: 0.1–5000
Settings	
Pressure Mode	(Application Mode: Intact Protein) Available options are: • Low Pressure (default): Optimal detection for intact protein mass and top-down detection of large fragment ions. • Standard Pressure: Optimal detection for small molecules, bottom-up and top-down protein experiments. • High Pressure: Optimal detection for native protein analysis experiments. TIP: You can change the default settings for Low Pressure and High Pressure on the Diagnostics pane of the Tune application.
Infusion Mode	Available options are: • Liquid Chromatography (default) Select Liquid Chromatography when the sample is introduced by LC (that is, precursors and precursor intensities change with time). • Infusion Select Infusion when injecting the sample by direct infusion (that is, precursors and precursor intensities do not change with time).

Parameter	Description
Expected LC Peak Width (s)	(Infusion Mode: Liquid Chromatography) The Expected LC Peak Width(s) is used to ensure proper calculations for AGC, depending on varying ion flux. A concept based on quick scans in the Astral detector determines the ion flux. These scans are performed 20 times per LC Peak width. Range: 1–1000
Mass Range	The available options are: • Normal: Last mass $\leq m/z$ 3000 • High: Last mass $\leq m/z$ 6000 NOTE: The application of high mass range mode will eventually cause scan speed limitation.
Advanced Peak Determination	Select the checkbox to apply the Advanced Peak Determination (APD) algorithm. This algorithm improves the determination of the charge states and monoisotopic m/z values of isotopic envelopes.
Mild Trapping	(Application Mode: Small Molecule) Select the checkbox if you want that ions are guided through the instrument with less energy to reduce fragmentation on MS1 level. However, a lower transmission may occur for certain compounds at the cost of sensitivity. Clear the checkbox for compounds that require highest transmission and ultimate sensitivity.
Default Charge State	The default charge state that the system uses to relate an m/z value to its mass if it cannot determine the charge state. The MS uses this charge state to calculate the normalized collision energy (NCE) for a given m/z value. Range: 1–100; default: 1 (Small Molecule), 2 (Peptide, Small Protein)
Orbitrap Lock Mass Correction	The internal calibrant source provides an internal reference mass that is analyzed in the same analytical scan of your sample measurements. Lock mass correction provides the most accurate corrections to the Orbitrap data. • Off: No lock mass correction is performed. • EASY-IC™: The internal EASY-IC source provides fluoranthene radical ions, which are used as

Parameter	Description
	reference masses for the positive and negative polarity lock mass correction. • User-defined Lock Mass: One or more <i>m/z</i> values can be defined as lock mass(es) to perform a lock mass correction during the run. When you select this option, the lock mass table appears. (Scroll down to view the table.) If no lock mass is found in one scan, the system will apply the last successful locking information to this scan. Time duration of last locking and lock mass correction are provided in the scan header of the individual scan.
Mode	(Orbitrap Lock Mass Correction: EASY-IC™ or User-defined Lock Mass) • Run Start (only with Lock Mass Correction: EASY-IC™) During the “prepare for run” phase, a lock mass correction is performed via a tSIM scan. Statistical manners are applied to exclude outliers and determine the best correction value. The correction is applied to all scans throughout the run. This mode enables highest acquisition speed. • Scan-to-Scan The lock mass correction is applied to every single scan. This mode delivers the highest mass accuracy. However, for the lock mass injection a certain overhead time is needed. The overhead time for injecting the lock mass (EASY-IC or User-defined Lock Mass) is typically 6 to 10 ms/scan. For User-defined Lock Mass, the Lock mass injection needs to be selected, to correct every scan (incl. ddMS²) in a Scan-to-Scan matter.

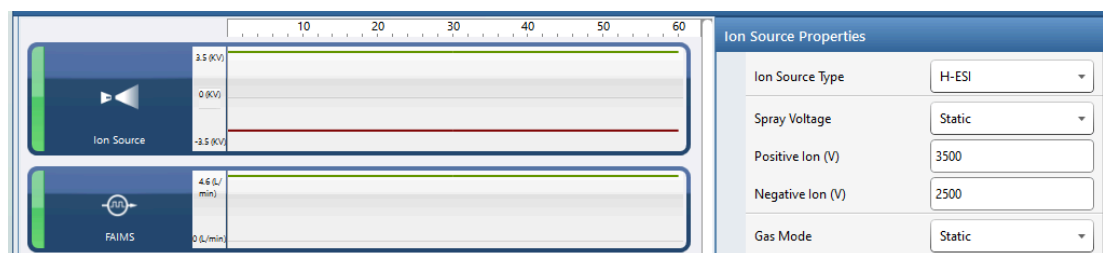
Parameter	Description
	Timed Applies the lock mass correction in the retention time window (start/stop) setting. The time window and the polarity is defined in the table that appears when you select this mode. The Timed mode provides the ability to refresh the lock mass correction multiple times during the data acquisition of a raw file (run), and therefore compromises high mass accuracy and high acquisition speed. The last successful locking information is applied to all subsequent scans until the next timed window applies and a lock mass correction is performed. If no lock mass is found, the last locking information will continue to apply. TIP: For Scan-to-Scan mode and Timed mode: When EASY-IC is chosen, you can decide to apply the detected lock mass correction of one scan to other scans by setting Use EASY-IC to On in the experiment properties of the reference scan and to Off in the scans to which lock mass correction is transferred.
Lock mass table	(Orbitrap Lock Mass Correction: User-defined Lock Mass or EASY-IC: Timed) Use the lock mass table to: - Specify user-defined lock masses - Specify time windows during which the lock mass correction (EASY-IC or User-defined Lock Mass) shall be applied Lists can be imported (or exported to) from a CSV, a TXT, or an XML file. <i>m/z</i>: The lock masses' <i>m/z</i>. Range: 40–6000 <i>t</i> start (only with Mode = Timed): The time (in minutes) to start the lock mass correction. Range: 0–Duration of active method; default: 0 <i>t</i> stop (only with Mode = Timed): The time (in minutes) to stop the lock mass correction.

Parameter	Description
	Range: 0.1–Duration of active method; default: 0 • Polarity: The polarity of the lock mass.
Current Lock Mass	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Use the list to select an available lock mass list. Select Save to save the active lock mass list. Select Save As to save the active lock mass list under another name. Select Delete to delete the active lock mass list.
Mass Tolerance (ppm)	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Defines the mass tolerance window (in $\pm$ ppm) for detecting the lock mass defined in the table. Range: 1–20; default: 15

5.2.2 Ion Source Properties pane and timeline

Use the Ion Source Properties pane and the timeline to set the ion source parameters. For the time-dependent spray voltage table, you can import the settings from (or export them to) a CSV, a TXT, or an XML file.

Figure Ion Source Properties (FAIMS enabled)



NOTE

If you installed the optional Thermo Scientific FAIMS Pro Duo system, the FAIMS parameters appear in this properties pane if you select a FAIMS Mode other than Not Installed.

TIP

You can copy () the ion source parameters from the Tune window and paste () them to the Method Editor.

Parameters in the Ion Source Properties pane

Parameter	Description
Ion Source Type	• NSI (nanoelectrospray ionization) • H-ESI (heated-electrospray ionization) • APCI (atmospheric pressure chemical ionization) • ESI (electrospray ionization) • OptiSpray
Spray Voltage Spray Current	• Static: The spray voltage or spray current does not change with time. • Time Dependent: The spray voltage or spray current changes stepwise with time. When you select Time Dependent, a table appears where you set the time and either the spray voltage (Ion Source Type: ESI, H-ESI, NSI, OptiSpray) or spray current (Ion Source Type: APCI). • Time (min): The time (in minutes) when the voltage settings change to the specified value. Range: 0 to method duration • Positive Ion Range: 0-3000 Volts (NSI, OptiSpray), 0-6000 Volts (H-ESI, ESI), 0-100 μA (APCI) • Negative Ion Range: 0-2500 Volts (NSI, OptiSpray), 0-5500 Volts (H-ESI, ESI), 0-100 μA (APCI)
Positive Ion (V)	The spray voltage (in volts) for the positive ion polarity mode. Range (ESI, H-ESI): 0–6000; default: 3400 Range (NSI, OptiSpray): 0–3000; default: 1800 (NSI), 1500 (OptiSpray)
Negative Ion (V)	The spray voltage (in volts) for the negative ion polarity mode. Range (ESI, H-ESI): 0–5500; default: 2300 Range (NSI, OptiSpray): 0–2500; default: 1500
Pos Ion Discharge Current (μ A)	(Ion Source Type: APCI) The ion discharge current (in microamperes) for the positive ion polarity mode.

Parameter	Description
	Range: 0–100; default: 5
Neg Ion Discharge Current (μA)	(Ion Source Type: APCI) The ion discharge current (in microamperes) for the negative ion polarity mode. Range: 0–100; default: 5
Gas Mode	- Static: The gas flows do not change with time. - Time Dependent: The gas flows change with time. When you select this option, a table appears where you can set the time (in minutes) and the flow rates for sheath gas, aux gas, and sweep gas. TIP: To maximize the table, select the  button in the top right corner.
Sheath Gas (Arbitrary)	(Ion Source Type: ESI, H-ESI, APCI, OptiSpray) The sheath gas flow rate (in arbitrary units). Range: 0–80
Aux Gas (Arbitrary)	(Ion Source Type: ESI, H-ESI, APCI) The auxiliary gas flow rate (in arbitrary units). Range: 0–25 NOTE: If the vaporizer temperature is set to values above 100 °C, the auxiliary gas is automatically set to 5 unless it is not set higher manually.
Sweep Gas (Arbitrary)	(FAIMS Mode: Not installed) The sweep gas flow rate (in arbitrary units). Range: 0–20
Ion Transfer Tube Temp (°C)	The temperature of the ion transfer tube (in °C). Range: 0 (ambient temperature) to 400
Vaporizer Temp (°C)	(Ion Source Type: H-ESI or APCI) The temperature of the vaporizer tube (in °C). Range: 0 (ambient temperature) to 550; default: 0
APPI Lamp	(Ion Source Type: ESI, H-ESI, or APCI) - Not in Use: The instrument method does not use the APPI lamp. - On: Switches on the installed APPI lamp.

Parameter	Description
	- Off: Switches off the installed APPI lamp. Default: Not in Use NOTE: - The Method Editor displays the APPI Lamp parameter under the following conditions: - You select a dedicated ion source type in the Instrument Configuration window. - You connect the APPI lamp's USB cable to the instrument. - The MS confirms the state of the APPI lamp before the data acquisition starts. If the ion source does not contain the APPI lamp, the data acquisition application displays an error message.
Cartridge Temp (°C)	(Ion Source Type: OptiSpray) The cartridge temperature range is dependent on the cartridge media. When the infusion cartridge is installed, the temperature is not be applied because there is no heater. Range: 35-60; default: 45 NOTE: When the mass spectrometer is off, the OptiSpray cartridge heater continues to hold the previously applied temperature.
Use Ion Source Settings from Tune	Uses the source settings that are currently set in the Tune application. Because the instrument method uses the source settings currently applied in the Tune application, the Method Editor application does not display the parameters.
FAIMS Mode	If you select a FAIMS Mode other than Not Installed, a FAIMS timeline is displayed in addition to the Ion Source timeline. Select one of these modes for running methods with FAIMS installed: - Not Installed (default) - Standard Resolution: Provides the best transmission mode by setting the inner and outer electrodes to 100 °C.

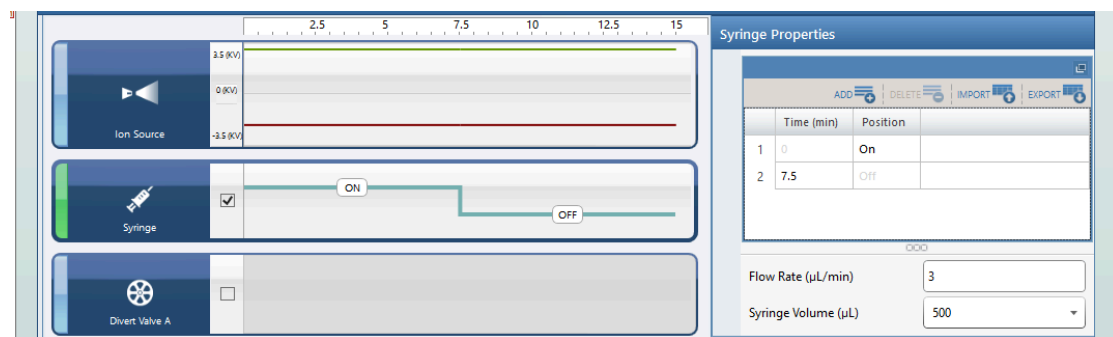
Parameter	Description
	- High Resolution: Provides twice the resolution of the standard resolution mode by setting the inner electrode to 70 °C and the outer electrode to 100 °C. - User Defined: Displays the fields to enter the electrode temperatures. NOTE: The electrode temperatures affect the ion separation. Although the ion transmission might be lower in the FAIMS high resolution mode, the peaks are narrower due to the electrodes' temperature differential.
Total Carrier Gas Flow	This parameter defines the flow of the carrier gas that propels the ions through the electrode space. The optimal gas flow for maximum transmission of ions depends on the type of ion transfer tube and is set as default for this field. In some cases, it might be beneficial to decrease or increase the carrier gas flow to optimize transmission or enhance the spray stability. The timing for dispersing the FAIMS and carrier nitrogen gas: - Static: The gas flow does not change with time. - Time Dependent: The gas flow changes with time as specified in the Total Carrier Gas Flow Table. - Time (min): When the gas flow rate changes to the specified value. Range: 0 to the maximum method duration (5000 maximum) - Gas (L/min): The flow rate for the gas. Range: 3.5–7.7; default: 4.6 This option is often used with higher gas flow rates at the beginning and the end of the chromatographic method to prevent liquid and undesired ions from entering the FAIMS electrode assembly.
Total Carrier Gas Flow (L/min)	(Ion Source Type: NSI, H-ESI, APCI, ESI; Total Carrier Gas Flow: Static) The flow rate for the FAIMS and carrier nitrogen gas.

Parameter	Description
	Range: 3.5–7.7; default: 4.6
FAIMS Inner Electrode Temp (°C)	(Ion Source Type: NSI, H-ESI, APCI, ESI; FAIMS Mode: User Defined) The maximum temperature of the FAIMS inner electrode (in °C). Range: 70–100; default: 100
FAIMS Outer Electrode Temp (°C)	(Ion Source Type: NSI, H-ESI, APCI, ESI; FAIMS Mode: User Defined) The maximum temperature for the FAIMS outer electrode (in °C). Range: 70–100; default: 100
Ion Source Properties icons	
Copy	Copies the ion source parameters to the clipboard.
	
Paste	Pastes the ion source parameters from the clipboard.
	

5.2.3 Syringe Properties pane and timeline

Select the Syringe checkbox to enable the table in the Syringe Properties pane. Then specify the time-dependence of the syringe pump. The Global Parameters page graphically displays the time dependence of the syringe pump on/off status as a timeline. For the syringe position table, you can import the settings from (or export them to) a CSV, a TXT, or an XML file.

Figure Syringe Properties



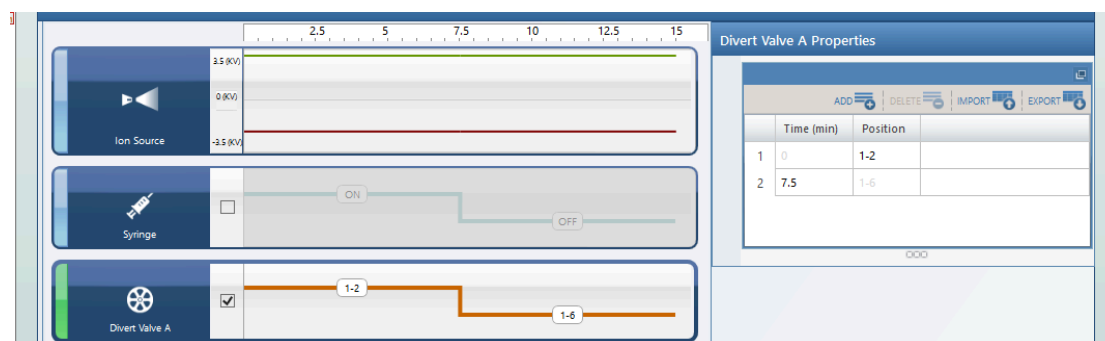
Parameters in the Syringe Properties pane

Parameter	Description
Syringe Pump position table	
Time (min)	The time (in minutes) when the syringe pump changes between On and Off, or Off and On. Range: 0 to the maximum method duration (5000 maximum); default: 0
Position	The syringe pump is On or Off at the corresponding time. Default: On
Flow Rate (µL/min)	The volume of solution per unit of time (in micro-liters per minute) that the syringe pump injects (the syringe pump flow rate). Range: 0.5 to 500 (<i>syringe volume</i> in µL)/min; default: 3
Syringe Volume (µL)	Values: 50, 100, 250, 500, 1000, 5000, and 10 000; default 500

5.2.4 Divert Valve Properties pane and timeline

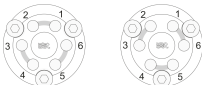
Select the Divert Valve checkbox to enable the table in the Divert Valve Properties pane. Then specify the time-dependence of the modular divert/inject valve. The Global Parameters page graphically displays the time dependence of the divert valve position as a timeline. For the divert valve position table, you can import the settings from (or export them to) a CSV, a TXT, or an XML file.

Figure Divert Valve Properties



Parameters in the Divert Valve Properties pane

Parameter	Description
Divert valve position table	

Parameter	Description
Time (min)	The time (in minutes) when the divert valve changes position. Range: 0 to the maximum method duration (5000 maximum); default: 0
Position	The divert valve position: 1-2 (default) or 1-6.
	 For additional information, refer to the instrument manuals.

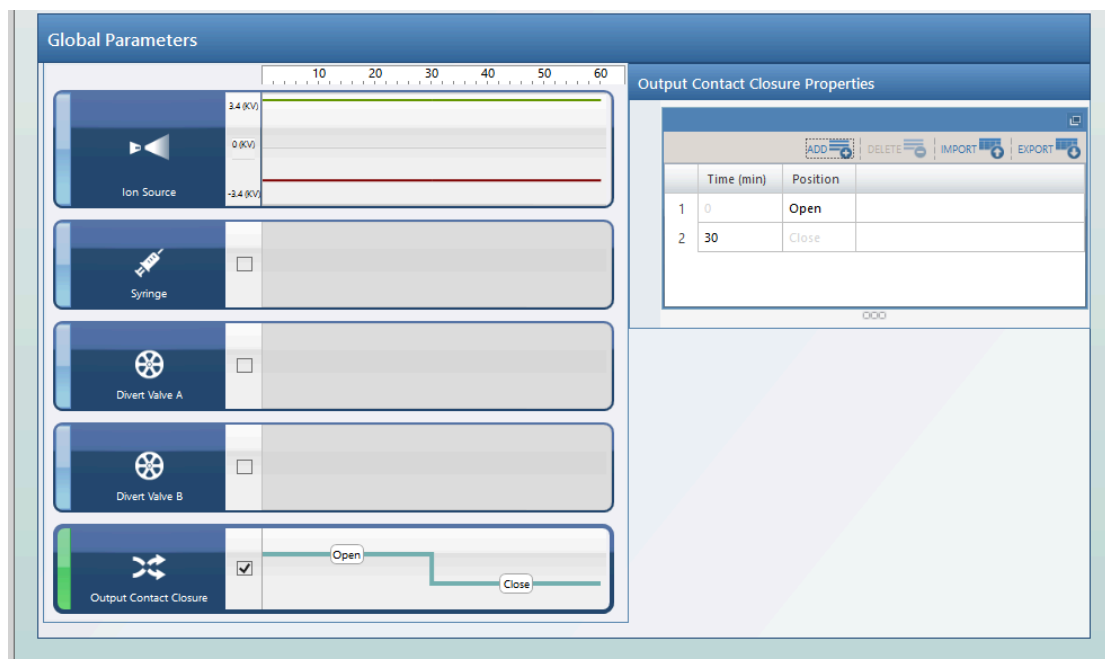
5.2.5 Output Contact Closure Properties pane and timeline

Select the Output Contact Closure checkbox to enable the table in the Output Contact Closure Properties pane. Then specify the time dependence of the contact closure on/off signal to an external device. The Global Parameters page graphically displays the time dependence of the contact closure on/off status as a timeline. For the contact closure position table, you can import the settings from (or export them to) a CSV, a TXT, or an XML file.

NOTE

You must configure contact closure in the Foundation Instrument Configuration window.

Figure Output Contact Closure Properties



Parameters in the Output Contact Closure Properties pane

Parameter	Description
Contact closure position table	
Time (min)	The time (in minutes) when the contact closure state changes. Range: 0 to the maximum method duration (5000 maximum); default: 0
Position	The contact closure signal is Closed or Open. Default: Open

5.3 Method Editor overview

The Method Editor is below the method timeline on the Scan Parameters page. This is where you build the MS instrument method, which can have several experiments, each with its own timeline.

5.3.1 Method Editor elements

The Method Editor displays the Standard View (Standard View toolbar, ) in a workflow tree diagram.

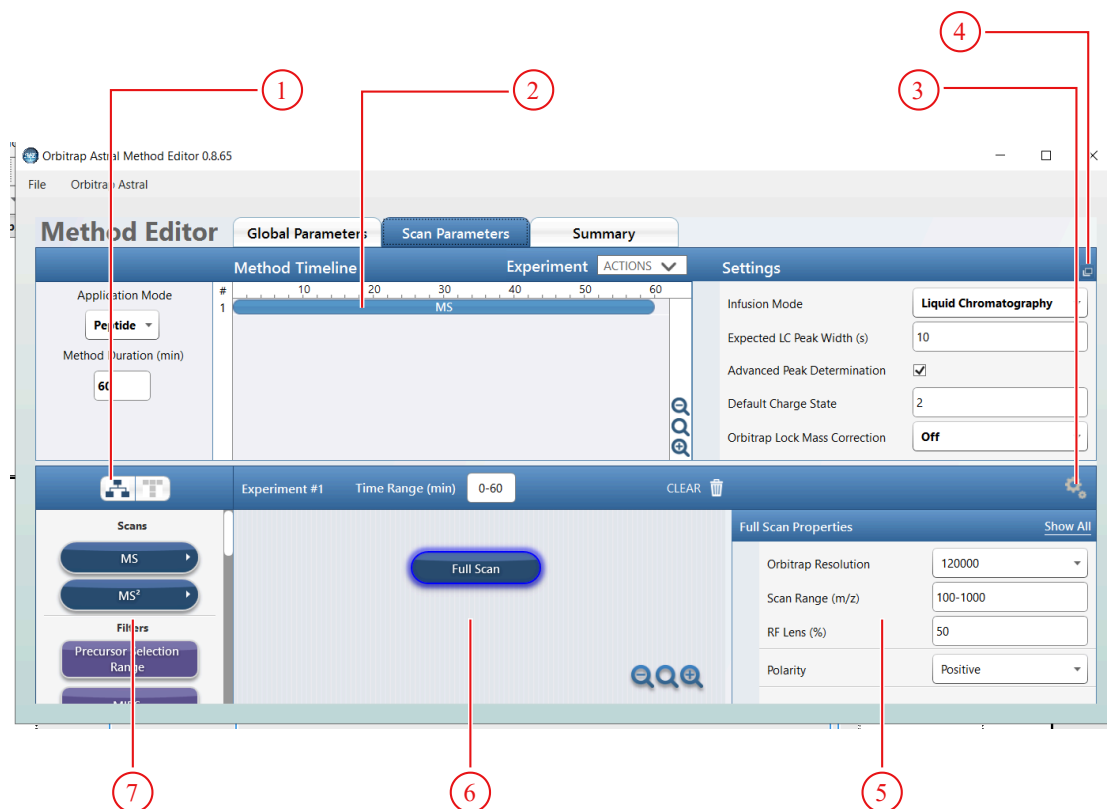
The Standard View experiments consist of scans and optional filters. (These workflow elements are also known as icons, objects, or nodes.) The first node of a workflow tree is the master scan. The master scan is the parent scan in a data-dependent experiment, or it can be the only scan in the experiment.

You can also build a workflow by selecting application-specific system templates from the

Templates View toolbar (). These templates describe market specific typical experiments.

All scan types and filter types have adjustable parameters that appear in the right-side properties pane.

Figure Scan Parameters page



No.	Description	No.	Description
1	Select the experiment toolbar: Standard or Templates.	5	Object properties pane
2	Experiment timeline	6	Workflow area
3	Preferences icon	7	Experiment objects for the selected toolbar (Standard View shown) that you drag to the workflow area.
4	Unlock and maximize the settings.		

5.3.2 Method timeline

You use the Method Timeline area (at the top of the window) to set the method duration and view the experiment timelines.

You can enter several experiments to the Method Timeline area.

On the Global Parameters and Scan Parameters pages, the Method Duration box displays the remaining experiment time (in minutes). Where experiments overlap, the MS cycles between them.

The Method Timeline area contains the following parameters and buttons.

Parameter	Description
Method Duration (min)	The time duration (in minutes) for the instrument method. Range: 0.1–5000; default: 60
Experiment Actions	• Add New Experiment • Delete Current Experiment
Buttons	
	• Scroll arrows: When you zoom in on an experiment, these arrows appear if the timeline exceeds the viewing area. • Zoom controls: Zoom Out, Reset Zoom, and Zoom In.

5.3.3 Experiment workflow area

Below the Method Timeline area, in the left-side pane of the Scan Parameters page, experiment objects (scan types and filter types, as applicable) are grouped into two experiment toolbars—Standard View (qualitative scans) and Templates View.

You build an experiment workflow by double-clicking or dragging an experiment object into the Method Editor, or by selecting a predefined experiment template that you can modify as needed.

5.3.3.1 Experiment controls

This table describes the parameters and buttons that are located above the workflow area.

Parameter or button	Description	
Experiment toolbars	Icon for the Standard View (	(Multiple experiments only) Selecting the left or right arrow displays a specific experiment in the workflow area.

Parameter or button	Description
Time Range (min)	The start and end times (in minutes) for the displayed experiment. The values must be in the specified method duration.
	Clears the contents of the selected experiment without removing the experiment timeline.
	Opens a new window that maximizes the Settings pane.
	The Preferences icon () controls the ON/OFF status of the following: • Enable Recommendations: Shows (or hides) tooltips that specify recommended values or ranges. • Enable Favorites: Shows (or hides) the stars that designate favorite parameters. • Restore Default Favorites: Resets all favorites to the factory setting. NOTE: This command is available only when Enable Favorites is selected. • Restore Advanced Template Messages: Restores the System Templates dialog box that tells you when a template has additional online information.

5.3.3.2 Standard View workflow objects

TIP

The selected workflow object has a purple border and shadow. Objects that are not selected have a white border. If a border color changes to yellow, the set value is not in the recommended range. If a border color changes to red, one or more of the object's parameters are out of the operational range. Point to the parameter box to see the correct range.

This table summarizes how to work with the Standard View workflow objects (scans and filters).

Task	Method
Add an object.	In the experiment toolbar, double-click an icon or drag it into the workflow area (center pane).
Copy an object.	• Right-click the object and select Copy. • (Scans) Drag the scan onto an open node. The object moves only if the new location is appropriate, and the set properties move with the object. • (Filters, triggers, and sorts) Press CTRL and drag the object onto an open node.
Paste an object.	• Right-click the object under which to paste the copied object, and then select Paste. • Copy an object from one experiment and paste it into another. NOTE: For scan objects, the paste function replaces the selected scan. For filter objects, the paste function adds the object to the workflow.
Move an object.	(Filters, triggers, and sorts only) Drag the object onto an open node.
Delete an object.	• Select the object and press DELETE. • Right-click the object and select Delete. • Point to the object, and then select X when it appears.
Restore the default settings.	Right-click the object and select Restore Defaults to reset the parameter to the default value.
Scans only	
Append a scan.	Right-click the scan node under which to add the copied scan, and then select Append .
Filters	
Cut a filter or trigger.	Right-click the object and select Cut .

5.3.4 Set the Method Editor preferences

This topic provides information about handling parameters.

Apply parameter recommendations

When you enter a value or range that is not recommended or outside the recommended range for the selected parameter, the Method Editor application changes the border color of the affected workflow objects and parameters to yellow. Point to a yellow parameter box to display the recommended value or range.

Procedure

1. Do one of the following:
 - Right-click the parameter box and select Apply Recommended Value.
 - You can choose to ignore the recommendation by choosing Ignore Recommendation. When you use the recommended value, the box's border color returns to normal.
2. Right-click the workflow object and select Restore Defaults, or right-click a specific parameter box and select Restore Default.

Add or remove parameters from the favorites list

By default, the Method Editor shows only the preset favorite parameters in the scan properties panes. For the filter types, all parameters are shown. To add or remove parameters from the favorites list, follow this procedure.

Procedure

1. In the properties pane title bar, select Show All if applicable.
All parameters for the scan type appear.
2. To add or remove a favorite, select the star adjacent to the appropriate parameter (blue is selected and gray is unselected).
3. To show the modified list of favorites, select Show Favorites.

5.3.5 Experiments and instrument methods

Start by creating an instrument method (METH file) with several experiments—one scan table or workflow diagram and one timeline per experiment.

TIP

You can also start with one of the system templates, which is a complete method that you can modify.

To view the system templates on the Scan Parameters page, select the Templates View icon () and select an application mode.

5.3.5.1 Create instrument methods

Start on the Global Parameters page.

Procedure

1. Set the parameters for the ion source, syringe, divert valve, and contact closure.
2. Set the Global System Parameters.
These parameters are located at the top left and top right of the window.
3. On the Scan Parameters page, do the following:
 - a. In the Time Range (min) box, enter the experiment's start and end times.
 - b. On the left side, select the Standard View () icon, and then double-click or drag a scan icon to the Place Scan Here node.

In Standard View, a graphical node (object) appears in the center pane with a purple border to indicate that it is the active node.
 - c. To enter a name for the experiment, double-click the experiment timeline. Enter the name of the experiment. Press <Enter>.
 - d. In the right-side properties pane, set the parameters as required.
4. (Optional in Standard View) To create a data-dependent scan after a Full or SIM master scan, do the following:
 - a. Double-click the MS² scan icon or drag it to the workflow area, and then set the parameters. The scan node's name becomes "ddMS²."
 - b. Add applicable filter types, and then set the parameters.

5.3.5.2 Add and delete experiments in a method

The instrument method can have several experiments.

Procedure

1. To add an experiment to the method, select Add New Experiment in the Experiment Actions list.
2. To delete an experiment from the method:
 - a. Go to the experiment by using the arrows above the workflow area.
 - b. In the Experiment Actions list, select Delete Current Experiment.

5.3.5.3 Save instrument methods

After creating an instrument method, save it.

Procedure

1. Select File > Save As.
2. In the Save As dialog box, enter an instrument method name (.meth extension), and then select Save.

5.3.5.4 Add custom method templates

If you want to save an instrument method as a template, save it in the Templates pane.

Procedure

1. After you build your template, select the Templates View icon ().
2. Select **Save As Template**.
3. Enter a template name and select **Save**.

The method template is added to the custom templates list.

Result

You can import the template at a later time by selecting **My Experiments** and selecting the template.

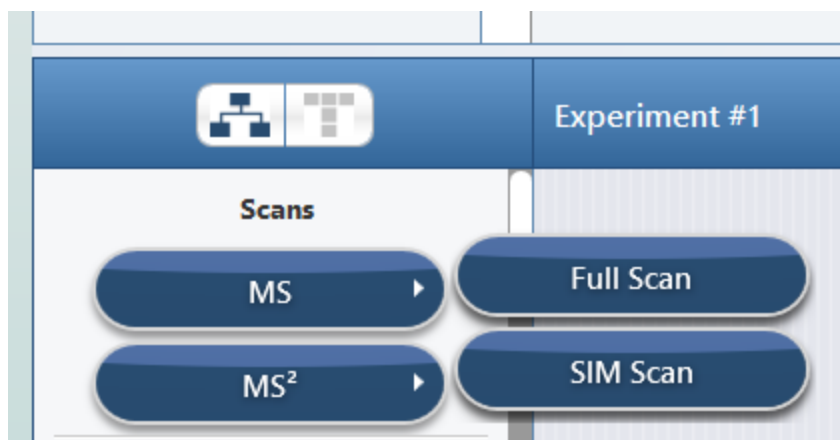
5.4 Standard view

On the Scan Parameters page, the Method Editor includes qualitative scan types in the left-side Standard View toolbar ().

5.4.1 Master scans

The master scan type is the first scan of a workflow. The following master scan types are supported: Full Scan and SIM Scan. You can create a data-dependent scan by appending the MS2 scan type after the Full Scan or SIM Scan type.

Figure Master scans



To add a Standard View scan type to the workflow

Procedure

1. Below the Method Duration box, select the Standard View icon ().
2. Double-click a scan type icon or drag it to the Place Scan Here node.
3. When you select a workflow node, its parameters appear in the right-side properties pane.

NOTE

Unavailable scan types have a dim background.

TIP

Point to a parameter name to display its tooltip.

Changed set values of given parameters appear in bold type. A red border around a parameter box indicates that the set value is out of range. Point to the parameter box to display the valid range. If you enter a value that is outside the recommended range, the box border color changes to yellow.

Select the Summary tab to view the instrument method settings.

Master scan rules

The Method Editor application enforces these master scan rules to prevent illogical instrument methods:

- No MS full scan after an MS master scan
- No scans, filters, or decisions after tMS2, and data-independent analysis (DIA) master scans

TIP

For additional rules, see the tooltip descriptions of the individual master scan types.

5.4.2 Full Scan Properties pane



The Full Scan Properties pane sets up a full-scan type experiment—that is, signal intensity versus m/z over the specified m/z range.

The MS full-scan master scan has these properties:

- A data-dependent MS/MS (ddMS²) scan type can follow an MS full scan.
- Any filter type can follow an MS full scan.

NOTE

Changed set values of given parameters appear in bold type. A red table cell or red border around a parameter box indicates that the set value is out of range. Point to the parameter box to display the valid range. If you enter a value that is outside the recommended range, the box border color changes to yellow.

Parameters in the Full Scan Properties pane

Parameter	Description
Orbitrap Resolution	The mass resolution of the Orbitrap analyzer, which is proportional to $1/\sqrt{m/z}$ of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the

Parameter	Description
	smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200. The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000, 180 000, 240 000, and 480 000.
Scan Range (m/z)	The range (in m/z) over which the mass analyzer detects peaks. Range: 40–6000 (8000 with BioPharma option); default: 100–1000 TIP: Because the system optimizes the transfer of ions for the defined mass range, we recommend that you restrict the scan range to the region of interest.
FAIMS Voltages	(Ion Source Type: NSI, H-ESI, APCI, ESI, OptiSpray; with FAIMS Mode enabled) Switches On (default) or Off the FAIMS voltages.
FAIMS CV (V)	(FAIMS Voltages: On) The optimized FAIMS compensation voltage. Range: –300 to 300; default 0 The observed CV values on this instrument are aligned with those of the other mass spectrometer platforms. This means the transmission for one m/z should be the same for the same CV setting on all platforms. For targeted CV experiments, it is recommended to use the value that was determined by a CV scan (performed in the Tune window) or optimized with single CV experiments in an HPLC run.
RF Lens (%)	Specify the value to control the RF amplitude applied to the ion funnel. Range: 0–200; default: 50 This value is a scaling factor applied to the relationship between the first mass and the RF amplitude. Decreasing the RF level decreases the transmission of high m/z ions through the ion funnel, increases the transmission of the low m/z ions, and might decrease the amount of fragmentation of fragile ions in the ion funnel. Increasing the RF level has the opposite effect.

Parameter	Description
	TIP: For optimal transmission, the default value provides a good starting point. A fine tuning of the MS is possible.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. - Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. - Custom: You can set the AGC Target in percent. TIP: Please refer to the system templates for application-specific suggestions.
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (DIA: 3000); default: 100 This is the normalized AGC Target value, it is represented as a percentage to aid in calculating the desired AGC Target. The base normalized value is different for scan types. The values for Orbitrap detector are as follows: Scan Type = Full Scan Normalized Base (100%) = 1e6
Absolute AGC Value	(AGC Target: Custom) The absolute AGC target value is a conversion of the AGC percentage target set in the method into absolute values. This read-only value will be automatically updated according to the scan type and the target percentage.
Maximum Injection Time (ms)	Maximum time in milliseconds that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 50
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. TIP: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral

Parameter	Description
	acquisition.
Data Type	Indicates how to collect data during the currently selected scan event: • Profile (default): Represents mass spectral peaks as point-to-point plots, with each point having an associated intensity value. • Centroid: Represents mass spectral peaks (as a bar graph) in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The normalized area of the peak provides the mass intensity data. Centroiding reduces the data file size while maintaining spectral quality.
Polarity	The polarity mode—Positive (default) or Negative—during a scan to detect ions of that polarity. NOTE: The polarity mode must be the same for all experiments within the method.
Source Fragmentation	Select the checkbox to switch on ion source CID. An offset voltage in the ion source accelerates the ions into the background gas. Collisions with the background gas might help in the desolvation of the ions and might increase sensitivity.
Energy (V)	(Source Fragmentation: On) The collision energy (in electron volts) for ion source fragmentation. Range: 1–135; default: 35 If you set the source fragmentation too high, fragmentation of ions might occur.
Use EASY-IC™	(Orbitrap Lock Mass Correction: EASY-IC™; Mode: Scan-to-Scan or Timed) If On is selected, it provides an internal reference mass that is used for mass correction during a run.
Lock Mass Injection	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Lock mass injection offers the possibility to inject a defined lock mass (if provided in the solvents or similar) to the ions of the analytical scan into the C-trap and ensures the detection

Parameter	Description
	and locking the chosen reference mass. If the user requests locking of the present scan, this function is essential. In general, a lock mass correction can only be performed, if the defined m/z value is detected in the performed Orbitrap scan. This means, the mass range of the scan of interest must provide the m/z value of the lock mass of interest. • If the Lock Mass Injection checkbox is selected, the lock mass injection will be performed for all scans in the method. • If the Lock Mass Injection checkbox is clear, you need to make sure that the lock mass is covered by at least one experiment/scan in the method. Example Full Scan (m/z 100–500) with ddMS², lock mass: 279.1591—the lock mass is covered by the mass range of the Full Scan and a correction will be applied to all other scans. The lock mass injection reduces the acquisition speed due to the associated overhead of about 6–10 ms/scan.
Scan Description	(Optional) User-defined scan description (maximum of 16 characters) appearing as a scan header in the raw data file. The scan description must be unique in the method. In the text field, you can use alphanumeric characters, commas, hyphens, parentheses, periods, plus signs, spaces, and underscores.

5.4.3 Targeted Selected Ion Monitoring (tSIM) Scan Properties pane



The Targeted SIM Scan Properties pane sets up a tSIM experiment, which is a master scan of the SIM scan type for which you know the precursor ions to monitor—that is, they are targeted or specified as a subset of the full scan range. You can import the settings of the mass list table from (or export them to) a CSV, a TXT, or an XML file.

NOTE

Changed set values of given parameters appear in bold type. A red table cell or red border around a parameter box indicates that the set value is out of range. Point to the parameter box to display the valid range. If you enter a value that is outside the recommended range, the box border color changes to yellow.

Parameters in the Targeted SIM Scan Properties pane

Parameter	Description
Isolation Window (m/z)	(SIM Window Mode: Center Mass) The isolation window (in m/z) for the precursor ions. The mass range for the peak is centered at the precursor mass and ranges from half of the isolation window to either side of the precursor mass. Range: 0.4-2000 (Orbitrap), 0.4-10 (Astral); default: 2 NOTE: A value that is too low may result in a loss of sensitivity due to lower transmission efficiency of a quadrupole at a narrower isolation window (physical property of quadrupole).
Isolation Offset	(SIM Window Mode: Center Mass) Applies an offset to the isolation window. The mass range for the peak is centered at the precursor mass plus the offset and ranges from half of the isolation window to either side of the precursor mass plus the offset. Available options are: • Off (default) • m/z Offset • Da Offset
Isolation Offset (m/z)	(Isolation Offset: m/z Offset) Specifies the offset value (in m/z) that is applied to the isolation and activation windows relative to that determined through the data-dependent criteria. For example, if the MS selects a peak at m/z 500 and you specify an m/z 2 width and m/z 0.5 offset, the isolation range is m/z 499.5–501.5 instead of m/z 499–501 (without the offset). Range: –200 to 200; default: 0
Isolation Offset (Da)	(Isolation Offset: Da Offset)

Parameter	Description
	Specifies the offset value (in Da units) that is applied to the isolation and activation windows relative to that determined through the data-dependent criteria. Range: –200 to 200; default: 0
SIM Window Mode	The available options are: - Center Mass supports the Exact Mass Calculator using the mass list table columns Formula, Adduct, and z. (default). - m/z Range allows to enter user-defined mass ranges. It blocks the Exact Mass Calculator and hides the columns Compound, Formula, Adduct, and z in the mass list table.
Detector Type	The available options are: - Astral: Select this detector type for highest sensitivity and speed. - Orbitrap: Select this detector type for highest resolution, mass accuracy and dynamic range.
Orbitrap Resolution	(Detector Type: Orbitrap) The mass resolution of the Orbitrap analyzer, which is proportional to $1/\sqrt{m/z}$ of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200. The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000, 180 000, 240 000, and 480 000.
FAIMS Voltages	(Ion Source Type: NSI, H-ESI, APCI, ESI, OptiSpray; with FAIMS Mode enabled) Switches On (default) or Off the FAIMS voltages.
FAIMS CV (V)	(FAIMS Voltages: On) The optimized FAIMS compensation voltage. Range: –300 to 300; default 0 The observed CV values on this instrument are aligned with those of the other mass spectrometer platforms. This means

Parameter	Description
	the transmission for one m/z should be the same for the same CV setting on all platforms. For targeted CV experiments, it is recommended to use the value that was determined by a CV scan (performed in the Tune window) or optimized with single CV experiments in an HPLC run.
RF Lens (%)	Specify the value to control the RF amplitude applied to the ion funnel. Range: 0–200; default: 50 This value is a scaling factor applied to the relationship between the first mass and the RF amplitude. Decreasing the RF level decreases the transmission of high m/z ions through the ion funnel, increases the transmission of the low m/z ions, and might decrease the amount of fragmentation of fragile ions in the ion funnel. Increasing the RF level has the opposite effect. TIP: For optimal transmission, the default value provides a good starting point. A fine tuning of the MS is possible.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. • Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. • Custom: You can set the AGC Target in percent. TIP: Please refer to the system templates for application-specific suggestions.
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (DIA: 3000); default: 100 This is the normalized AGC Target value, it is represented as a percentage to aid in calculating the desired AGC Target. The base normalized value is different for scan types. The values for Orbitrap detector are as follows: Scan Type = tSIM Normalized Base (100%) = 1e5

Parameter	Description
	The values for Astral detector are as follows: Scan Type = tSIM Normalized Base (100%) = 1e3
Absolute AGC Value	(AGC Target: Custom) The absolute AGC target value is a conversion of the AGC percentage target set in the method into absolute values. This read-only value will be automatically updated according to the scan type and the target percentage.
Maximum Injection Time (ms)	Maximum time in milliseconds that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 10 (Astral), 50 (Orbitrap)
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. NOTE: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral acquisition.
Data Type	Indicates how to collect data during the currently selected scan event: • Profile (default): Represents mass spectral peaks as point-to-point plots, with each point having an associated intensity value. • Centroid: Represents mass spectral peaks (as a bar graph) in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The normalized area of the peak provides the mass intensity data. Centroiding improves mass spectral data quality, obtains better mass assignments, and reduces the data file size. IMPORTANT: The Data Type Profile is not available when the Detector Type is Astral.
Polarity	The polarity mode—Positive (default) or Negative—during a scan to detect ions of that polarity.

Parameter	Description
	IMPORTANT: The polarity mode must be the same for all experiments within the method.
Source Fragmentation	Select the checkbox to switch on ion source CID. An offset voltage in the ion source accelerates the ions into the background gas. Collisions with the background gas might help in the desolvation of the ions and might increase sensitivity.
Energy (V)	(Source Fragmentation: On) Specifies the collision energy voltage for ion source fragmentation. Range: 1–135; default: 35 If you set the source fragmentation energy too high, fragmentation of ions might occur.
Use EASY-IC™	(Orbitrap Lock Mass Correction: EASY-IC™; Mode: Scan-to-Scan or Timed) If On is selected, it provides an internal reference mass that is used for mass correction during a run.
Loop Control	(Multiple experiments only) Indicates when the current experiment ends and the next one begins. • N: Runs each cycle on the next <i>N</i> targeted scans. • Time: Run each cycle for the specified time • All (default): Runs each cycle on the full list of spectra. This feature is useful for a long mass list. It ensures that scans from each experiment in the timeline are run at regular intervals. During each cycle, the current scan is run for <i>N</i> targets or as many scans that can be run during the specified cycle time. In targeted experiments, the scan cycle repeats through the mass list from where it last left off. With this feature, you can add another experiment without having to split up the targeted list.
Time (sec)	(Loop Control: Time) The duration in seconds of each scan loop. Range: 0.2–100,000; default: 3
N (Number of Spectra)	(Loop Control: N)

Parameter	Description
	The number of spectra for each scan loop. Range: 1–999; default: 20
Lock Mass Injection	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Lock mass injection offers the possibility to inject a defined lock mass (if provided in the solvents or similar) to the ions of the analytical scan into the C-trap and ensures the detection and locking the chosen reference mass. If the user requests locking of the present scan, this function is essential. In general, a lock mass correction can only be performed, if the defined m/z value is detected in the performed Orbitrap scan. This means, the mass range of the scan of interest must provide the m/z value of the lock mass of interest. • If the Lock Mass Injection checkbox is selected, the lock mass injection will be performed for all scans in the method. • If the Lock Mass Injection checkbox is clear, you need to make sure that the lock mass is covered by at least one experiment/scan in the method. Example Full Scan (m/z 100–500) with ddMS², lock mass: 279.1591—the lock mass is covered by the mass range of the Full Scan and a correction will be applied to all other scans. The lock mass injection reduces the acquisition speed due to the associated overhead of about 6–10 ms/scan.
Scan Description	(Optional) User-defined scan description (maximum of 16 characters) appearing as a scan header in the raw data file. The scan description must be unique in the method. In the text field, you can use alphanumeric characters, commas, hyphens, parentheses, periods, plus signs, spaces, and underscores.
Time Mode	Enables scheduled or unscheduled scans. For scheduled scans, you can specify the retention time and the window size or the start time and the end time. The available options are: • Start/End Time: t start (min) and t stop (min) available in scan table

Parameter	Description
	- Retention Time Window (default): RT Time (min) and Window (min) available in scan table - Unscheduled

Mass list table

TIP

To specify a parameter value for a specific precursor ion, select  to add that parameter to the mass list table. Select  to remove the parameter from the table.

Parameter	Description
m/z range	(SIM Window Mode: m/z Range) Specify the mass range over which the analyzer detects. Range: 40–6000, default: 523.265-525.265 Orbitrap: First mass: 40–2500; last mass: >first mass–6000 Astral: Mass difference between first mass and last mass: 0.4–10 TIP: For optimum operation, it is recommended that Last Mass < 15 × First Mass.
Compound	(Optional) You can give each entry a name. This name will be recorded to the raw file.
Formula	(Optional) You can enter either a chemical formula, peptide sequence, or amino acids. If more than one formula type for your input is possible, the Method Editor application chooses the type with the highest priority (chemical formula > peptides > amino acids). Enter a prefix directly with the formula to make sure that the right type is applied. Peptide sequences can be marked by the prefix “p:” (for example, p:MRFA with m/z = 524.26496), amino acids by the prefix “a:” (for example, a:MRFA with m/z = 506.25440). You can also include any isotopes that differ from the most abundant. For example, methanol with a C-13 isotope would be defined as [13]CH3OH.

Parameter	Description
	When you use this field, you must specify an adduct.
Adduct	- Positive polarity mode: no adduct (default), +H, +Na, +K, or +NH₄ - Negative polarity mode: no adduct (default), –H, +Cl, +OH, or +HCOO This field is disabled when no formula is used.
Center Mass (m/z)	(SIM Window Mode: Center Mass) The precursor mass (in <i>m/z</i>). If the Formula field is filled, the MS automatically calculates the <i>m/z</i>. The precursor mass is the center mass of the isolation range. The isolation range is one-half of the isolation width applied to either side of the precursor mass. The allowable range of values for the first mass of the isolation range: 40–2500; default: 524.2649.
z	(SIM Window Mode: m/z Range) The expected charge state of the precursor ion. Range: 0–100; default: 1 TIP: Use zero (oder 0) to let the instrument ignore the charge state.
RT Time (min)	(Time Mode: Retention Time Window) The retention time (RT) (in minutes). Range: 0–999; default: 30
Window (min)	(Time Mode: Retention Time Window) The window width (in minutes), centered around the defined Retention Time. For example, when Retention Time is set to 10 and Window is set to 2, acquisition of spectra for the target will only occur between the 9 and 11 minutes. Range: 0–999; default: 60
t start (min)	(Time Mode: Start/End Time) The time (in minutes) to start acquiring spectra for the defined target. Range: 0–60; default: 0

Parameter	Description
t stop (min)	(Time Mode: Start/End Time) The time (in minutes) to stop acquiring spectra for the defined target. Range: 0–60; default: 60
Isolation Window (<i>m/z</i>)	(SIM Window Mode: Center Mass) The isolation window (in <i>m/z</i>) for the precursor ions. The mass range for the peak is centered at the precursor mass and ranges from half of the isolation window to either side of the precursor mass. Range: 0.4-2000 (Orbitrap), 0.4-10 (Astral); default: 2 NOTE: A value that is too low may result in a loss of sensitivity due to lower transmission efficiency of a quadrupole at a narrower isolation window (physical property of quadrupole).

5.4.4 Targeted MS2 (tMS2) Scan Properties pane



The Targeted MS2 (tMS2) Scan Properties pane sets up a targeted MS/MS experiment, which can generate SRM scan type data. The scan is “targeted” because you know which precursors to monitor. You specify the precursors in the mass list table whose settings you can import from (or export to) a CSV, a TXT, or an XML file.

NOTE

Changed set values of given parameters appear in bold type. A red table cell or red border around a parameter box indicates that the set value is out of range. Point to the parameter box to display the valid range. If you enter a value that is outside the recommended range, the box border color changes to yellow.

Parameters in the Targeted MS² Scan Properties pane

Parameter	Description
Isolation Window (<i>m/z</i>)	The isolation window (in <i>m/z</i>) for the precursor ions.

Parameter	Description
	The mass range for the peak is centered at the precursor mass and ranges from half of the isolation window to either side of the precursor mass Range: 0.4–2000; default: 2
	NOTE: A value that is too low may result in a loss of sensitivity due to lower transmission efficiency of a quadrupole at a narrower isolation window (physical property of quadrupole).
Isolation Offset	Applies an offset to the isolation window. The mass range for the peak is centered at the precursor mass plus the offset and ranges from half of the isolation window to either side of the precursor mass plus the offset. Available options are: • Off (default) • m/z Offset • Da Offset
Isolation Offset (m/z)	(Isolation Offset: m/z Offset) Specifies the offset value (in <i>m/z</i>) that is applied to the isolation and activation windows relative to that determined through the data-dependent criteria. For example, if the MS selects a peak at <i>m/z</i> 500 and you specify an <i>m/z</i> 2 width and <i>m/z</i> 0.5 offset, the isolation range is <i>m/z</i> 499.5–501.5 instead of <i>m/z</i> 499–501 (without the offset). Range: –200 to 200; default: 0
Isolation Offset (Da)	(Isolation Offset: Da Offset) Specifies the offset value (in Da units) that is applied to the isolation and activation windows relative to that determined through the data-dependent criteria. For example, if the instrument selects a peak at Da 500 and you specify a Da 2 width with a Da 0.5 offset, the isolation range is Da 499.5–501.5 instead of Da 499–501 (without the offset). Range: –200 to 200; default: 0
Reported Mass	(Isolation Offset: m/z Offset or Da Offset) Specifies which mass is written to the raw file. The available options are: • Offset Mass (default) • Original Mass Original mass is the mass that was detected (that is, the mass that would be written to the raw file if isolation offset were set to Off). Offset mass is the center mass of the executed isolation.
Collision Energy Type	The available options are: • Normalized (default): Collision energy is set in percentage and applies an energy dependent on mass and charge state.

Parameter	Description
	• Absolute: Collision energy is set independent of the mass and charge of the precursor ion.
HCD Collision Energy (% or V)	The MS injects isolated precursor ions into the IRM and the ion processor (for Astral when using single Collision energy value) with an HCD collision energy (as a percentage or independent of mass and charge).
HCD Collision Energies (% or V)	The MS injects isolated precursor ions into the IRM in one to five steps, each with a different HCD collision energy (as a percentage or independent of mass and charge). The MS then mass analyzes all groups of ions in a single scan. A high HCD collision energy results in more ion activation, which generally leads to more fragmentation. A low HCD collision energy results in less ion activation, which generally leads to less fragmentation. With a single value, only one collision energy is applied. If more than one value is defined, the instrument will perform Stepped Collision Energy. Use commas to separate several energy values. The ion processor and the IRM are calibrated to generate the same MS² spectra. Range: 1–200; default: 30
Detector Type	The available options are: • Astral: Select this detector type for highest sensitivity and speed. • Orbitrap: Select this detector type for highest resolution, mass accuracy and dynamic range.
TMT	(Detector Type: Astral) Set TMT to On when using the TMT workflow. The available options are: • On • Off (default) TIP: For a better TMT performance, clear the Advanced Peak Determination checkbox when TMT is set to On.
Orbitrap Resolution	(Detector Type: Orbitrap) The mass resolution of the Orbitrap analyzer, which is proportional to 1/sqrt of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200. The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000, 180 000, 240 000, and 480 000.
Scan Range Mode	Select the scan range mode to determine the scan range. The available options are:

Parameter	Description																					
	Auto: The last mass (lm) is automatically set to (Precursor + IsoWidth / 2) × Charge × 1.02 + 20.0. The first mass (fm) is set such as to satisfy the Factor 5–10–15 rule for the lm:fm ratio to ensure the best retainment of all masses in the IRM. It is subject to the corresponding <i>m/z</i> lower limit: <table><tr><th>Last mass <i>m/z</i> [Th]</th><th>Factor of 5–10–15 Rule</th><th>First mass <i>m/z</i> [Th]</th></tr><tr><td>≤ 200</td><td>5</td><td>40</td></tr><tr><td>200–250</td><td>5</td><td>lm / 5</td></tr><tr><td>250–500</td><td>10</td><td>50</td></tr><tr><td>500–1500</td><td>10</td><td>lm / 10</td></tr><tr><td>1500–2250</td><td>15</td><td>150</td></tr><tr><td>≥ 2250</td><td>15</td><td>lm / 15</td></tr></table>	Last mass <i>m/z</i> [Th]	Factor of 5–10–15 Rule	First mass <i>m/z</i> [Th]	≤ 200	5	40	200–250	5	lm / 5	250–500	10	50	500–1500	10	lm / 10	1500–2250	15	150	≥ 2250	15	lm / 15
Last mass <i>m/z</i> [Th]	Factor of 5–10–15 Rule	First mass <i>m/z</i> [Th]																				
≤ 200	5	40																				
200–250	5	lm / 5																				
250–500	10	50																				
500–1500	10	lm / 10																				
1500–2250	15	150																				
≥ 2250	15	lm / 15																				
	- Define First Mass: Explicitly define the first mass and the system automatically sets the last mass. - Define <i>m/z</i> Range: Explicitly define the first and last mass. Because the system optimizes the transfer of ions for the defined scan range, it is recommended to restrict the scan range to the region of interest. (Application Mode: Small Molecule) Auto-Extended: The instrument calculates the scan range starting from the precursor <i>m/z</i>. This scan range mode complies with the Factor 15 rule with the first mass <i>m/z</i> range 40–2500.																					
	TIP: Because the system optimizes the transfer of ions for the defined scan range, it is recommended to restrict the scan range to the region of interest.																					
First Mass (<i>m/z</i>)	(Detector Type: Orbitrap; Scan Range Mode: Define First Mass) The scan range's First Mass <i>m/z</i> value. Range: 40–2500; default: 120																					
	TIP: Please be aware that too low first masses may discriminate higher fragment masses.																					
Scan Range (<i>m/z</i>)	(Scan Range Mode: Define <i>m/z</i> Range) The values for the Scan Range (<i>m/z</i>). Range: 40–6000 (Astral); 40–6000 (8000 with BioPharma option) (Orbitrap); default: 100–1000																					

Parameter	Description
	TIP: For optimal results, restrict the scan range to the region of interest. In particular for the Astral analyzer, last mass should be set lower than m/z 3000 unless the experiment requires a higher last mass setting. Setting last mass higher than m/z 3000 may slow down acquisition speed.
FAIMS Voltages	(Ion Source Type: NSI, H-ESI, APCI, ESI, OptiSpray; with FAIMS Mode enabled) Switches On (default) or Off the FAIMS voltages.
FAIMS CV (V)	(FAIMS Voltages: On) The optimized FAIMS compensation voltage. Range: –300 to 300; default 0 The observed CV values on this instrument are aligned with those of the other mass spectrometer platforms. This means the transmission for one m/z should be the same for the same CV setting on all platforms. For targeted CV experiments, it is recommended to use the value that was determined by a CV scan (performed in the Tune window) or optimized with single CV experiments in an HPLC run.
RF Lens (%)	Specify the value to control the RF amplitude applied to the ion funnel. Range: 0–200; default: 50 This value is a scaling factor applied to the relationship between the first mass and the RF amplitude. Decreasing the RF level decreases the transmission of high m/z ions through the ion funnel, increases the transmission of the low m/z ions, and might decrease the amount of fragmentation of fragile ions in the ion funnel. Increasing the RF level has the opposite effect. TIP: For optimal transmission, the default value provides a good starting point. A fine tuning of the MS is possible.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. - Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. - Custom: You can set the AGC Target in percent. TIP: Please refer to the system templates for application-specific suggestions.
Absolute AGC Value	(AGC Target: Custom) The absolute AGC target value is a conversion of the AGC percentage target set in the method into absolute values. This read-only value will be automatically updated according to the scan type and the target percentage.

Parameter	Description
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (MS2, DIA: 3000); default: 100 This is the normalized AGC Target value, it is represented as a percentage to aid in calculating the desired AGC Target. The base normalized value is different for scan types. The values for Orbitrap detector are as follows: Scan Type = tMS2 Normalized Base (100%) = 1e5 The values for Astral detector are as follows: Scan Type = tMS2 Normalized Base (100%) = 1e4
Maximum Injection Time (ms)	Maximum time in milliseconds that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 10 (Astral), 50 (Orbitrap)
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. NOTE: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral acquisition.
Data Type	Indicates how to collect data during the currently selected scan event: • Profile (default): Represents mass spectral peaks as point-to-point plots, with each point having an associated intensity value. • Centroid: Represents mass spectral peaks (as a bar graph) in terms of two parameters: The centroid (the weighted center of mass) and the intensity. The normalized area of the peak provides the mass intensity data. Centroiding reduces the data file size while maintaining spectral quality. IMPORTANT: The Data Type Profile is not available when the Detector Type is Astral.
Polarity	The polarity mode—Positive (default) or Negative—during a scan to detect ions of that polarity. IMPORTANT: The polarity mode must be the same for all experiments within the method.
Source Fragmentation	Select the checkbox to switch on ion source CID. An offset voltage in the ion source accelerates the ions into the background gas. Collisions with the background gas might help in the desolvation of the ions and might increase sensitivity.
Energy (V)	(Source Fragmentation: On)

Parameter	Description
	Specifies the collision energy voltage for ion source fragmentation. Range: 1–135; default: 35 If you set the source fragmentation energy too high, fragmentation of ions might occur.
Use EASY-IC™	(Orbitrap Lock Mass Correction: EASY-IC™; Mode: Scan-to-Scan or Timed) If On is selected, it provides an internal reference mass that is used for mass correction during a run.
Loop Control	(Multiple experiments only) Indicates when the current experiment ends and the next one begins. • N: Runs each cycle on the next <i>N</i> targeted scan(s) • Time: Run each cycle for the specified time • All (default): Runs each cycle on the full list of spectra This feature is useful for a long mass list. It ensures that scans from each experiment in the timeline are run at regular intervals. During each cycle, the current scan is run for <i>N</i> targets or as many scans that can be run during the specified cycle time. In targeted experiments, the scan cycle repeats through the mass list from where it last left off. With this feature, you can add another experiment without having to split up the targeted list.
N (Number of Spectra)	(Loop Control: N) The number of spectra for each scan loop. Range: 1–999; default: 20
Time (sec)	(Loop Control: Time) The duration in seconds of each scan loop. Range: 0.2–100000; default: 3
Time Mode	Enables scheduled or unscheduled scans. For scheduled scans, you can specify the retention time and the window size or the start time and the end time. The available options are: • Start/End Time: <i>t</i> start (min) and <i>t</i> stop (min) available in scan table • Retention Time Window (default): RT Time (min) and Window (min) available in scan table • Unscheduled
Lock Mass Injection	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Lock mass injection offers the possibility to inject a defined lock mass (if provided in the solvents or similar) to the ions of the analytical scan into the C-trap and ensures the detection and locking the chosen reference mass. If the user requests locking of the present scan, this function is essential.

Parameter	Description
	In general, a lock mass correction can only be performed, if the defined m/z value is detected in the performed Orbitrap scan. This means, the mass range of the scan of interest must provide the m/z value of the lock mass of interest. • If the Lock Mass Injection checkbox is selected, the lock mass injection will be performed for all scans in the method. • If the Lock Mass Injection checkbox is clear, you need to make sure that the lock mass is covered by at least one experiment/scan in the method. Example Full Scan (m/z 100–500) with ddMS², lock mass: 279.1591—the lock mass is covered by the mass range of the Full Scan and a correction will be applied to all other scans. The lock mass injection reduces the acquisition speed due to the associated overhead of about 6–10 ms/scan.
Scan Description	(Optional) User-defined scan description (maximum of 16 characters) appearing as a scan header in the raw data file. The scan description must be unique in the method. In the text field, you can use alphanumeric characters, commas, hyphens, parentheses, periods, plus signs, spaces, and underscores.

Mass list table

TIP

To specify a parameter value for a specific precursor ion, select  to add that parameter to the mass list table. Select  to remove the parameter from the table.

Parameter	Description
Compound	(SIM Window Mode: Center Mass) You can give each entry a name. This name will be recorded to the raw file.
Formula	(SIM Window Mode: Center Mass) You can enter either a chemical formula, peptide sequence, or amino acids. If more than one formula type for your input is possible, the Method Editor application chooses the type with the highest priority (chemical formula > peptides > amino acids). Enter a prefix directly with the formula to make sure

Parameter	Description
	that the right type is applied. Peptide sequences can be marked by the prefix “p:” (for example, p:MRFA with m/z = 524.26496), amino acids by the prefix “a:” (for example, a:MRFA with m/z = 506.25440). You can also include any isotopes that differ from the most abundant. For example, methanol with a C-13 isotope would be defined as [13]CH3OH. When you use this field, you must specify an adduct.
Adduct	(SIM Window Mode: Center Mass) - Positive polarity mode: no adduct (default), +H, +Na, +K, or +NH₄ - Negative polarity mode: no adduct (default), –H, +Cl, +OH, or +HCOO This field is disabled when no formula is used.
m/z	The precursor mass (in m/z). If the Formula field is filled, the MS automatically calculates the m/z. Range: 1–100000
z	(SIM Window Mode: m/z Range) The expected charge state of the precursor ion, which is used to determine the collision energy or reaction times. Range: 0–100; default: 1 Use zero (oder 0) to let the instrument ignore the charge state.
RT Time (min)	(Time Mode: Retention Time Window) The retention time (RT) (in minutes). Range: 0–999; default: 30
Window (min)	(Time Mode: Retention Time Window) The window width (in minutes), centered around the defined Retention Time. For example, when Retention Time is set to 10 and Window is set to 2, acquisition of spectra for the target will only occur between the 9 and 11 minutes. Range: 0–999; default: 60
t start (min)	(Time Mode: Start/End Time) The time (in minutes) to start acquiring spectra for the defined target. Range: 0–60; default: 0

Parameter	Description
t stop (min)	(Time Mode: Start/End Time) The time (in minutes) to stop acquiring spectra for the defined target. Range: 0–60; default: 60

5.4.5 Data-Independent Analysis (DIA) Properties pane

DIA Scan

The Data-Independent Analysis (DIA) Properties pane sets up a DIA experiment, which scans, isolates, and fragments all ions in a series of precursor isolation windows that span a defined precursor mass range. You can export the settings of the mass list table to a CSV, a TXT, or an XML file.

The DIA master scan has these properties:

- The precursor masses come from a calculated list or come from an incremented isolation window.
- No filters or scans can follow the DIA scan.

The entries in the DIA m/z window table are created automatically, depending on your settings for Precursor Mass Range, Isolation Window, and Window Overlap; or can be user-defined.

NOTE

Changed set values of given parameters appear in bold type. A red table cell or red border around a parameter box indicates that the set value is out of range. Point to the parameter box to display the valid range. If you enter a value that is outside the recommended range, the box border color changes to yellow.

Parameters in the Data-Independent Analysis Properties pane

Parameter	Description
Precursor Mass Range (m/z)	(DIA Window Type: Auto) The precursor ion mass range (in m/z) used for DIA scans. Range: 40–2500; default: 100–1100

Parameter	Description
	NOTE: The number of rows in the DIA <i>m/z</i> Window table automatically adjusts after you set the values for the precursor mass range and the Isolation Window (<i>m/z</i>).
DIA Window Type	The following options are available: • Auto: You cannot edit the entries in the table cells. Add/Import/Delete operations are not possible for the mass list table. • User Defined: You can edit the entries in the table cells. You can delete mass ranges and you can create overlapping mass ranges. Add/Import/Delete operations for the mass list table are possible.
Isolation Window (<i>m/z</i>)	(DIA Window Type: Auto) The isolation window (in <i>m/z</i>) for the precursor ions. The mass range for the peak is centered at the precursor mass and ranges from half of the isolation window on either side of the precursor mass. Range: 0.4–800; default: 4 NOTE: A value that is too low may result in a loss of sensitivity due to lower transmission efficiency of a quadrupole at a narrower isolation window (physical property of quadrupole).
Window Overlap (<i>m/z</i>)	(DIA Window Type: Auto) The overlap value for the isolation window (in <i>m/z</i>). Range: 0–Isolation Window <i>m/z</i> max. value; default: 0
Window Placement Optimization	(DIA Window Type: Auto and Application Mode: Peptide) Optimize for +1 or +2 ions only. Window Placement Optimization considers the sum formulas of peptide samples and resulting mass defects. Calculated <i>m/z</i> windows are shifted accordingly. The available options are: • On • Off (default) NOTE: The combination of Window Overlap with Window Placement Optimization is not recommended.
Number of Scan Events	(DIA Window Type: Auto) The (read-only) number of scan windows, with one scan event per window. The number of scan events is equal to the mass range size, divided by the isolation window size.
DIA Window Mode	The available options are: • Center Mass displays or allows to set a <i>m/z</i> with a Window width • <i>m/z</i> Range displays the <i>m/z</i> range or allows to enter user-defined mass ranges

Parameter	Description
Collision Energy Type	The available options are: • Normalized (default): Collision energy is set in percentage and applies an energy dependent on mass and charge state. • Absolute: Collision energy is set independent of the mass and charge of the precursor ion.
HCD Collision Energy (% or V)	The MS injects isolated precursor ions into the IRM and the ion processor (for Astral when using single Collision energy value) with an HCD collision energy (as a percentage or independent of mass and charge).
HCD Collision Energies (% or V)	The MS injects isolated precursor ions into the IRM in one to five steps, each with a different HCD collision energy (as a percentage or independent of mass and charge). The MS then mass analyzes all groups of ions in a single scan. A high HCD collision energy results in more ion activation, which generally leads to more fragmentation. A low HCD collision energy results in less ion activation, which generally leads to less fragmentation. With a single value, only one collision energy is applied. If more than one value is defined, the instrument will perform Stepped Collision Energy. Use commas to separate several energy values. The ion processor and the IRM are calibrated to generate the same MS² spectra. Range: 1–200; default: 30
Detector Type	The available options are: • Astral: Select this detector type for highest sensitivity and speed. • Orbitrap: Select this detector type for highest resolution, mass accuracy and dynamic range.
TMT	(Detector Type: Astral) Set TMT to On when using the TMT workflow. The available options are: • On • Off (default) TIP: For a better TMT performance, clear the Advanced Peak Determination checkbox when TMT is set to On.
Orbitrap Resolution	(Detector Type: Orbitrap) The mass resolution of the Orbitrap analyzer, which is proportional to 1/sqrt of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200.

Parameter	Description
	The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000, 180 000, 240 000, and 480 000.
FAIMS Voltages	(Ion Source Type: NSI, H-ESI, APCI, ESI, OptiSpray; with FAIMS Mode enabled) Switches On (default) or Off the FAIMS voltages.
FAIMS CV (V)	(FAIMS Voltages: On) The optimized FAIMS compensation voltage. Range: –300 to 300; default 0 The observed CV values on this instrument are aligned with those of the other mass spectrometer platforms. This means the transmission for one m/z should be the same for the same CV setting on all platforms. For targeted CV experiments, it is recommended to use the value that was determined by a CV scan (performed in the Tune window) or optimized with single CV experiments in an HPLC run.

Scan Range Mode Select the scan range mode to determine the scan range. The available options are:

- Auto: The last mass (m_l) is automatically set to $(\text{Precursor} + \text{IsoWidth} / 2) \times \text{Charge} \times 1.02 + 20.0$. The first mass (m_f) is set such as to satisfy the Factor 5–10–15 rule for the $m_l:m_f$ ratio to ensure the best retainment of all masses in the IRM. It is subject to the corresponding m/z lower limit:

Last mass m/z [Th]	Factor of 5–10–15 Rule	First mass m/z [Th]
≤ 200	5	40
200–250	5	$m_l / 5$
250–500	10	50
500–1500	10	$m_l / 10$
1500–2250	15	150
≥ 2250	15	$m_l / 15$

- Define First Mass: Explicitly define the first mass and the system automatically sets the last mass.
- Define m/z Range: Explicitly define the first and last mass.

Because the system optimizes the transfer of ions for the defined scan range, it is recommended to restrict the scan range to the region of interest.

Parameter	Description
	(Application Mode: Small Molecule) Auto-Extended: The instrument calculates the scan range starting from the precursor m/z. This scan range mode complies with the Factor 15 rule with the first mass m/z range 40–2500.
First Mass (m/z)	(Scan Range Mode: Define First Mass) The scan range's First Mass m/z value. Range: 40–2500; default: 120 NOTE: Please be aware that too low first masses may discriminate higher fragment masses.
Scan Range (m/z)	(Scan Range Mode: Define m/z range) The values for the Scan Range (m/z). Range: 40–6000 (Astral); 40–6000 (8000 with BioPharma option)(Orbitrap); default: 100–1000 TIP: For optimal results, restrict the scan range to the region of interest. In particular for the Astral analyzer, last mass should be set lower than m/z 3000 unless the experiment requires a higher last mass setting. Setting last mass higher than m/z 3000 may slow down acquisition speed.
RF Lens (%)	Specify the value to control the RF amplitude applied to the ion funnel. Range: 0–200; default: 50 This value is a scaling factor applied to the relationship between the first mass and the RF amplitude. Decreasing the RF level decreases the transmission of high m/z ions through the ion funnel, increases the transmission of the low m/z ions, and might decrease the amount of fragmentation of fragile ions in the ion funnel. Increasing the RF level has the opposite effect. TIP: For optimal transmission, the default value provides a good starting point. A fine tuning of the MS is possible.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. - Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. - Custom: You can set the AGC Target in percent. TIP: Please refer to the system templates for application-specific suggestions.
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (MS2, DIA: 3000); default: 100

Parameter	Description
	This is the normalized AGC Target value, it is represented as a percentage to aid in calculating the desired AGC Target. The base normalized value is different for scan types. The values for Orbitrap detector are as follows: Scan Type = DIA Normalized Base (100%) = 1e5 The values for Astral detector are as follows: Scan Type = DIA Normalized Base (100%) = 1e4
Absolute AGC Value	(AGC Target: Custom) The absolute AGC target value is a conversion of the AGC percentage target set in the method into absolute values. This read-only value will be automatically updated according to the scan type and the target percentage.
Maximum Injection Time (ms)	Maximum time in milliseconds that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 10 (Astral), 50 (Orbitrap)
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. NOTE: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral acquisition.
Data Type	Indicates how to collect data during the currently selected scan event: • Profile (default): Represents mass spectral peaks as point-to-point plots, with each point having an associated intensity value. • Centroid: Represents mass spectral peaks (as a bar graph) in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The normalized area of the peak provides the mass intensity data. Centroiding reduces the data file size while maintaining spectral quality. IMPORTANT: The Data Type Profile is not available when the Detector Type is Astral.
Polarity	The polarity mode—Positive (default) or Negative—during a scan to detect ions of that polarity. IMPORTANT: The polarity mode must be the same for all experiments within the method.
Source Fragmentation	Select the checkbox to switch on ion source CID. An offset voltage in the ion source accelerates the ions into the background gas. Collisions with the background gas might help in the desolvation of the ions and might increase sensitivity.

Parameter	Description
Energy (V)	(Source Fragmentation: On) Specifies the collision energy voltage for ion source fragmentation. Range: 1–135; default: 35 If you set the source fragmentation energy too high, fragmentation of ions might occur.
Use EASY-IC™	(Orbitrap Lock Mass Correction: EASY-IC™; Mode: Scan-to-Scan or Timed) If On is selected, it provides an internal reference mass that is used for mass correction during a run.
Loop Control	(Multiple experiments only) Indicates when the current experiment ends and the next experiment begins. • N: Runs each cycle on the next <i>N</i> targeted scans. • Time: Run each cycle for the specified time • All (default): Runs each cycle on the full list of spectra. This feature is useful for a long mass list. It ensures that scans from each experiment in the timeline are run at regular intervals. During each cycle, the current scan is run for <i>N</i> targets or as many scans that can be run during the specified cycle time. In targeted experiments, the scan cycle repeats through the mass list from where it last left off. With this feature, you can add another experiment without having to split up the targeted list.
N (Number of Spectra)	(Loop Control: N) The number of spectra for each scan loop. Range: 1–999; default: 20
Time (sec)	(Loop Control: Time) The duration in seconds of each scan loop. Range: 0.2–100,000; default: 3
Lock Mass Injection	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Lock mass injection offers the possibility to inject a defined lock mass (if provided in the solvents or similar) to the ions of the analytical scan into the C-trap and ensures the detection and locking the chosen reference mass. If the user requests locking of the present scan, this function is essential. In general, a lock mass correction can only be performed, if the defined <i>m/z</i> value is detected in the performed Orbitrap scan. This means, the mass range of the scan of interest must provide the <i>m/z</i> value of the lock mass of interest. • If the Lock Mass Injection checkbox is selected, the lock mass injection will be performed for all scans in the method. • If the Lock Mass Injection checkbox is clear, you need to make sure that the lock mass is covered by at least one experiment/scan in the method.

Parameter	Description
	Example Full Scan (m/z 100–500) with ddMS², lock mass: 279.1591—the lock mass is covered by the mass range of the Full Scan and a correction will be applied to all other scans. The lock mass injection reduces the acquisition speed due to the associated overhead of about 6–10 ms/scan.
Scan Description	(Optional) User-defined scan description (maximum of 16 characters) appearing as a scan header in the raw data file. The scan description must be unique in the method. In the text field, you can use alphanumeric characters, commas, hyphens, parentheses, periods, plus signs, spaces, and underscores.

Parameters in the DIA m/z Window table

Parameter	Description
m/z range	(DIA Window Mode: m/z Range) The mass ranges depend on your settings for Precursor Mass Range, Isolation Window, and Window Overlap. Range: first mass: 40–2500; last mass: >first mass–6000
Center Mass m/z	(DIA Window Mode: Center Mass) The precursor mass is the Center Mass of the isolation range. The isolation range is one half of the Window Width applied to either side of the Center Mass. Range: 40–2500
Window Width (m/z)	(DIA Window Mode: Center Mass) Specifies the isolation window for the precursor ions. Range: 0.4–800

5.4.6 All Ion Fragmentation (AIF) Scan Properties pane



The All Ion Fragmentation Properties pane sets up an AIF experiment. The AIF master scan uses an MS/MS scan with a wide isolation window to fragment all precursor ions in the ion routing multipole.

NOTE

Changed set values of given parameters appear in bold type. A red table cell or red border around a parameter box indicates that the set value is out of range. Point to the parameter box to display the valid range. If you enter a value that is outside the recommended range, the box border color changes to yellow.

Parameters in the All Ion Fragmentation Properties pane

Parameter	Description
Isolation Range (<i>m/z</i>)	The isolation range (as an <i>m/z</i> value). Range: 40–2500; default: 100–1000
Collision Energy Type	The available options are: • Normalized (default): The accelerating voltage (as a percentage). • Absolute: Collision energy is set independent of the mass and charge of the precursor ion.
HCD Collision Energy (% or V) HCD Collision Energies (% or V)	The MS injects isolated precursor ions into the IRM and the ion processor (for Astral when using single Collision energy value) with an HCD collision energy (as a percentage or independent of mass and charge). The MS injects isolated precursor ions into the IRM in one to five steps, each with a different HCD collision energy (as a percentage or independent of mass and charge). The MS then mass analyzes all groups of ions in a single scan. A high HCD collision energy results in more ion activation, which generally leads to more fragmentation. A low HCD collision energy results in less ion activation, which generally leads to less fragmentation. With a single value, only one collision energy is applied. If more than one value is defined, the instrument will perform Stepped Collision Energy. Use commas to separate several energy values. The ion processor and the IRM are calibrated to generate the same MS² spectra. Range: 1–200; default: 30
Detector Type	The available options are: • Astral: Select this detector type for highest sensitivity and speed.

Parameter	Description
	Orbitrap: Select this detector type for highest resolution, mass accuracy and dynamic range.
Orbitrap Resolution	(Detector Type: Orbitrap) The mass resolution of the Orbitrap analyzer, which is proportional to $1/\sqrt{m/z}$ of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200. The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000, 180 000, 240 000, and 480 000.
Scan Range (m/z)	The values for the Scan Range (m/z). Range: 40–6000 (Astral); 40–6000 (8000 with BioPharma option) (Orbitrap); default: 100–1000 NOTE: For optimal results, restrict the scan range to the region of interest. In particular for the Astral analyzer, last mass should be set lower than m/z 3000 unless the experiment requires a higher last mass setting. Setting last mass higher than m/z 3000 may slow down acquisition speed.
FAIMS Voltages	(Ion Source Type: NSI, H-ESI, APCI, ESI, OptiSpray; with FAIMS Mode enabled) Switches On (default) or Off the FAIMS voltages.
FAIMS CV (V)	(FAIMS Voltages: On) The optimized FAIMS compensation voltage. Range: –300 to 300; default 0 The observed CV values on this instrument are aligned with those of the other mass spectrometer platforms. This means the transmission for one m/z should be the same for the same CV setting on all platforms. For targeted CV experiments, it is recommended to use the value that was determined by a CV scan (performed in the Tune window) or optimized with single CV experiments in an HPLC run.
RF Lens (%)	Specify the value to control the RF amplitude applied to the ion funnel.

Parameter	Description
	Range: 0–200; default: 50 This value is a scaling factor applied to the relationship between the first mass and the RF amplitude. Decreasing the RF level decreases the transmission of high m/z ions through the ion funnel, increases the transmission of the low m/z ions, and might decrease the amount of fragmentation of fragile ions in the ion funnel. Increasing the RF level has the opposite effect. TIP: For optimal transmission, the default value provides a good starting point. A fine tuning of the MS is possible.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. • Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. • Custom: You can set the AGC Target in percent. TIP: Please refer to the system templates for application-specific suggestions.
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (MS2, DIA: 3000); default: 100 This is the normalized AGC Target value, it is represented as a percentage to aid in calculating the desired AGC Target. The base normalized value is different for scan types. The values for Orbitrap detector are as follows: Scan Type = AIF Normalized Base (100%) = 1e6 The values for Astral detector are as follows: Scan Type = AIF Normalized Base (100%) = 1e4
Absolute AGC Value	(AGC Target: Custom) The absolute AGC target value is a conversion of the AGC percentage target set in the method into absolute values. This read-only value will be automatically updated according to the scan type and the target percentage.

Parameter	Description
Maximum Injection Time (ms)	Maximum time in milliseconds that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 10 (Astral), 50 (Orbitrap)
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. NOTE: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral acquisition.
Data Type	Indicates how to collect data during the currently selected scan event: • Profile (default): Represents mass spectral peaks as point-to-point plots, with each point having an associated intensity value. • Centroid: Represents mass spectral peaks (as a bar graph) in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The normalized area of the peak provides the mass intensity data. Centroiding reduces the data file size while maintaining spectral quality. IMPORTANT: The Data Type Profile is not available when the Detector Type is Astral.
Polarity	The polarity mode—Positive (default) or Negative—during a scan to detect ions of that polarity. IMPORTANT: The polarity mode must be the same for all experiments within the method.
Source Fragmentation	Select the checkbox to switch on ion source CID. An offset voltage in the ion source accelerates the ions into the background gas. Collisions with the background gas might help in the desolvation of the ions and might increase sensitivity.
Energy (V)	(Source Fragmentation: On)

Parameter	Description
	Specifies the collision energy voltage for ion source fragmentation. Range: 1–135; default: 35 If you set the source fragmentation energy too high, fragmentation of ions might occur.
Use EASY-IC™	(Orbitrap Lock Mass Correction: EASY-IC™; Mode: Scan-to-Scan or Timed) If On is selected, it provides an internal reference mass that is used for mass correction during a run.
Lock Mass Injection	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Lock mass injection offers the possibility to inject a defined lock mass (if provided in the solvents or similar) to the ions of the analytical scan into the C-trap and ensures the detection and locking the chosen reference mass. If the user requests locking of the present scan, this function is essential. In general, a lock mass correction can only be performed, if the defined m/z value is detected in the performed Orbitrap scan. This means, the mass range of the scan of interest must provide the m/z value of the lock mass of interest. • If the Lock Mass Injection checkbox is selected, the lock mass injection will be performed for all scans in the method. • If the Lock Mass Injection checkbox is clear, you need to make sure that the lock mass is covered by at least one experiment/scan in the method. Example Full Scan (m/z 100–500) with ddMS², lock mass: 279.1591—the lock mass is covered by the mass range of the Full Scan and a correction will be applied to all other scans. The lock mass injection reduces the acquisition speed due to the associated overhead of about 6–10 ms/scan.
Scan Description	(Optional) User-defined scan description (maximum of 16 characters) appearing as a scan header in the raw data file. The scan description must be unique in the method. In the text field, you can use alphanumeric characters, commas,

Parameter	Description
	hyphens, parentheses, periods, plus signs, spaces, and underscores.

5.4.7 Data-dependent scans

A data-dependent scan is a real-time, automated scan that use specified criteria to select one or more ions of interest for subsequent analysis. When you move the scan icon to the workflow area, the Method Editor application populates the parameter values in the properties pane.

5.4.7.1 Data-Dependent Properties pane

The Data-Dependent Properties pane appears when you select one of these boxes in the workflow diagram: Number of Scans or Scans Per Outcome. The Data-Dependent Properties pane specifies the following data-dependent parameters:

- Data-Dependent Mode
- Time between Master Scans (sec)
- Number of Dependent Scans

When you build an MS (full) scan method, the Cycle Time or Scans Per Outcome becomes unavailable, depending on whether there are multiple scan outcomes and which time or scan box you select.

The following table summarizes the data-dependent options as they apply toward single or multiple scan outcomes from an MS scan.

Select box	Data-Dependent Mode options
Cycle Time (sec) 	• Cycle Time: Acquires as many dependent scans as possible in the specified cycle time (Top Speed) before continuing on to the next experiment. This mode is optimal for achieving a regular frequency of master scans. Time between Master Scans (sec): Range: 0.1–100000; default: 1.2
Number of Scans 	• Number of Scans: Acquires up to the defined number of dependent scans ($\pm$TopN).

Select box**Data-Dependent Mode options**

- Scans per Outcome: Acquires up to the defined number of dependent scans (Top N) per branch in the experiment.

Range: 1–100, default: 20 (Orbitrap); 1–500, default: 100 (Astral)

5.4.7.2 Data-Dependent SIM Scan (ddSIM) Properties pane



A Data-Dependent SIM (ddSIM) scan is a selected ion monitoring (SIM) scan type scan that the system performs on ions that satisfy the data-dependent rules.

Parameters in the Data-Dependent SIM Scan (ddSIM) Properties pane

Parameter	Description
Scan Priority	Specify the scan priority for the selected data-dependent scan node. Priority 1 is highest priority. You can set the priority level for each branch in any order. The priority number appears on the dependent scan nodes unless they have the same priority value. Within one priority level, real-time scan scheduling can supersede explicit left-to-right scan ordering. • If scans have the same priority value (default), the MS executes the scans serially (from left to right) until one or more branches are exhausted. • If scans have different priority values, the MS executes all the highest priority scans before executing the next priority level, and so on. When there are many high-priority scans (Number of Scans mode) or the high-priority scans use the entire cycle time (Cycle Time mode), the lower-priority scans do not execute at all. Range: 1–99

Parameter	Description
	NOTE: This property is available only when there are multiple outcomes.
Isolation Window (m/z)	The isolation window (in m/z) for the precursor ions. The mass range for the peak is centered at the precursor mass and ranges from half of the isolation window to either side of the precursor mass. Range: 0.4–10 (Astral), 2000 (Orbitrap); default: 2 NOTE: A value that is too low may result in a loss of sensitivity due to lower transmission efficiency of a quadrupole at a narrower isolation window (physical property of quadrupole).
Detector Type	The available options are: • Astral: Select this detector type for highest sensitivity and speed. • Orbitrap: Select this detector type for highest resolution, mass accuracy and dynamic range. IMPORTANT: The detector type must be the same in all dependent scans.
Orbitrap Resolution	(Detector Type: Orbitrap) The mass resolution of the Orbitrap analyzer, which is proportional to $1/\sqrt{m/z}$ of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200. The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000, 180 000, 240 000, and 480 000.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. • Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. • Custom: You can set the AGC Target in percent.

Parameter	Description
	TIP: Please refer to the system templates for application-specific suggestions.
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (MS2, DIA: 3000); default: 100 This is the normalized AGC Target value, it is represented as a percentage to aid in calculating the desired AGC Target. The base normalized value is different for scan types. The values for Orbitrap detector are as follows: Scan Type = ddSIM Normalized Base (100%) = 1e5 The values for Astral detector are as follows: Scan Type = ddSIM Normalized Base (100%) = 1e3
Absolute AGC Value	(AGC Target: Custom) The absolute AGC target value is a conversion of the AGC percentage target set in the method into absolute values. This read-only value will be automatically updated according to the scan type and the target percentage.
Maximum Injection Time (ms)	Maximum time in milliseconds that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 10 (Astral), 50 (Orbitrap)
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. NOTE: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral acquisition.
Data Type	Indicates how to collect data during the currently selected scan event: • Profile (default): Represents mass spectral peaks as point-to-point plots, with each point having an associated intensity value.

Parameter	Description
	Centroid: Represents mass spectral peaks (as a bar graph) in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The normalized area of the peak provides the mass intensity data. Centroiding reduces the data file size while maintaining spectral quality.
	IMPORTANT: The Data Type Profile is not available when the Detector Type is Astral.
Lock Mass Injection	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Lock mass injection offers the possibility to inject a defined lock mass (if provided in the solvents or similar) to the ions of the analytical scan into the C-trap and ensures the detection and locking the chosen reference mass. If the user requests locking of the present scan, this function is essential. In general, a lock mass correction can only be performed, if the defined m/z value is detected in the performed Orbitrap scan. This means, the mass range of the scan of interest must provide the m/z value of the lock mass of interest. - If the Lock Mass Injection checkbox is selected, the lock mass injection will be performed for all scans in the method. - If the Lock Mass Injection checkbox is clear, you need to make sure that the lock mass is covered by at least one experiment/scan in the method. Example Full Scan (m/z 100–500) with ddMS², lock mass: 279.1591—the lock mass is covered by the mass range of the Full Scan and a correction will be applied to all other scans. The lock mass injection reduces the acquisition speed due to the associated overhead of about 6–10 ms/scan.
Scan Description	(Optional) User-defined scan description (maximum of 16 characters) appearing as a scan header in the raw data file. The scan description must be unique in the method. In the text field, you can use alphanumeric characters, commas,

Parameter	Description
	hyphens, parentheses, periods, plus signs, spaces, and underscores.

5.4.7.3 Data-Dependent MS² Properties pane



The Data-Dependent MS² scan is a real-time, automated MS/MS (ddMS²) scan that uses specified criteria to select one or more ions of interest for subsequent analysis.

A data-dependent scan can follow Full Scan or SIM master scans or another data-dependent scan.

TIP

Point to a parameter name to display its description.

Changed set values of given parameters appear in bold type. A red border around a parameter box indicates that the set value is out of range. Point to the parameter box to display the valid range. If you enter a value that is outside the recommended range, the box border color changes to yellow.

Parameters in the Data-Dependent MS² Scan Properties pane

Parameter	Description
Scan Priority	Specify the scan priority for the selected data-dependent scan node. Priority 1 is highest priority. You can set the priority level for each branch in any order. The priority number appears on the dependent scan nodes unless they have the same priority value. Within one priority level, real-time scan scheduling can supersede explicit left-to-right scan ordering. • If scans have the same priority value (default), the MS executes the scans serially (from left to right) until one or more branches are exhausted. • If scans have different priority values, the MS executes all the highest priority scans before executing the next priority level, and so on. When there are many high-priority scans (Number of Scans mode) or the high-priority scans use the entire cycle time (Cycle Time mode), the lower-priority scans do not execute at all. Range: 1–99

Parameter	Description
	NOTE: This property is available only when there are multiple outcomes.
Isolation Window (m/z)	The isolation window (in m/z) for the precursor ions. The mass range for the peak is centered at the precursor mass and ranges from half of the isolation window to either side of the precursor mass. Range: 0.4–2000; default: 2 NOTE: A value that is too low may result in a loss of sensitivity due to lower transmission efficiency of a quadrupole at a narrower isolation window (physical property of quadrupole).
Isolation Offset	Applies an offset to the isolation window. The mass range for the peak is centered at the precursor mass plus the offset and ranges from half of the isolation window to either side of the precursor mass plus the offset. Available options are: • Off (default) • m/z Offset • Da Offset
Isolation Offset (m/z)	(Isolation Offset: m/z Offset) Specifies the offset value (in m/z) that is applied to the isolation and activation windows relative to that determined through the data-dependent criteria. For example, if the MS selects a peak at m/z 500 and you specify an m/z 2 width and m/z 0.5 offset, the isolation range is m/z 499.5–501.5 instead of m/z 499–501 (without the offset). Range: –200 to 200; default: 0
Isolation Offset (Da)	(Isolation Offset: Da Offset) Specifies the offset value (in Da units) that is applied to the isolation and activation windows relative to that determined through the data-dependent criteria. For example, if the instrument selects a peak at Da 500 and you specify a Da 2 width with a Da 0.5 offset, the isolation range is Da 499.5–501.5 instead of Da 499–501 (without the offset). Range: –200 to 200; default: 0
Reported Mass	(Isolation Offset: m/z Offset or Da Offset) Specifies which mass is written to the raw file. The available options are: • Offset Mass (default) • Original Mass Original mass is the mass that was detected (that is, the mass that would be written to the raw file if isolation offset were set to Off). Offset mass is the center mass of the executed isolation.

Parameter	Description
Collision Energy Type	The available options are: • Normalized (default): Collision energy is set in percentage and applies an energy dependent on mass and charge state. • Absolute: Collision energy is set independent of the mass and charge of the precursor ion.
HCD Collision Energy (% or V)	The MS injects isolated precursor ions into the IRM and the ion processor (for Astral when using single Collision energy value) with an HCD collision energy (as a percentage or independent of mass and charge). The MS injects isolated precursor ions into the IRM in one to five steps, each with a different HCD collision energy (as a percentage or independent of mass and charge). The MS then mass analyzes all groups of ions in a single scan. A high HCD collision energy results in more ion activation, which generally leads to more fragmentation. A low HCD collision energy results in less ion activation, which generally leads to less fragmentation. With a single value, only one collision energy is applied. If more than one value is defined, the instrument will perform Stepped Collision Energy. Use commas to separate several energy values. The ion processor and the IRM are calibrated to generate the same MS² spectra. Range: 1–200; default: 30
Detector Type	The available options are: • Astral: Select this detector type for highest sensitivity and speed. • Orbitrap: Select this detector type for highest resolution, mass accuracy and dynamic range. IMPORTANT: The detector type must be the same in all dependent scans.
Orbitrap Resolution	(Detector Type: Orbitrap) The mass resolution of the Orbitrap analyzer, which is proportional to 1/sqrt of the mass-to-charge ratio. Mass resolution is defined as the observed m/z value divided by the smallest difference $\Delta m/z$ for two ions that can be separated: $m/z / \Delta m/z$. Spectra acquired at higher Orbitrap Resolution allow greater resolution in m/z but take longer to acquire. The selected resolution is specified for a peak at m/z 200. The available options are: 7500, 15 000, 30 000, 45 000, 60 000, 120 000, 180 000, 240 000, and 480 000.
TMT	(Detector Type: Astral) Set TMT to On when using the TMT workflow. The available options are: • On

Parameter	Description																					
	Off (default) TIP: For a better TMT performance, clear the Advanced Peak Determination checkbox when TMT is set to On.																					
Scan Range Mode	Select the scan range mode to determine the scan range. The available options are: - Auto: The last mass (lm) is automatically set to (Precursor + IsoWidth / 2) × Charge × 1.02 + 20.0. The first mass (fm) is set such as to satisfy the Factor 5–10–15 rule for the lm:fm ratio to ensure the best retainment of all masses in the IRM. It is subject to the corresponding <i>m/z</i> lower limit: <table><thead><tr><th>Last mass <i>m/z</i> [Th]</th><th>Factor of 5–10–15 Rule</th><th>First mass <i>m/z</i> [Th]</th></tr></thead><tbody><tr><td>≤ 200</td><td>5</td><td>40</td></tr><tr><td>200–250</td><td>5</td><td>lm / 5</td></tr><tr><td>250–500</td><td>10</td><td>50</td></tr><tr><td>500–1500</td><td>10</td><td>lm / 10</td></tr><tr><td>1500–2250</td><td>15</td><td>150</td></tr><tr><td>≥ 2250</td><td>15</td><td>lm / 15</td></tr></tbody></table> - Define First Mass: Explicitly define the first mass and the system automatically sets the last mass. - Define <i>m/z</i> Range: Explicitly define the first and last mass. Because the system optimizes the transfer of ions for the defined scan range, it is recommended to restrict the scan range to the region of interest. (Application Mode: Small Molecule) Auto-Extended: The instrument calculates the scan range starting from the precursor <i>m/z</i>. This scan range mode complies with the Factor 15 rule with the first mass <i>m/z</i> range 40–2500.	Last mass <i>m/z</i> [Th]	Factor of 5–10–15 Rule	First mass <i>m/z</i> [Th]	≤ 200	5	40	200–250	5	lm / 5	250–500	10	50	500–1500	10	lm / 10	1500–2250	15	150	≥ 2250	15	lm / 15
Last mass <i>m/z</i> [Th]	Factor of 5–10–15 Rule	First mass <i>m/z</i> [Th]																				
≤ 200	5	40																				
200–250	5	lm / 5																				
250–500	10	50																				
500–1500	10	lm / 10																				
1500–2250	15	150																				
≥ 2250	15	lm / 15																				
First Mass (<i>m/z</i>)	(Scan Range Mode: Define First Mass) The scan range's First Mass <i>m/z</i> value. Range: 40–2500; default: 120 NOTE: Please be aware that too low first masses may discriminate higher fragment masses.																					
Scan Range	(Scan Range Mode: Define <i>m/z</i> Range) The values for the Scan Range (<i>m/z</i>).																					

Parameter	Description
(m/z)	Range: 40–6000 (Astral); 40–6000 (8000 with BioPharma option) (Orbitrap); default: 100–1000 TIP: For optimal results, restrict the scan range to the region of interest. In particular for the Astral analyzer, last mass should be set lower than m/z 3000 unless the experiment requires a higher last mass setting. Setting last mass higher than m/z 3000 may slow down acquisition speed.
AGC Target	The Automatic Gain Control (AGC) controls the ions that are injected into the mass analyzer. The AGC target value defines the number of ions to be collected in the IRM. • Standard (default): The system sets the recommended target (good starting point) in an automated fashion per scan type. • Custom: You can set the AGC Target in percent. TIP: Please refer to the system templates for application-specific suggestions.
Normalized AGC Target (%)	(AGC Target: Custom) Specifies the Automatic Gain Control (AGC) target. This is a percentage representing the maximum number of charges to accumulate for a given analysis. Range: 0.001–1000 (MS2, DIA: 3000); default: 100 This is the normalized AGC Target value, it is represented as a percentage to aid in calculating the desired AGC Target. The base normalized value is different for scan types. The values for Orbitrap detector are as follows: Scan Type = ddMS2 Normalized Base (100%) = 1e5 The values for Astral detector are as follows: Scan Type = ddMS2 Normalized Base (100%) = 1e4
Absolute AGC Value	(AGC Target: Custom) The absolute AGC target value is a conversion of the AGC percentage target set in the method into absolute values. This read-only value will be automatically updated according to the scan type and the target percentage.
Maximum Injection Time (ms)	Maximum time in milliseconds that is allowed to accumulate ions in the IRM until the AGC Target is reached. Range: 1–500; default: 10 (Astral), 50 (Orbitrap)
Microscans	Number of scans to average in a given spectrum. Range: 1–10; default: 1 A microscan is one ion injection followed by ion detection. The MS sums microscans to produce one scan, which improves the signal-to-noise ratio of the mass spectral data. NOTE: The overall scan time increases linearly with the number of microscans, significantly slowing the rate of spectral acquisition.

Parameter	Description
Data Type	Indicates how to collect data during the currently selected scan event: • Profile (default): Represents mass spectral peaks as point-to-point plots, with each point having an associated intensity value. • Centroid: Represents mass spectral peaks (as a bar graph) in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The normalized area of the peak provides the mass intensity data. Centroiding reduces the data file size while maintaining spectral quality. IMPORTANT: The Data Type Profile is not available when the Detector Type is Astral.
Use EASY-IC™	(Orbitrap Lock Mass Correction: EASY-IC™; Mode: Scan-to-Scan or Timed) If On is selected, it provides an internal reference mass that is used for mass correction during a run.
Lock Mass Injection	(Orbitrap Lock Mass Correction: User-defined Lock Mass) Lock mass injection offers the possibility to inject a defined lock mass (if provided in the solvents or similar) to the ions of the analytical scan into the C-trap and ensures the detection and locking the chosen reference mass. If the user requests locking of the present scan, this function is essential. In general, a lock mass correction can only be performed, if the defined m/z value is detected in the performed Orbitrap scan. This means, the mass range of the scan of interest must provide the m/z value of the lock mass of interest. • If the Lock Mass Injection checkbox is selected, the lock mass injection will be performed for all scans in the method. • If the Lock Mass Injection checkbox is clear, you need to make sure that the lock mass is covered by at least one experiment/scan in the method. Example Full Scan (m/z 100–500) with ddMS², lock mass: 279.1591—the lock mass is covered by the mass range of the Full Scan and a correction will be applied to all other scans. The lock mass injection reduces the acquisition speed due to the associated overhead of about 6–10 ms/scan.
Scan Description	(Optional) User-defined scan description (maximum of 16 characters) appearing as a scan header in the raw data file. The scan description must be unique in the method. In the text field, you can use alphanumeric characters, commas, hyphens, parentheses, periods, plus signs, spaces, and underscores.

5.4.8 Filters

The filters and triggers determine which ions pass or do not pass to the next stage (node) of mass analysis. You can add an appropriate filter to the experiment above or below a scan node. The Method Editor displays its properties pane on the right side. Here, you can modify the filter parameters.

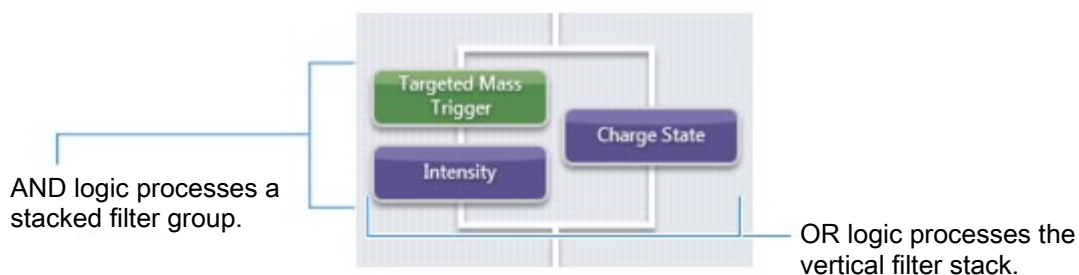
IMPORTANT

Filters and triggers are only available below scans using the Orbitrap analyzer.

Branching filter logic

Depending on its location in the experiment, a filter acts as either a prefilter (applied toward all outcomes [branches] that follow) or a postfilter (applied toward a specific outcome [branch]). If an experiment does not have multiple outcomes after a filter, the filter acts as a prefilter only. If you add dependent scans after a filter, the original filter does not move and acts as a prefilter. The figure shows how the Method Editor uses Boolean AND/OR logic to process filter groups located above a data-dependent scan.

Figure Example of branching filter logic



NOTE

The Method Editor deactivates all filters that are not appropriate for the workflow.

No further scan can follow an Astral scan. Thus, also no filters can follow an Astral scan.

Summary of filter usage

The table summarizes which filter types are available to append to the various scan types with the Orbitrap analyzer.

Filter types	Scan types					
	MS	tSIM	tMS2	DIA	ddSIM	ddMS2
Precursor Selection Range filter	✓	✓	–	–	✓	✓

Filter types	Scan types					
Monoisotopic Precursor Selection (MIPS) filter	✓	✓	–	–	✓	✓
Intensity filter	✓	✓	–	–	✓	✓
Precursor Fit filter	✓	✓	–	–	✓	✓
Charge State filter	✓	✓	–	–	✓	✓
Dynamic Exclusion filter	✓	✓	–	–	✓	✓
Targeted Inclusion - Targeted Mass filter	✓	✓	–	–	✓	✓
Targeted Inclusion - Targeted Mass Difference filter	✓	✓	–	–	✓	✓
Targeted Inclusion - Targeted Isotopic Ratio filter	✓	✓	–	–	✓	✓
Targeted Exclusion - Targeted Mass Exclusion filter	✓	✓	–	–	✓	✓
Apex Detection filter	✓	–	–	–	–	–
Isotope Exclusion filter	✓	✓	–	–	–	–
Triggers						
Targeted Mass Trigger	✓	✓	–	–	✓	✓

5.4.8.1 Precursor Selection Range filter

Precursor Selection Range

Use the Precursor Selection Range filter to select precursors from a narrower range than the scanned range. You can set a mass-to-charge ratio selection range (m/z). You can add a precursor selection range filter after any scan.

This filter is optional. If the filter is not used, the mass range defaults to the scan range of the master scan,

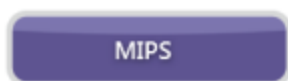
NOTE

You cannot set the precursor selection range outside of the scan range.

Parameters in the Precursor Selection Range Properties pane

Parameter	Description
Mass Range (m/z)	The precursor selection mass range in which to perform the data-dependent scans. The MS ignores all precursor peaks that are outside of this mass range. Range: Up to the scan range of the Master Scan or the Data Dependent Scan which this Filter is appended. Range: 100–1000

5.4.8.2 Monoisotopic Precursor Selection (MIPS) filter



Use the Monoisotopic Precursor Selection (MIPS) filter to restrict data-dependent scans to those where the monoisotopic precursor can be determined. The monoisotopic peak m/z is written to the raw file while the isolation is centered around the most abundant peak in the isotopic cluster.

MIPS will only function for precursors with resolved isotopes and defined charge states. When using MIPS in combination with Dynamic Exclusion, all isotopes are added to the exclusion list once a single isotope is selected for data-dependent analysis.

TIP

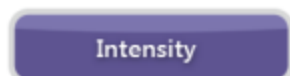
To exclude isotopes, add the Isotope Exclusion Filter to the workflow and select the Exclude Isotopes parameter in the properties pane.

Parameter in the MIPS Properties pane

Parameter	Description
Monoisotopic Peak Determination	• Small Molecule: Uses a heuristic algorithm, referred to as the “non-standard” monoisotopic assignment algorithm. • Peptide: Uses a combination of the non-standard and averagine algorithms. • Intact Protein: Uses the “averagine” monoisotopic assignment algorithm, which is based on matching

Parameter	Description
	isotope abundances to the average molecule, a fictitious peptide that has the average number of the various amino acid residues for a given molecular weight.
Relax restrictions when too few precursors are found	The instrument fragments ion clusters that have a charge state assignment but not a monoisotopic m/z assignment: Targets with monoisotopic m/z assignment are triggered with higher priority than targets with no monoisotopic m/z assignment. Targets are sorted by intensity.

5.4.8.3 Intensity filter



Use the Intensity filter to select only ions with signal intensities above the signal intensity threshold or in an intensity range.

Parameter	Description
Filter Type	- Intensity Threshold: The minimum intensity required to trigger a data-dependent scan. Intensity threshold is most often used to prevent data-dependent scans from triggering on low intensity species that will require long injection times and may not provide high quality spectra. Range: 0 to 1.0×10^{20}; default: 5.0×10^3 - Intensity Range: Defines a specific range of precursor intensities that will trigger data-dependent scans. Minimum value: 0–1.0×10^{20}; default: 5.0×10^3 Maximum value: 0–1.0×10^{20}; default: 1.0×10^{20}

5.4.8.4 Precursor Fit filter

Precursor Fit

The Precursor Fit filter provides a comparison of the theoretical precursor ion isotopic envelope to the observed envelope. The Precursor Fit (normalized between the theoretical and observed envelopes in the Fit Window) must be above the Fit Threshold to pass the filter and trigger a data-dependent scan. This is used to prevent dependent scans when there is a disagreement between the theoretical precursor envelope and what is observed inside the isolation window.

Select master scan (Full Scan or tSIM scan). Then append a data-dependent scan (ddMS2 or ddSIM).

Parameters in the Charge State Properties pane

Parameter	Description
Fit Threshold (%)	Determines where the dependent scan does not trigger until above a specified purity threshold. Range: 0–100; default: 70
Fit Window (m/z)	The fit window is recommended to be the same size as the isolation window of the dependent scan. If there are multiple dependent scans for single precursor filter, the recommendation should be that the fit window is the same size as the largest isolation window. Range: 0.4–2000; default: 2

NOTE

If the isolation window size in the dependent scan(s) does not match the fit window size, the fit calculation will not match that of what is isolated. Isolation Offset should not be used when using the Theoretical Precursor Fit Filter.

5.4.8.5 Charge State filter

Charge State

Use the Charge State filter to restrict data-dependent scans to those that fit the specified charge state requirements. You can specify a single charge state (for example, 2) or a charge state range (for

example, 2-6) to select precursors for data-dependent scans. Additionally, you can choose to trigger peaks whose charge state the mass spectrometer cannot determine.

Parameters in the Charge State Properties pane

Parameter	Description
Include charge state(s)	The charge states that pass the filter. Range: 1–100; default: 2–6
Include undetermined charge states	If selected, the application also triggers on peaks that do not have an assigned charge state. Causes for charge state to be undetermined include unresolved isotopic clusters (that is, intact proteins), overlapping isotopic clusters, and low intensities (that is, poor ion statistics). Set the default charge state in the Global MS parameters. This default charge state will be used to calculate the normalized collision energy for peaks with undetermined charge states. This default charge state will also be used to determine the scan range for dependent scans of precursors with undetermined charge state.

5.4.8.6 Dynamic Exclusion filter

Dynamic Exclusion

Use the Dynamic Exclusion filter to prevent the instrument from triggering a data-dependent scan after it has already been triggered. Dynamic exclusion is valuable if there are a lot of co-eluting analytes, matrix components or contaminants.

If you use the Dynamic Exclusion filter and if the instrument performs a data-dependent scan on an ion, that ion moves to the pre-exclusion list (a temporary reject mass list) for a user-specified period of time.

It is recommended to properly set the parameter Expected LC Peak Width(s) to optimally use the Dynamic Exclusion filter.

Parameters in the Dynamic Exclusion Properties pane

Parameter	Description
Share dynamic exclusion list with other selected dynamic exclusion filters	Select this checkbox to share these properties globally with other selected Dynamic Exclusion filters. Any modification to one selected filter will automatically apply these changes to the other selected filters. When the method is executed, a shared dynamic exclusion list will be used. When multiple experiments are used, a precursor excluded in one experiment will also be excluded in the next experiment, reducing redundant acquisitions. NOTE: This checkbox is available only when multiple Dynamic Exclusion filters are present.
Dynamic Exclusion Mode	The available options are: • Auto (default): Exclusion time is calculated dependent on the Expected LC Peak Width (s) set value. The therein set value is multiplied by a factor of 1.5 (Application Mode: Small Molecule) or 2.5 (Application Mode: Peptide or Intact Protein). Auto Mass Tolerance is 10 ppm low and 10 ppm high. • Custom: Select this option for a more detailed user interface setting.
Exclude after n times	(Dynamic Exclusion Mode: Custom) The number of times (counts) that an ion may be selected for a data-dependent scan before it goes on the dynamic exclusion list. Range: 1–100; default: 1
If occurs within (s)	(Dynamic Exclusion Mode: Custom) Duration in which <i>n</i> events must occur for the precursor to be added to the exclusion list. Range: 1–1000 NOTE: This parameter is shown only when Exclude after n times >1.
Exclusion Duration (s)	(Dynamic Exclusion Mode: Custom) The time in seconds that an ion stays on the dynamic exclusion list. Range: 1–6000; default: 60

Parameter	Description
Mass Tolerance	(Dynamic Exclusion Mode: Custom) Defines how close a measured mass must be to the mass in the mass list to be considered the same mass. - ppm (default): The low and high values of the mass tolerance width in parts per million (ppm) of the reference mass. You can specify the mass tolerance of the reference mass: first select the ppm option, and then enter the desired values in the Low box and High box. The mass tolerance is expressed in parts per million (ppm). When you select ppm, the size of the window around the reference mass is proportional to the m/z of the reference mass. m/z: The low and high values of the mass tolerance width in m/z. You can specify the mass tolerance width as a m/z value first by selecting the m/z option, and then enter the appropriate values in the Low box and High box. When you select m/z, the size of the window around the reference mass remains the same, regardless of the m/z of the reference mass.
Low	(Dynamic Exclusion Mode: Custom) The low side of the mass tolerance width window. The low value defines how far below the reference mass the window starts on the low-mass side of the reference peak. Range, ppm: 0.01–1 000 000.00; default: 10 Range, m/z: 0.01–100.00; default: 0.5
High	(Dynamic Exclusion Mode: Custom) The high side of the exclusion mass width window. The high value defines how far above the reference mass the mass tolerance window extends on the high-mass side of the reference peak. Range, ppm: 0.01–1 000 000.00; default: 10 Range, m/z: 0.01–100.00; default: 0.5
Exclude isotopes	(Dynamic Exclusion Mode: Custom) Excludes the ion cluster isotopes. (Default)

Parameter	Description
	When you select this check box, the Dynamic Exclusion filter groups the peaks together as a cluster and returns only the most intense ion, which is then targeted by subsequent dependent scans. Otherwise, the filter adds only the targeted m/z to the dynamic exclusion list. NOTE: If both a Dynamic Exclusion Filter and the Isotope Exclusion Filter are present between the same set of scans, this checkbox is selected and grayed out.
Perform dependent scan on single charge state per precursor only	(Dynamic Exclusion Mode: Custom) Performs a data-dependent scan on the most intense charge state of an ion (if you sort by intensity), or highest charge state of an ion (if you sort by charge state), and then put its mass, not m/z, on the exclusion list. For example, if an ion has charge states 2, 3, and 4, and 2 is the most intense, the MS performs a data-dependent scan on charge state 2, and then puts its mass, not its m/z, on the exclusion list. The MS does not perform data-dependent scans on charge states 3 and 4. NOTE: This checkbox is not available if the Charge State filter is used and the Include undetermined charge states checkbox is selected.
Exclude Within Cycle	(Dynamic Exclusion Mode: Custom) Select this checkbox to have each m/z value added to the exclusion list immediately after it is first analyzed. This will exclude it from other data-dependent scans that it would have qualified for in the same cycle.

5.4.8.7 Targeted Inclusion - Targeted Mass filter



Use the Targeted Mass filter to set up a inclusion list that selects only the masses of ions that are in the target mass list. You create a target mass list by entering masses and charges in the mass list table. You can also import the settings from (or export them to) a CSV, a TXT, or an XML file.

Parameters in the Targeted Mass Properties pane

Parameter	Description
Mass List Type	The available options are: • m/z & z (default) • m/z • M and z range
Time Mode	Enables scheduled or unscheduled scans. For scheduled scans, you can specify the retention time and the window size or the start time and the end time. The available options are: • Start/End Time: t start (min) and t stop (min) available in scan table • Retention Time Window (default): RT Time (min) and Window (min) available in scan table • Unscheduled
Include Intensity Threshold	When selected, an additional column Intensity Threshold appears in the mass list table. You can define individual intensity thresholds per Targeted Mass. If the measured intensity is less than this threshold value, the ion is not considered for data-dependent analysis.
Mass Tolerance	Defines how close a measured mass must be to the mass in the mass list to be considered the same mass. • ppm: The low and high values of the mass tolerance width in parts per million (ppm) of the reference mass. This option is appropriate for the Orbitrap mass analyzer because the peak width is proportional to the m/z. You can specify the mass tolerance of the reference mass: first select the ppm option, and then enter the desired values in the Low box and High box. The mass tolerance is expressed in parts per million (ppm). When you select ppm, the size of the window around the reference mass is proportional to the m/z of the reference mass. • m/z (default): The low and high values of the mass tolerance width in m/z. This option is appropriate

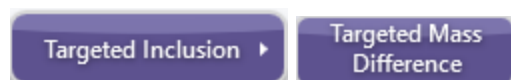
Parameter	Description
	for the ion trap mass analyzer because the peak width is constant. You can specify the mass tolerance width in m/z: first select the m/z option, and then enter the desired values in the Low box and High box. When you select m/z, the size of the window around the reference mass remains the same regardless of the m/z of the reference mass.
Low	The low side of the mass tolerance width window. The low value defines how far below the reference mass the window starts on the low-mass side of the reference peak. Range, ppm: 0.01–1 000 000; default: 10 Range, m/z: 0.01–100; default: 0.5
High	The high side of the mass tolerance width window. The high value defines how far above the reference mass the mass tolerance window extends on the high-mass side of the reference peak. Range, ppm: 0.01–1 000 000; default: 10 Range, m/z: 0.01–100; default: 0.5
Ignore charge state requirement in the table for ions with undetermined charge state	(Not available for “m/z” mass list type) When selected, ions that have an unassigned charge state in the Master Scan do not need to meet the charge state requirement in the mass list table.
Set Collision Energy per Compound	When selected, an additional column HCD Collision Energy appears in the mass list table. The Collision Energy Type comes from the data-dependent scan below.
Perform dependent scan on most intense ion if no targets are found	Select the checkbox to perform data-dependent scans on the most intense ions if none of the target ions were found. If there is enough cycle time, a data-dependent experiment is performed on other ions that are present in the master scan in order of intensity (Pick others, if idle).

Mass List table

Parameter	Description
	Use the mass list table to create a mass list for targeted inclusion. The maximum number of entries is 150 000 records per file.
Compound	(Optional) You can give each entry a name. This name will be recorded to the raw file.
Formula	(Mass List Type: <i>m/z</i> & <i>z</i> and <i>M</i> & <i>z</i> range) (Optional) You can enter either a chemical formula, peptide sequence, or amino acids. If more than one formula type for your input is possible, the Method Editor application chooses the type with the highest priority (chemical formula > peptides > amino acids). Enter a prefix directly with the formula to make sure that the right type is applied. Peptide sequences can be marked by the prefix "p:" (for example, p:MRFA with <i>m/z</i> = 524.26496), amino acids by the prefix "a:" (for example, a:MRFA with <i>m/z</i> = 506.25440). You can also include any isotopes that differ from the most abundant. For example, methanol with a C-13 isotope would be defined as [13]CH3OH. When you use this field, you must specify an adduct.
Adduct	(Mass List Type: <i>m/z</i> & <i>z</i>) - Positive ion mode: no adduct (default), +H, +Na, +K, and +NH₄ - Negative ion mode: no adduct (default), -H, +Cl, +OH, and +HCOO This field is disabled when no formula is used.
<i>m/z</i>	(Mass List Type: <i>m/z</i> or <i>m/z</i> & <i>z</i>) The mass spectrometer triggers on the <i>m/z</i> with the specified mass, regardless of its charge. Range: 1–100 000; default: 524.265 (positive polarity mode) and 523.2582 (negative polarity mode)
<i>z</i>	(Mass List Type: <i>m/z</i> & <i>z</i>) The molecular ion charge value. Range: 0–24; default: 1
<i>M</i>	(Mass List Type: <i>M</i> & <i>z</i> range) The molecular ion mass value. Range: 0–100 000; default: 523.2577

Parameter	Description
z range	(Mass List Type: M & z range) The molecular ion charge range. Range: 1–24;default: 2–5
Intensity Threshold	(Include Intensity Threshold: On) The signal intensity value that the specified <i>m/z</i> value must meet or exceed to be considered for triggering an MS2. Default: 0
t start (min)	(Time Mode: Start/End Time) The time (in minutes) to start acquiring spectra for the defined target. Range: 0–60; default: 0
t stop (min)	(Time Mode: Start/End Time) The time (in minutes) to stop acquiring spectra for the defined target. Range: 0–60; default: 60

5.4.8.8 Targeted Inclusion - Targeted Mass Difference filter



Use the Targeted Mass Difference filter to set up an inclusion filter. For subsequent scans, this filter selects only ions that have one or two corresponding peaks with mass differences in the target mass list. There are several options that determine which ions to use for subsequent scans.

You create a target mass list by entering mass differences, not the *m/z*, in the mass list table. You can also import the settings from (or export them to) a CSV, a TXT, or an XML file.

Parameters in the Targeted Mass Difference Properties pane

Parameter	Description
Number of Precursors in the Targeted Group	(<i>n</i> = 2) Proceeds to the next scan if the mass spectrometer finds two peaks in the spectrum that have a mass difference, Delta M1 (in Daltons). (<i>n</i> = 3) Proceeds to the next scan if the MS finds three peaks: the two peaks with the lowest mass

Parameter	Description
	separated by Delta M1, and the lowest and highest peaks separated by Delta M2. Values: $n = 2$ or 3; default: $n=2$
	NOTE: The system calculates the m/z of the ions.
Partner Intensity Range Relative to the Most Intense Precursor (%)	The intensity ratio (as a percentage) of the partner ion signal intensity to the most intense precursor ion signal intensity that must be exceeded to trigger subsequent analysis. Range: 0–100; default: 10–100
Mass Tolerance	Defines how close a measured mass must be to the mass in the mass list to be considered the same mass. - ppm (default): The low and high values of the mass tolerance width in parts per million (ppm) of the reference mass. This option is appropriate for the Orbitrap mass analyzer because the peak width is proportional to the m/z. You can specify the mass tolerance of the reference mass: first select the ppm option, and then enter the desired values in the Low box and High box. The mass tolerance is expressed in parts per million (ppm). When you select ppm, the size of the window around the reference mass is proportional to the m/z of the reference mass. m/z: The low and high values of the mass tolerance width in m/z. This option is appropriate for the ion trap mass analyzer because the peak width is constant. You can specify the mass tolerance width in m/z: first select the m/z option, and then enter the desired values in the Low box and High box. When you select m/z, the size of the window around the reference mass remains the same regardless of the m/z of the reference mass.
Low	The low side of the mass tolerance width window. The low value defines how far below the reference mass the window starts on the low-mass side of the reference peak.

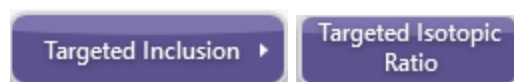
Parameter	Description
	Range, ppm: 0.01–1 000 000; default: 10 Range, m/z: 0.01–100; default: 0.5
High	The high side of the mass tolerance width window. The high value defines how far above the reference mass the mass tolerance window extends on the high-mass side of the reference peak. Range, ppm: 0.01–1 000 000; default: 10 Range, m/z: 0.01–100; default: 0.5
Perform Subsequent Scan On	For $n = 2$, the options are as follows: • Most Intense Ion in the Pair (default) • Lowest m/z Ion in the Pair • Highest m/z Ion in the Pair • Both Ions in the Pair For $n = 3$, the options are as follows: • Most Intense Ion in the Triplet (default) • Lowest m/z Ion in the Triplet • All Ions in the Triplet
Charge State Requirement	• Ions must be the same charge state (default): Ions have a mass delta peak or peaks with the same charge state as the target peak. • Ions may be of different charge states: No charge state requirement.

Mass List table

Parameter	Description
	Use the mass list table to create a mass list for targeted mass difference inclusion. The maximum number of entries is 30,000.
Compound	(Optional) You can give each entry a name. This name will be recorded to the raw file.
Delta M1	The mass difference (in Daltons) between the peaks in the doublet (for $n = 2$) or between the lowest mass peak and second lowest mass peak in the triplet (for $n = 3$).

Parameter	Description
	Range: 0–2000; default: 8.01
Delta M2	(Number of Precursors in the Targeted Group: 3) The mass difference (in Daltons) between the lowest and highest mass peaks in the triplet, with Delta M2 ≥ Delta M1. Range: 0–2000; default: 16.02

5.4.8.9 Targeted Inclusion - Targeted Isotopic Ratio filter



Use the Targeted Isotopic Ratio filter to set up an inclusion filter that triggers the next data-dependent scan only if the system detects an ion with the isotopic mass difference and ratio from the list. You can also import the settings from (or export them to) a CSV, a TXT, or an XML file.

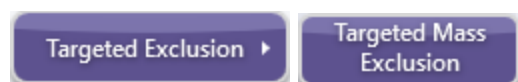
Parameters in the Targeted Isotopic Ratio Properties pane

Parameter	Description
Mass List Table	
Compound	(Optional) You can give each entry a name. This name will be recorded to the raw file.
Delta M	The mass difference (in Daltons) between the monoisotopic peak and the next isotopic peak. Range: –2000 to +2000; default: 1.1
Expected Ratio	The isotopic intensity ratio (dimensionless). Range: 0–1; default: 0.8
Ratio and Mass Tolerances	
Ratio Tolerance (%)	Defines how close (as a percentage) an isotopic ratio must be to the expected ratio in the mass list to be considered the same ratio. Range: 0.001–100; default: 5
	NOTE: The ratio tolerance is a percentage. For example, a ratio of 0.8 and a tolerance of 5 imply that the ratio must be in

Parameter	Description
	$0.8 \pm 5/100 = 0.75$ to 0.85 to be considered the same ratio as 0.8.
Mass Tolerance	Defines how close a measured mass must be to the mass in the mass list to be considered the same mass. - ppm (default): The low and high values of the mass tolerance width in parts per million (ppm) of the reference mass. This option is appropriate for the Orbitrap mass analyzer because the peak width is proportional to the m/z. You can specify the mass tolerance of the reference mass: first select the ppm option, and then enter the desired values in the Low box and High box. The mass tolerance is expressed in parts per million (ppm). When you select ppm, the size of the window around the reference mass is proportional to the m/z of the reference mass. m/z: The low and high values of the mass tolerance width in m/z. This option is appropriate for the ion trap mass analyzer because the peak width is constant. You can specify the mass tolerance width in m/z: first select the m/z option, and then enter the desired values in the Low box and High box. When you select m/z, the size of the window around the reference mass remains the same regardless of the m/z of the reference mass.
Low	The low side of the mass tolerance width window. The low value defines how far below the reference mass the window starts on the low-mass side of the reference peak. Range, ppm: 0.01–1 000 000; default: 10 Range, m/z: 0.01–100; default: 0.5
High	The high side of the mass tolerance width window. The high value defines how far above the reference mass the mass tolerance window extends on the high-mass side of the reference peak.

Parameter	Description
	Range, ppm: 0.01–1 000 000; default: 10 Range, m/z : 0.01–100; default: 0.5

5.4.8.10 Targeted Exclusion - Targeted Mass Exclusion filter



Use the Targeted Mass Exclusion filter to exclude ions from being triggered in a data-dependent experiment. You create a mass list by entering masses and charges in the mass list table. You can also import the settings from (or export them to) a CSV, a TXT, or an XML file.

Parameters in the Targeted Mass Exclusion Properties pane

Parameter	Description
Mass List Type	The available options are: m/z & z (default) m/z M and z range
Time Mode	Enables scheduled or unscheduled scans. For scheduled scans, you can specify the retention time and the window size or the start time and the end time. The available options are: - Start/End Time: t start (min) and t stop (min) available in scan table - Retention Time Window (default): RT Time (min) and Window (min) available in scan table - Unscheduled
Include Intensity Threshold	When selected, an additional column Intensity Threshold appears in the mass list table. You can define individual intensity thresholds per Targeted Mass. If the measured intensity is less than this threshold value, the ion is not considered for data-dependent analysis.
Exclusion mass width	(Custom dynamic exclusion mode)

Parameter	Description
	Defines how close a measured mass must be to the mass in the mass list to be considered the same mass. - ppm (default): The low and high values of the mass tolerance width in parts per million (ppm) of the reference mass. You can specify the mass tolerance of the reference mass: first select the ppm option, and then enter the desired values in the Low box and High box. The mass tolerance is expressed in parts per million (ppm). When you select ppm, the size of the window around the reference mass is proportional to the m/z of the reference mass. m/z: The low and high values of the mass tolerance width in m/z. You can specify the mass tolerance width as an m/z value first by selecting the m/z option, and then enter the appropriate values in the Low box and High box. When you select m/z, the size of the window around the reference mass remains the same, regardless of the m/z of the reference mass.
Low	The low side of the mass tolerance width window. The low value defines how far below the reference mass the window starts on the low-mass side of the reference peak. Range, ppm: 0.01–1 000 000.00; default: 10 Range, m/z: 0.01–100.00; default: 0.5
High	The high side of the mass tolerance width window. The high value defines how far above the reference mass the mass tolerance window extends on the high-mass side of the reference peak. Range, ppm: 0.01–1 000 000.00; default: 10 Range, m/z: 0.01–100.00; default: 0.5
Exclude isotopic cluster	The method execution considers isotope peaks of an identified isotope cluster depending on the status of this checkbox: When not selected: the method execution considers each m/z value individually for targeted

Parameter	Description
	exclusion. When selected: the method execution excludes all isotope cluster peaks for targeted exclusion when a single isotope peak m/z of this cluster was identified.
Ignore Charge State Requirement in the Table for Ions with Undetermined Charge State	(Not available for “m/z” mass list type) When selected, ions that have an unassigned charge state in the master scan, do not need to meet the charge state requirement in the mass list table.

Mass List table

Parameter	Description
Compound	(Optional) You can give each entry a name. This name will be recorded to the raw file.
Formula	(Mass List Type: m/z & z and M & z range) (Optional) You can enter either a chemical formula, peptide sequence, or amino acids. If more than one formula type for your input is possible, the Method Editor application chooses the type with the highest priority (chemical formula > peptides > amino acids). Enter a prefix directly with the formula to make sure that the right type is applied. Peptide sequences can be marked by the prefix “p:” (for example, p:MRFA with m/z = 524.26496), amino acids by the prefix “a:” (for example, a:MRFA with m/z = 506.25440). You can also include any isotopes that differ from the most abundant. For example, methanol with a C-13 isotope would be defined as $[^{13}\text{C}]\text{CH}_3\text{OH}$. When you use this field, you must specify an adduct.
Adduct	(Mass List Type: m/z & z) - Positive polarity mode: no adduct (default), +H, +Na, +K, or +NH₄ - Negative polarity mode: no adduct (default), –H, +Cl, +OH, or +HCOO This field is disabled when no formula is used.

Parameter	Description
<i>m/z</i>	(Mass List Type: <i>m/z</i> or <i>m/z</i> & <i>z</i>) The precursor mass (in <i>m/z</i>). If the Formula field is filled, the MS automatically calculates the <i>m/z</i> . Range: 4–2500; default: 524.265
<i>z</i>	(Mass List Type: <i>m/z</i> & <i>z</i>) The expected charge state of the precursor ion, which is used to determine the collision energy or reaction times. Range: 0–100; default: 1 TIP: Use zero (or 0) to let the instrument ignore the charge state.
M	(Mass List Type: M & <i>z</i> range) The molecular mass (in Daltons). Range: 0–100000; default: 523.2577
<i>z</i> range	(Mass List Type: M & <i>z</i> range) The molecular ion charge range. Range: 1–24; default: 2–5
t start (min)	(Time Mode: Start/End Time) The time (in minutes) to start acquiring spectra for the defined target. Range: 0–60; default: 0
t stop (min)	(Time Mode: Start/End Time) The time (in minutes) to stop acquiring spectra for the defined target. Range: 0–60; default: 60
RT Time (min)	(Time Mode: Retention Time Window) The retention time (RT) (in minutes). Range: 0–999; default: 30
Window (min)	(Time Mode: Retention Time Window) The window width (in minutes), centered around the defined Retention Time. For example, when Retention Time is set to 10 and Window is set to 2, acquisition of spectra for the target will only occur between the 9 and 11 minutes. Range: 0–999; default: 60

5.4.8.11 Apex Detection filter

Apex Detection

The Apex Detection filter sets up a chromatographic peak apex detection to trigger data-dependent scans. You obtain the highest quality data by deferring data acquisition by triggering near the apex of a chromatographic peak.

The use of apex detection provides numerous benefits for data-dependent LC/MS/MS, including the following:

- Improved MS/MS quality and signal levels
- Shorter ion injection times
- Reduced MS/MS acquisition of chemical background
- Less dependence on static or dynamic exclusion lists
- Reduced re-triggering of scans on tailing chromatographic peaks

Parameters in the Apex Detection Properties pane

Parameter	Description
Desired Apex Window (%)	The apex triggering window (as a percentage of the peak width). The apex is expected at half the value of the expected LC peak width as specified under Global Parameters > Settings. The apex window is symmetrically placed around the expected apex. The MS fits Gaussian functions to the extracted ion chromatograms and, from the functions and the expected peak width, determines whether the chromatogram is in the triggering window. If it is, the MS performs a data-dependent scan on that ion. The threshold defines the start of the peak. Range: 0–100; default: 30

5.4.8.12 Isotope Exclusion filter

Isotope Exclusion

The Isotope Exclusion filter excludes isotopes of a selected precursor from also being selected for data-dependent analysis in a given cycle. This is used to reduce redundant MS/MS acquisitions.

Parameters in the Isotope Exclusion Properties pane

Parameter	Description
Isotope Exclusion	The available options are: - Assigned: Excludes only redundant isotopes that were grouped together into the same isotopic envelope on-the-fly. - Assigned and unassigned: Excludes also a defined m/z window around precursors that the system was not able to assign isotopes for.
Exclusion window (m/z) for unassigned isotopes	(Isotope Exclusion: Assigned and unassigned) Defines the exclusion window (m/z) for the precursors with unassigned isotopes. This will exclude the entire window when the isotopes are not assigned, and may result in exclusion of other precursors if they are also present in the window. Use this option when isolation windows are narrow and the resultant isotopic distribution is altered or when non-standard isotopic distribution are expected. Range: 0–20; default: 3

5.4.9 Targeted Mass Trigger

NOTE

The Targeted Mass Trigger is only functional when it is appended to ddSIM and ddMS2 Orbitrap scans.



The Targeted Mass Trigger appends both to master scans to trigger precursor ions and to data dependent scans to trigger product ions.

- If the scan is a master scan, the next scan is triggered only if the system detects a precursor ion in the list and if all other conditions that are selected are met.

- If the scan is a data dependent scan, the next scan is triggered only if the system detects a product ion in the list and if all other conditions that are selected are met. This next scan is performed on the same precursor ion found in the master scan, but usually choose different data dependent properties (example: a higher resolution setting, an isolation offset, or a different collision energy).

You can also import the settings of the mass list table from (or export them to) a CSV, a TXT, or an XML file.

Parameters in the Targeted Mass Trigger Properties pane

Parameter	Description
Mass List Type	• M (shown when the trigger appends to a master scan) • m/z • m/z & z • M & z range
Time Mode	Enables scheduled or unscheduled scans. For scheduled scans, you can specify the retention time and the window size or the start time and the end time. The available options are: • Start/End Time: t start (min) and t stop (min) available in scan table • Retention Time Window (default): RT Time (min) and Window (min) available in scan table • Unscheduled
Conditions	
In addition to finding a product ion in the list, the mass analyzer must satisfy the following conditions, if selected, for the next scan to occur.	
Use Group IDs	When selected, you can group entries in the mass list table by a unique identifier Group ID. Groups are subsets of the whole list. When determining the trigger conditions, groups are considered independently of each other.
Group ID must equal Isolation (m/z)	(Use Group IDs = Yes) When selected, the Group ID must match the isolation m/z of the parent scan to trigger a dependent scan.

Parameter	Description
	If a group matches all the other conditions of this trigger, it must also match the isolation m/z of the parent scan within 25 ppm to trigger. This is used to correlate groups to their parent by the parent's mass.
Trigger Only with Detection of at Least N Ions from the Group	(Use Group IDs = Yes) Performs a data-dependent scan of the precursor ion only if it detects at least n product ions in the mass list with the same Group ID, where n is equal to or less than the number of rows in the mass list table. Range: $n = 1-1000000$; default: 3
Trigger Only with Detection of at Least N Ions from the List	(Use Group IDs = No) Performs a data-dependent scan of the precursor ion only if it detects at least n product ions in the mass list, where n is equal to or less than the number of rows in the mass list table.
n :	Range: $n = 1-1000000$; default: 3
Ion(s) must be within top n most intense	Performs a data-dependent scan of the precursor ion only if it detects a product ion in the mass list, and if the product ion peak is in the TopN of all peak intensities.
n :	Range: $n = 1-1\ 000\ 000$; default: 10
Ion(s) must be above the threshold (relative intensity, %)	Performs a data-dependent scan of the precursor ion only if it detects a product ion, and if the product ion intensity is greater than the threshold intensity relative to the most intense ion. Range: 1–100; default: 30
Ignore Charge State Requirements in the table for Ions with Undetermined Charge State	(Not available for “m/z” mass list type) Enables triggering on unassigned charge states. When the MS detects the correct m/z, it assumes that the charge state is as defined in the table. The filter assumes that unassigned charge states are the same as the targeted ion charge state for this and subsequent scans.

Mass List table

Parameter	Description
Compound	(Optional) You can give each entry a name. This name will be recorded to the raw file.
Formula	(Optional) (Mass List Type: M, m/z & z, or M & z range) You can enter either a chemical formula, peptide sequence, or amino acids. If more than one formula type for your input is possible, the Method Editor application chooses the type with the highest priority (chemical formula > peptides > amino acids). Enter a prefix directly with the formula to make sure that the right type is applied. Peptide sequences can be marked by the prefix "p:" (for example, p:MRFA with m/z = 524.26496), amino acids by the prefix "a:" (for example, a:MRFA with m/z = 506.25440). You can also include any isotopes that differ from the most abundant. For example, methanol with a C-13 isotope would be defined as [13]CH3OH. When you use this field, you must specify an adduct.
Adduct	(Mass List Type: m/z & z) - Positive ion mode: no adduct (default), +H, +Na, +K, and +NH4 - Negative ion mode: no adduct (default), -H, +Cl, +OH, and +HCOO This field is disabled when no formula is used.
m/z	(Mass List Type: m/z or m/z & z) The m/z value. Range: 40–2500; default: 523.2577 (positive polarity mode, m/z and m/z & z options), 523.2582 (negative polarity mode, m/z and m/z & z options)
z loss z	(Mass List Type: M or m/z & z) The molecular ion charge state. Range: 0–100; default: 1
M	(Mass List Type: M or M & z range) The molecular mass (in Daltons). Range: 0–100 000; default: 523.2577
z range	(Mass List Type: M & z range) The molecular ion charge range.

Parameter	Description
	Range: 1–24; default: 2–5
Group ID	(Use Group IDs: Yes) Range: 1–2000
Mass Tolerance	Defines how close a measured mass must be to the mass in the mass list to be considered the same mass. - ppm (default): The low and high values of the mass tolerance width in parts per million (ppm) of the reference mass. This option is appropriate for the Orbitrap mass analyzer because the peak width is proportional to the m/z. You can specify the mass tolerance of the reference mass: first select the ppm option, and then enter the desired values in the Low box and High box. The mass tolerance is expressed in parts per million (ppm). When you select ppm, the size of the window around the reference mass is proportional to the m/z of the reference mass. m/z: The low and high values of the mass tolerance width in m/z. This option is appropriate for the ion trap mass analyzer because the peak width is constant. You can specify the mass tolerance width in m/z: first select the m/z option, and then enter the desired values in the Low box and High box. When you select m/z, the size of the window around the reference mass remains the same regardless of the m/z of the reference mass.
Low	The low side of the mass tolerance width window. The low value defines how far below the reference mass the window starts on the low-mass side of the reference peak. Range, ppm: 0.01–1 000 000; default: 25 Range, m/z: 0.01–100; default: 0.5
High	The high side of the mass tolerance width window. The high value defines how far above the reference mass the mass tolerance window extends on the high-mass side of the

Parameter	Description
	reference peak.
	Range, ppm: 0.01–1 000 000; default: 25
	Range, m/z : 0.01–100; default: 0.5

5.4.10 Alternate precursor sorts

Here, you can specify the order with which precursors are selected for MS2 experiments.

Sort by intensity


 A green rectangular button with rounded corners and a subtle gradient, containing the text "Sort by Intensity" in white.

The instrument sorts the precursor ions according to their intensity priority. The following options are available:

- Most Intense: the instrument sorts the precursor ions from highest to lowest intensity.
- Least Intense: the instrument sorts the precursor ions from lowest to highest intensity.

Sort by charge


 A green rectangular button with rounded corners and a subtle gradient, containing the text "Sort by Charge" in white.

The instrument sorts the precursor ions according to their charge state priority. The following options are available:

- Lowest Charge State: the instrument sorts the precursor ions from lowest to highest charge state (priority is given to the ions with the lowest charge state).
- Highest Charge State: the instrument sorts the precursor ions from highest to lowest charge state (priority is given to the ions with the highest charge state). (Default)

The Sort by Charge filter is combinable with either the Sort by Intensity filter or the Sort by m/z filter. To do so, add the Sort by Charge filter to the workflow area first.

Sort by m/z


 A green rectangular button with rounded corners and a subtle gradient, containing the text "Sort by m/z " in white.

The instrument sorts the precursor ions according to their mass-to-charge priority. The following options are available:

- Highest m/z : the instrument sorts the precursor ions from highest to lowest m/z .
- Lowest m/z : the instrument sorts the precursor ions from lowest to highest m/z . (Default)

5.5 Use the experiment templates

Use the left-side Templates View toolbar () to select predefined, application-specific system templates. You can then modify the experiment parameters in the Properties pane. You can also create a custom workflow template and save it as an EXP file.

5.5.1 Save custom templates

You can modify system templates and save them under new names.

Procedure

1. In the Templates View toolbar, select **Save as Template** and browse to the destination folder.
2. Enter a name for the template and select **Save**.
3. In the Save Template dialog box, enter a template name (.exp extension), and select **Save**. A popup message “Save Method Template” appears.

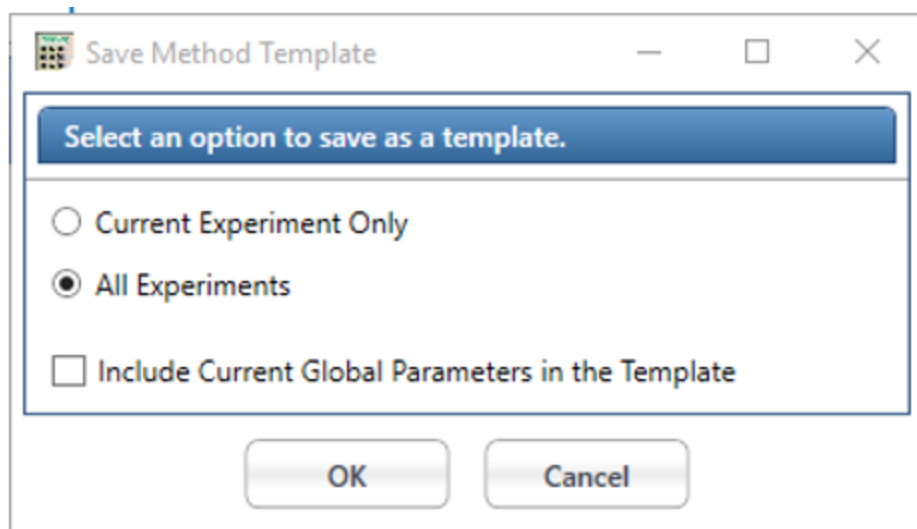


Figure Save Method Template dialog box

4. Select an option to save a template:

- **Current Experiment Only:** If there are two or more experiments in the method timeline, only the marked experiment is saved as a Method Template.
- **All Experiments:** If there are two or more experiments in the method timeline, all experiments in the method timeline are saved as a Method Template.

5. (Optional) Select the **Include Current Global Parameters in the Template** checkbox. The default state is not selected (Only the Scan Parameters of the method are saved). When selected (active), Scan Parameters and Global Parameters (such as ion source settings) are saved in the template.

Import a custom template

Procedure

1. Point to **My Experiments** to display the available templates.
2. Double-click a system template button or drag it to the **Place Scan Here** node.

5.5.2 System templates

The System Template folders typically contain multiple system templates, which are divided in template categories.

Use a System Template

Procedure

1. Select the left-side Templates View toolbar () to display the available template categories.
2. Select the triangle at the right part of the template category button to display the available system templates.
3. Double-click a system template button or drag it to the Place Scan Here node.
4. If the Global Settings Replacement Options dialog box appears, select an option:
 - Use source settings from the template
 - Keep current source settings

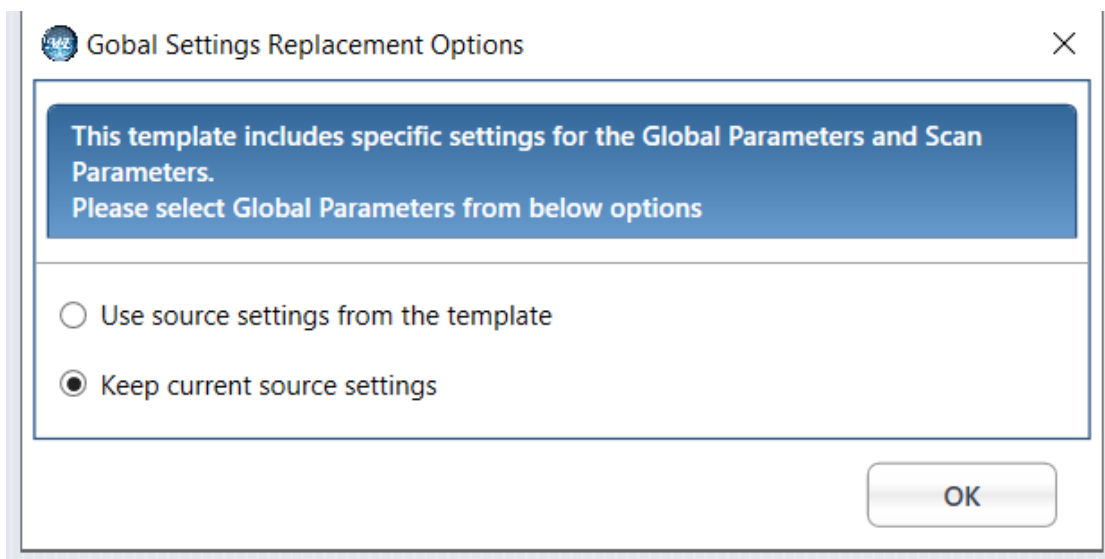


Figure Global Settings Replacement Options dialog box

5. Select **OK** to close the dialog box.

Result

The workflow area displays the selected experiment.

Available system templates

When you select an application mode on the Global Parameters or Scan Parameters page, the Method Editor changes the list of displayed system templates.

5.6 Summary page

The Summary page displays the parameters for mass spectrometer setup, syringe pump, divert valves, and contact closure that you specified on the Global Parameters page and the Scan Parameters page.

Button	Description
	Print a summary report.
	Find a word on the Document View.
	Enter the text in the field. Select the down arrow to select a find option. Select the left/right arrows to find the text on the Document View.

Button	Description
	Select the display mode of the Document View: • Page Mode • Two Page Mode • Scroll Mode
	Use the slider or select the minus/plus signs to zoom in or out the Document View. Or, position the mouse pointer in the pane, press and hold the <Ctrl> key, and roll the mouse wheel forward to zoom in or backward to zoom out.

Display the Summary page

Procedure

1. In the Method Editor application window, select the Summary tab.
2. Select the tabs to display the data in a Document View or a Tree View.

Customer feedback

You are encouraged to report errors or omissions in the text or index. Send an email to the Technical Documentation at documentation.bremen@thermofisher.com. The PDF versions of our manuals allow adding comments with Adobe Acrobat Reader or other freely available PDF reader programs.

You can also send feedback about our product documentation through the Thermo Scientific Help Portal.

Procedure

1. Go to docs.thermofisher.com.
2. Enter the product name in the search bar and press ENTER (or select the search icon).
3. From the search results list, select a product manual or topic.
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Glossary

APCI source

Contains the APCI probe assembly, APCI manifold, and API stack.

API source

The sample interface between the liquid chromatograph (LC) and the mass spectrometer (MS).

Asymmetric Track Lossless analyzer (Astral)

A high-resolution accurate mass (HRAM) analyzer that shares some of the operational principles of other types of mass analyzers, such as ion trap, time-of-flight, and Orbitrap analyzers.

atmospheric pressure chemical ionization (APCI)

A soft ionization technique done in an ion source operating at atmospheric pressure. Electrons from a corona discharge initiate the process by ionizing the mobile phase vapor molecules. A reagent gas forms, which efficiently produces positive and negative ions of the analyte through a complex series of chemical reactions.

atmospheric pressure ionization (API)

Ionization performed at atmospheric pressure by using atmospheric pressure chemical ionization (APCI), electrospray ionization (ESI), or nanoelectrospray ionization (nanoESI or NSI).

atmospheric pressure photoionization (APPI)

A soft ionization technique in which an ion is generated from a molecule when it interacts with a photon from a light source.

autosampler

A device that automatically loads collected samples (as for spectroscopic or chromatographic analysis) into a laboratory instrument.

centroid data

Data used to represent mass spectral peaks in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The data is displayed as a bar graph. The normalized area of the peak provides the mass intensity data.

chromatogram

A single sample injection will result in a plot of detector response versus time. This plot is referred to as a chromatogram.

collision-induced dissociation (CID)

A method of fragmentation where molecular ions are accelerated to high-kinetic energy and then allowed to collide with neutral gas molecules such as helium or nitrogen. The collisions break the bonds and fragment the ions into smaller pieces.

contact closure connection

The cable connection is from the external peripheral device to the mass spectrometer contact closure pins (Start In and Ground). The external device sends the contact closure (start) signal to the mass spectrometer.

Data Dependent experiment

A real-time, automated experiment that uses specified criteria to select one or more ions of interest for subsequent analysis, such as MS/MS or ZoomScan.

data-independent acquisition (DIA)

A data-independent acquisition mode in which all precursor ions are fragmented.

divert/inject valve

A modular valve that can be plumbed as a divert valve to control the flow direction or as a loop injector for flow injection analysis.

electrospray ionization (ESI)

A soft ionization technique operating at atmospheric pressure. Ions are generated in solution and a high voltage is applied to generate small droplets that are then evaporated until all ions are in the gas phase.

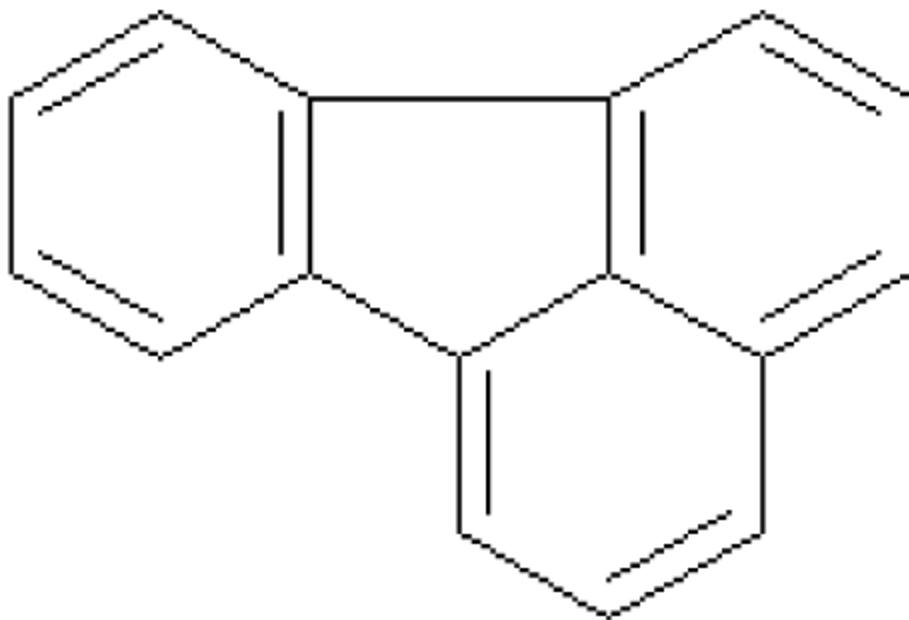
Extensible Markup Language (XML)

A general-purpose markup language that is used to facilitate the sharing of data across different information systems, particularly via the Internet.

FlexMix

Calibration solution that is a mixture of 16 highly pure, ionizable components (mass ranges: 50 to 3,000 m/z), which is designed for both positive and negative ionization calibration.

fluoranthene



A reagent anion that is used in an electron transfer dissociation (ETD) experiment. This compound is not only used as reagent ion but also as internal calibration compound.

fragment ion

A charged dissociation product of an ionic fragmentation. Such an ion can dissociate further to form other charged molecular or atomic species of successively lower formula weights.

full-scan type

Provides a full mass spectrum of each analyte or parent ion. With the full-scan type, the mass analyzer is scanned from the first mass to the last mass without interruption. Also known as single-stage full-scan type.

handshake

A signal that acknowledges that communication can take place.

heated-electrospray ionization (H-ESI)

A type of atmospheric pressure ionization that converts ions in solution into ions in the gas phase by using electrospray (ESI) in combination with heated auxiliary gas.

higher energy collision-induced dissociation (HCD)

Collision-induced dissociation that occurs in a high-pressure cell in the MS detector. In HCD, the ion optics accelerate precursor ions into a highpressure cell, where they collide with nitrogen gas. The collisions cause the ions to fragment into product ions. The ion optics then send the product ions to the mass analyzer for mass analysis. HCD produces triple quadrupole-like product ion mass spectra.

high-Field Asymmetric waveform Ion Mobility Spectroscopy (FAIMS)

An optional device for separating ions at atmospheric pressure. FAIMS provides ion separation by taking advantage of compound-dependent changes in ion mobility at high electric field strengths.

internal lock mass

A lock that is analyzed during the same MS experiment as your sample and is contained in the sample solution or infused into the LC flow during the experiment. Internal lock masses provide the most accurate corrections to the data.

isolation width

The baseline width of a window for a mass peak (or peak cluster) of interest for an MS/MS or MSⁿ scan.

log file

A text file, with a .log file extension, that is used to store lists of information.

mass-to-charge ratio (m/z)

Used to denote the quantity formed by dividing the mass of an ion (in u or Da) by the number of charges carried by the ion. For example, for the ion $C_7H_7^{2+}$, $m/z=45.5$.

matrix-assisted laser desorption/ionization (MALDI)

A method of ionizing proteins where a direct laser beam is used to facilitate vaporization and ionization while a matrix protects the biomolecule from being destroyed by the laser.

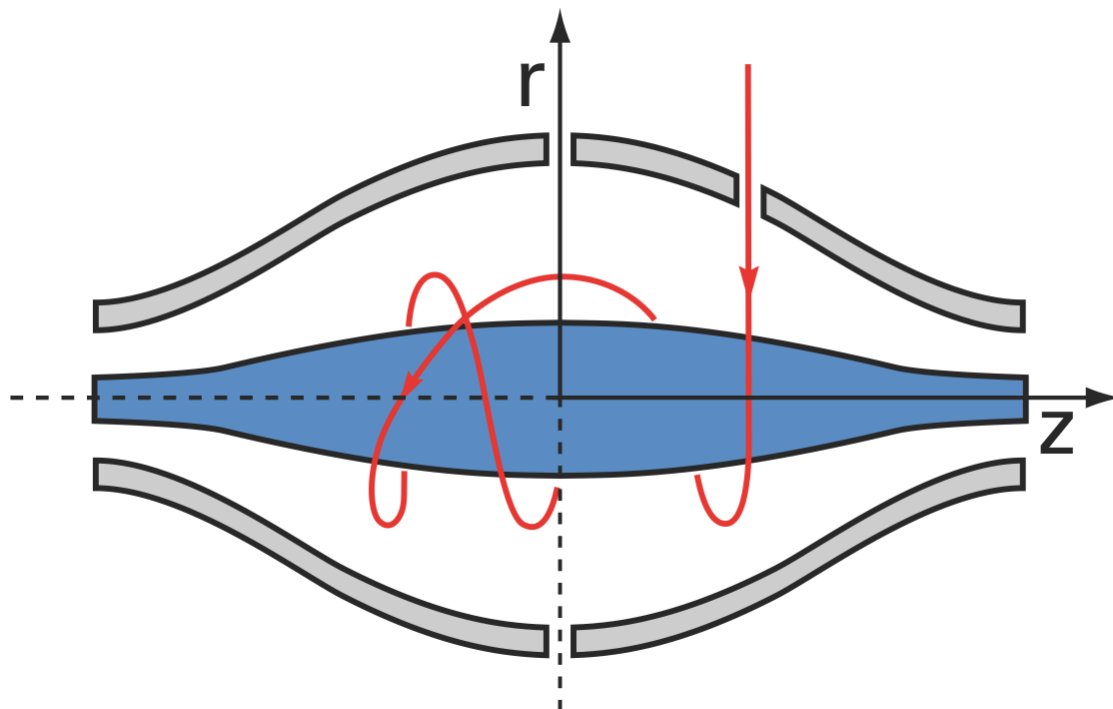
microscan

One mass analysis (ion injection and storage or scan-out of ions) followed by ion detection. Microscans are summed, to produce one scan, to improve the signal-to-noise ratio of the mass spectral data. The number of microscans per scan is an important factor in determining the overall scan time.

nanoelectrospray ionization (nano ESI or NSI)

A type of electrospray (ESI) that accommodates very low flow rates of sample and solvent on the order of 1 to 20 nL/min (for static nanospray) or 100 to 1,000 nL/min (for dynamic nanoelectrospray, also called nanoESI nanoLC gradient separation).

Orbitrap mass analyzer (OT)



The Orbitrap™ mass analyzer consists of a spindle-shape central electrode surrounded by a pair of bell-shaped outer electrodes. Ions inside the mass analyzer orbit in stable trajectories around the central electrode with harmonic oscillations along it. Two detection electrodes record an image current of the ions as they undergo harmonic oscillations. A Fourier transformation extracts different harmonic frequencies from the image current. An ion's mass-to-charge ratio m/z is related to the frequency f of its harmonic oscillations and to the instrumental constant k by: $m/z = k/f^2$

polarity

The polarity of an ion can be either positive or negative: 1. positive ion: An atom, radical, molecule, or molecular moiety that has lost one or more electrons or has been protonated, thus acquiring an electrically positive charge. The charge state becomes +1, +2, +3, and so on, depending on the number of electrons lost or protons gained. 2. negative ion: An atom, radical, molecule, or molecular moiety that has gained one or more electrons, acquiring an electrically negative charge. The charge state becomes -1, -2, -3, and so on, depending on the number of electrons acquired.

product mass

The mass-to-charge ratio of a product ion. The location of the center of a target production peak in mass-to-charge ratio (m/z) units.

profile data

Data representing mass spectral peaks as point-to-point plots, with each point having an associated intensity value.

resolution

The ability to distinguish between two points on the wavelength or mass axis.

selected ion monitoring (SIM)

A scan type where the mass spectrometer acquires and records ion current following the isolation of a list of singular mass-to-charge ratio values.

selected reaction monitoring (SRM)

A scan type with two stages of mass analysis and in which a particular reaction or set of reactions, such as the fragmentation of an ion or the loss of a neutral moiety, is monitored. In SRM a limited number of product ions is monitored.

signal-to-noise ratio (S/N)

The ratio of the signal height (S) to the noise height (N). The signal height is the baseline corrected peak height. The noise height is the peak-to-peak height of the baseline noise.

source CID

A technique for fragmenting ions in an atmospheric pressure ionization (API) source. Collisions occur between the ion and the background gas, which increase the internal energy of the ion and stimulate its dissociation.

syringe pump

A device that delivers a solution from a syringe at a specified rate.

tandem mass spectrometry

An analytical technique involving two stages of mass analysis. Initially, ions formed in the ion source are analyzed by an initial analyzer. In the subsequent stage, these mass-selected ions are fragmented, and the resulting ionic fragments are further mass analyzed. It is also called as MS/MS or MS².

XML (Extensible Markup Language)

A general-purpose markup language that is used to facilitate the sharing of data across different information systems, particularly via the Internet.

Index

- Icons
- A
- B
- C
- D
- E
- F
- G
- H
- I
- L
- M
- O
- P
- R
- S
- T
- W

Icons

- *.exp
- *.exp

- *.meth
- *.raw
- *.raw

A

- About dialog box
- About dialog box
- acquisition settings box
- advanced peak determination (APD)
- Advanced Template Messages
- AGC target
- AGC target
- AGC target
- AGC target
- AGC target
- AGC target
- AGC target
- AGC target
- alerts console
- analog inputs, external devices
- APCI discharge current and voltage, optimizing
- APPI lamp
- APPI lamp
- application mode
- application mode
- Automatic Gain Control (AGC)
- Automatic Gain Control (AGC)

- Automatic Gain Control (AGC)
- Automatic Gain Control (AGC)
- Automatic Gain Control (AGC)
- Automatic Gain Control (AGC)
- Automatic Gain Control (AGC)
- Automatic Gain Control (AGC)
- auxiliary gas flow rate
 - optimizing
 - range

B

- background flow, of ion source gases
- BioPharma license
- buttons
 - centroid and profile modes
 - divert/inject valve
 - spectrum averaging

C

- calibration
 - checkpoints
 - event messages
- Calibration pane
- calibration solution
- calibrations
 - parameters
 - spray stability

- spray stability
- Camera pane
- carrier gas flow
- Charge State filter
- checkpoints
- Chromatogram view
- color indicators
- color indicators
- contact closure
- contact closure
- contact closure
- contaminants
- Control pane
- custom template
- customer feedback

D

- data acquisition
- Data Dependent Mode options
- data type
- data-dependent scan
- Data-Dependent SIM (ddSIM) scan
- Data-Independent Analysis (DIA)
- ddMS2 scan
- default charge state
- Define Scan pane
- detector type

- detector type
- detector type
- detector type
- detector type
- detector type
- detector type
- Diagnostics pane
- divert valve
- divert valve
- Dynamic Exclusion filter

E

- electrode temperatures
- electrospray voltage, setting
- error
- errors, in the document
- exclusion list
- experiment name
- experiment templates
- experiment workflow
- experiment workflow
- exporting data

F

- FAIMS compensation voltage
- FAIMS compensation voltage
- FAIMS compensation voltage

- FAIMS compensation voltage
- FAIMS compensation voltage
- FAIMS compensation voltage
- FAIMS Mode, setting (optional device)
- FAIMS Mode, setting (optional device)
- Favorites
- Favorites
- filter types
 - apex detection
 - description
 - dynamic exclusion
 - targeted inclusion
 - targeted inclusion
 - targeted inclusion
 - targeted mass exclusion
- flow rate (syringe), properties table

G

- Global Settings Replacement Options dialog box

H

- HCD collision energy
- HCD collision energy
- HCD collision energy
- HCD collision energy
- HCD collision energy
- Help Portal

- H-ESI, heater temperature
- History pane

I

- icons
 - Application Mode
 - operational status
 - system power
- importing data
- importing data; exporting data
- importing data; exporting data
- Instrument Configuration window
- Intensity filter
- ion polarity mode
- ion source
 - APCI discharge current, setting
 - auxiliary gas, setting
 - LC flow rates, setting
 - selecting type
 - sheath gas, setting
 - sweep gas, setting
 - timeline
- Ion Source page
- ion source temperatures
- ion transfer tube (capillary) temperature, setting
- isolation offset
- isolation offset

- isolation offset
- isolation window, setting
- Isotope Exclusion filter

L

- LC Peak width
- License dialog box
- licenses
- licenses
- lock mass correction
- lock mass injection
- lock mass injection
- lock mass injection
- lock mass injection
- lock mass injection
- lock mass injection
- lock mass injection
- lock masses, user-defined for internal calibration
- loop control

M

- manuals
- mass range
- mass range
- mass resolution
- mass resolution
- mass resolution

- mass resolution
- mass resolution
- mass resolution
- mass resolution
- mass resolution
- mass table, buttons
- master scan, description
- method duration
- Method Editor
 - elements of
 - Global Parameters
 - overview
- method timeline
- mild trapping
- monitor mass accuracy, icon
- Monoisotopic Precursor Selection (MIPS) filter
- My Experiments, saved template

O

- Optimization page
- Options menu
- OptiSpray
 - cartridge controls
 - cartridge temperature
 - diagnostics
- OptiSpray ion source

P

- panes
 - Calibration
 - Define Scan
 - Favorites
 - Ion Source, Optimization page
 - Status
- parameters
 - Calibration pane, skip spray stability evaluation
 - Charge State Properties pane
 - filter types
 - Ion Source page
 - recommendations
- patents
- PDF files, of manuals
- Plot Chromatogram dialog box
- Plot Chromatogram dialog box
- polarity mode
- Precursor Fit filter
- precursor mass
- precursor mass
- Precursor Selection Range filter
- precursors, sort order
- prefilter and postfilter, description
- pressure mode
- procedures
 - adding a method template to the custom templates list

- adding a qual view scan
- adding an experiment to the method
- alerts console, using
- chromatogram, plotting
- Control pane, viewing
- deleting an experiment from the method
- favorite state, applying
- favorite state, deleting
- favorite state, loading
- favorite state, renaming
- favorite state, saving
- History pane, viewing or modifying change records
- ion source parameters, optimizing
- mass accuracy and signal stability, monitoring
- reports, generating
- saving an instrument method
- spectrum, zooming in/out
- Status pane, viewing options
- product documentation

R

- raw data files
 - importing
 - naming format
- raw file
- readbacks, color code
- readbacks, color code

- Report Generation Options dialog box
- Report Generation Options dialog box
- reports
- reports for calibrations, diagnostics, and optimization
- RF level
- RF level
- RF level
- RF level
- RF level
- RF level

S

- Save Template dialog box
- scan description
- scan parameter descriptions
- scan parameter descriptions
- scan priority
- scan priority
- scan range mode
- scan types
 - all ion fragmentation (AIF) scan
 - Full Scan
 - MS, selecting
 - MS2 scan, targeted
 - SIM scan, targeted
- schematic drawing, of the instrument
- serial number

- serial number
- sheath gas flow rate
 - optimizing
 - range
- SIM Window Mode
- source gas pressures
- Spectrum view
 - header information
 - shortcut menu commands
- spray stability
- spray stability
- spray voltage
 - optimizing
- Status pane
- Stepped Collision Energy
- Stepped Collision Energy
- Stepped Collision Energy
- Stepped Collision Energy
- Summary page
- Summary page
- sweep gas
- sweep gas
- sweep gas flow rate
 - optimizing
 - range
- syringe pump
- syringe pump
- syringe pump

- syringe volume, properties table
- system
 - changing settings status, button
 - disconnected status, button
 - error status, button
 - initializing status, button
 - ion source off status, button
 - normal status, button
- system templates

T

- Targeted Mass Difference filter
- Targeted Mass Exclusion filter
- Targeted Mass filter
- Targeted Mass Trigger
- Targeted SIM Scan
- third-party manuals
- toolbars
 - Qual View
 - Templates View
- tooltips
- tooltips
- total ion current, optimizing
- training
- Tune application
 - overview
 - scan type

- usage
- tune parameters
 - compound dependent
 - mass dependent
- Tune preferences
- Tune Preferences dialog box
- Tune, using ion source settings

W

- website
- workflow
 - experiment templates
- workflow area
- workflow objects